Université Gaston Berger de Saint-Louis

UFR SCIENCES APPLIQUEES ET DE TECHNOLOGIE(SAT)
SECTION D'INFORMATIQUE
MASTER SCIENCE ET TECHNOLOGIE

RESEAU, SECURITE ET SYSTEMES DISTRIBUES(R2SD)

PROJET FINAL RESEAUX HAUT DEBIT

RESEAUX HAUT DEBIT

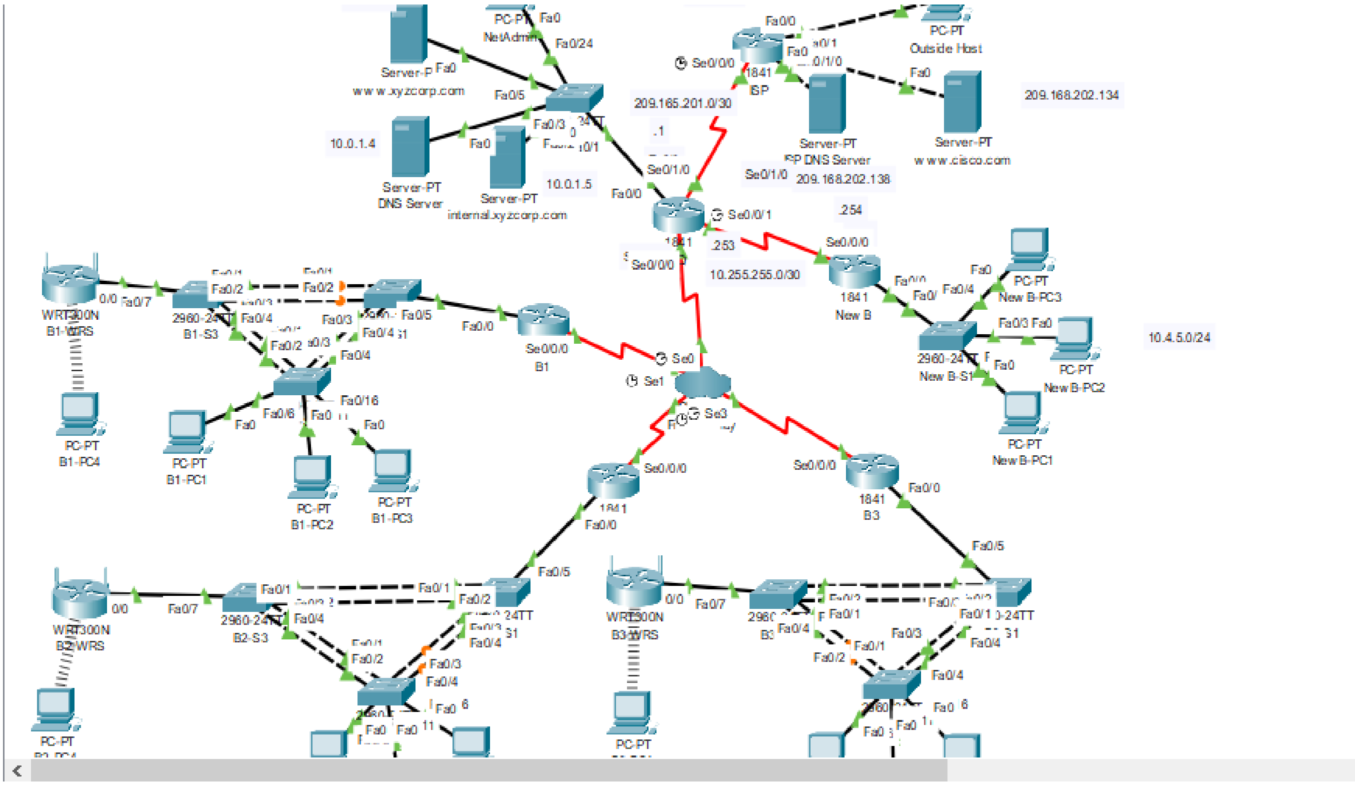
Sous la direction

M. Cheikh Sidy CISSE

Présenté par

Serigne Omar SENE

TOPOLOGIE



Objectifs pédagogiques

Configurer le protocole Frame Relay dans une topologie Hub and Spoke

Configurer l'encapsulation PPP avec authentification CHAP et PAP

Configurer la traduction d'adresses de réseau statique et dynamique

Configurer le routage statique et le routage par défaut

Tâche 1 : configuration du protocole Frame Relay dans une topologie Hub and Spoke

Étape 1 : configuration du cœur de réseau Frame Relay.

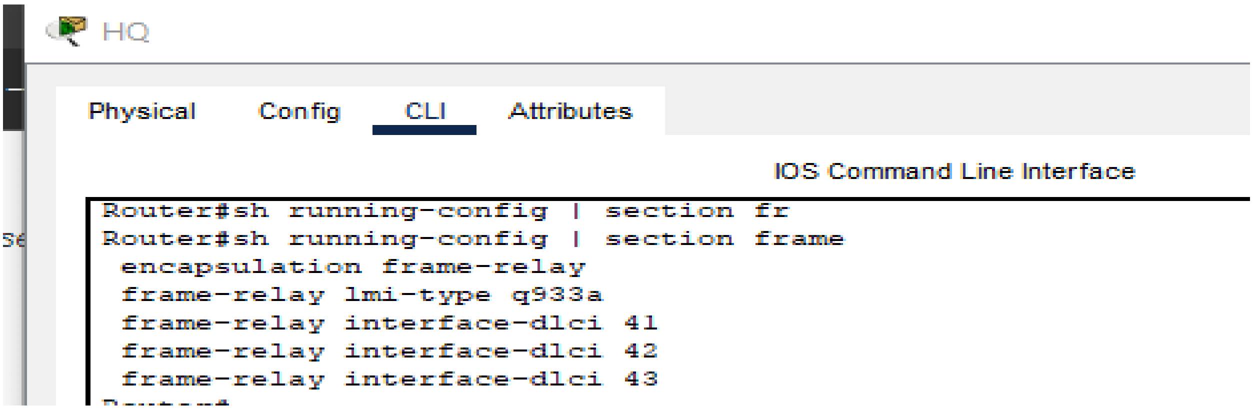
Utilisez les tables d'adressage et respectez les consignes suivantes :

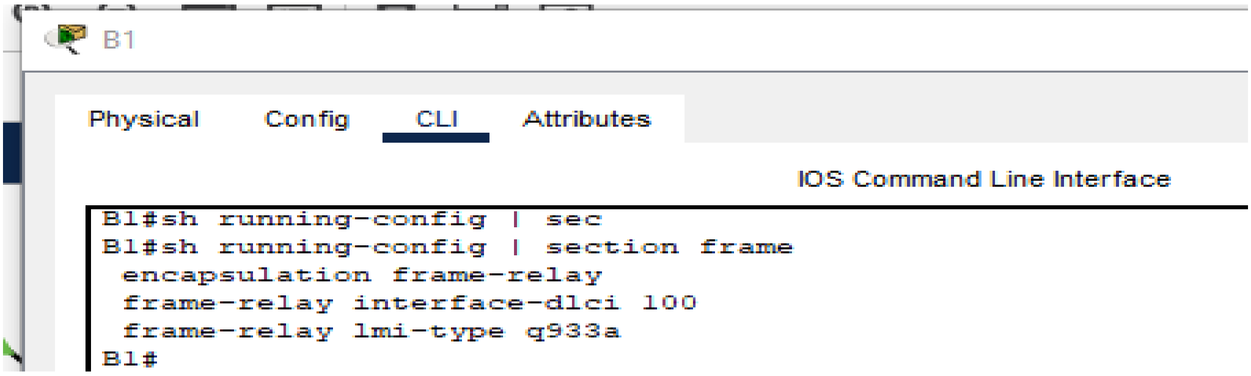
HQ est le routeur concentrateur (hub). B1, B2 et B3 sont les rayons (spokes).

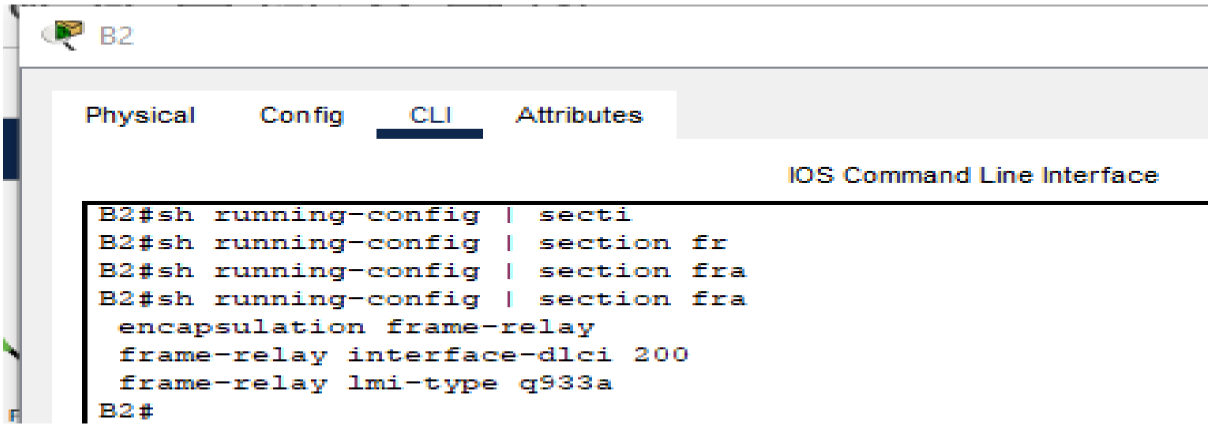
HQ utilise une sous-interface point à point pour chaque routeur Branch.

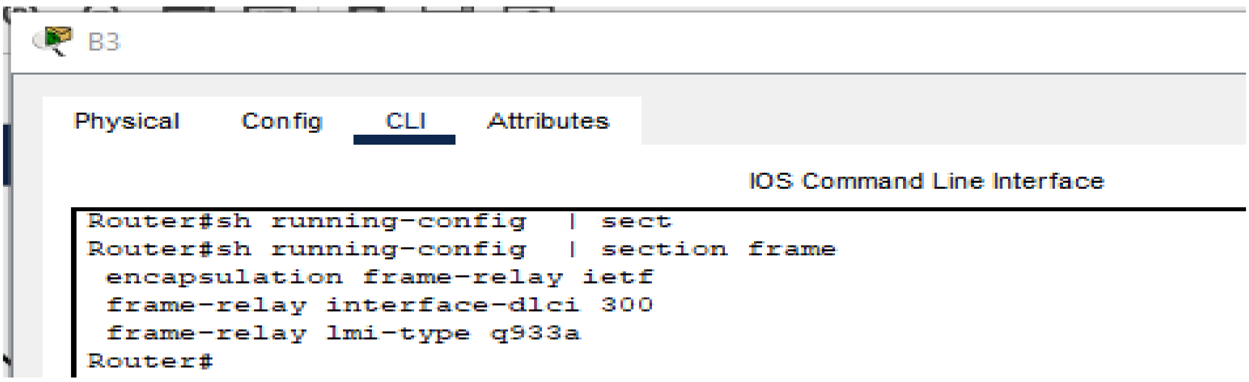
Il est nécessaire d'activer manuellement l'encapsulation IETF sur B3.

Le type LMI doit être configuré manuellement sur la valeur q933a pour HQ, B1 et B2. B3 utilise le protocole ANSI

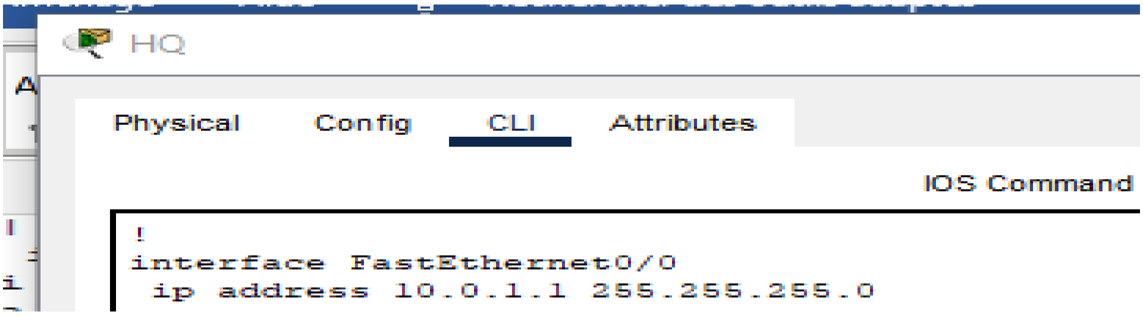

HQ
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router#sh running-config | section fr
Router#sh running-config | section frame
encapsulation frame-relay
frame-relay lmi-type q933a
frame-relay interface-dlci 41
frame-relay interface-dlci 42
frame-relay interface-dlci 43


B1
Physical
Config
CLI
Attributes
IOS Command Line Interface
B1#sh running-config | sec
B1#sh running-config | section frame
encapsulation frame-relay
frame-relay interface-dlci 100
frame-relay lmi-type q933a
B1#

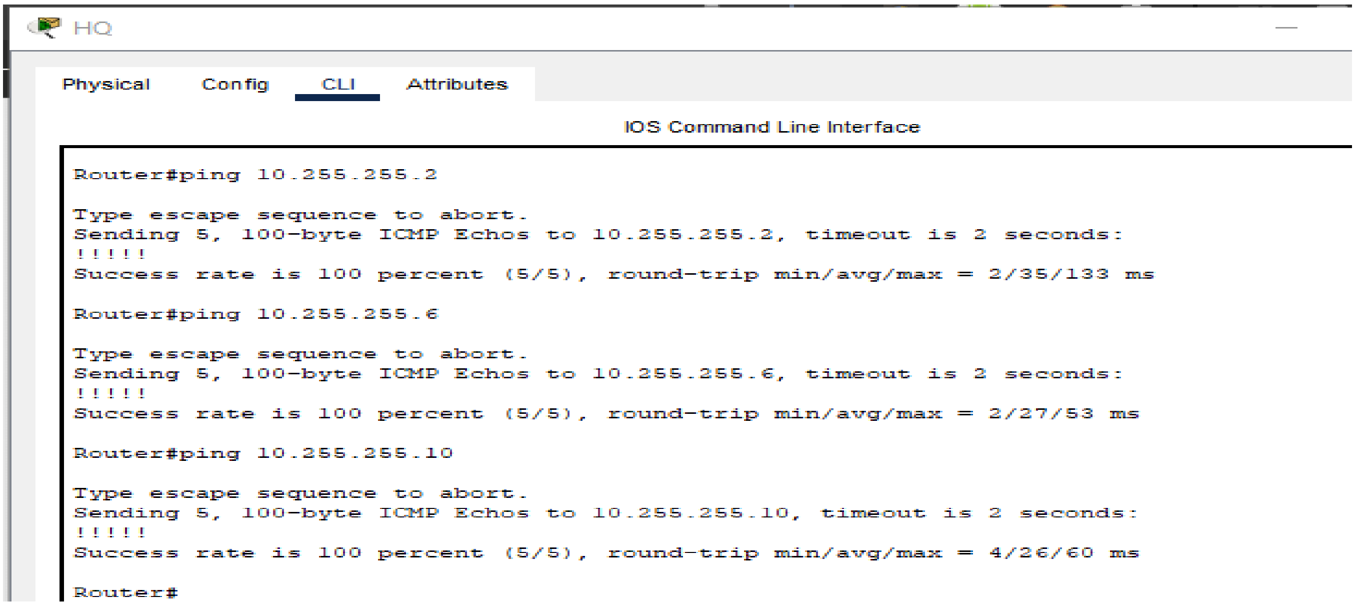

B2
Physical
Config
CLI
Attributes
IOS Command Line Interface
B2#sh running-config | secti
B2#sh running-config | section fr
B2#sh running-config | section fra
B2#sh running-config | section fra
encapsulation frame-relay
frame-relay interface-dlci 200
frame-relay lmi-type q933a
B2#


B3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router#sh running-config | sect
Router#sh running-config | section frame
encapsulation frame-relay ietf
frame-relay interface-dlci 300
frame-relay lmi-type q933a
Router#

Étape 2 : configuration de l'interface de réseau local sur HQ

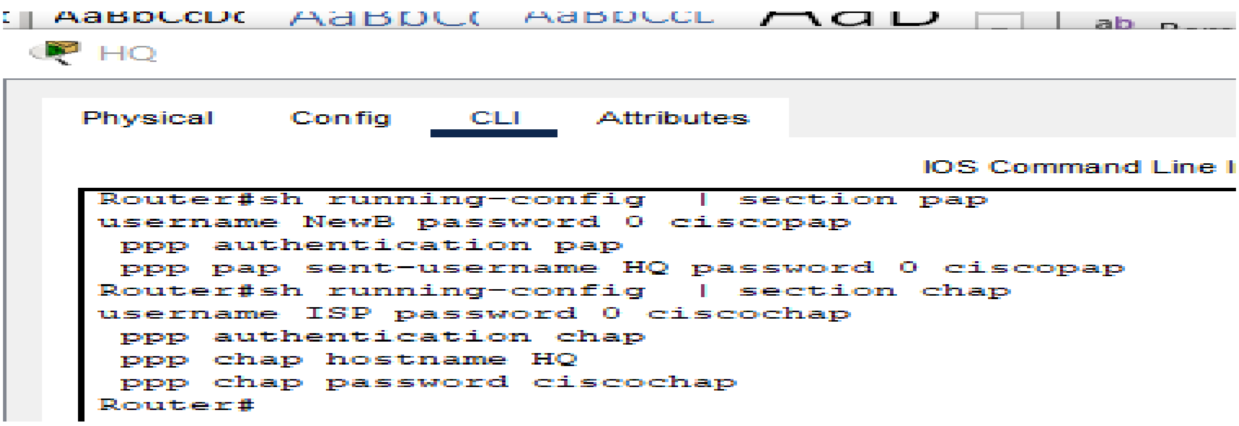


Étape 3 : vérification de l'aboutissement de l'envoi de requêtes ping vers chaque routeur Branch, à partir de HQ



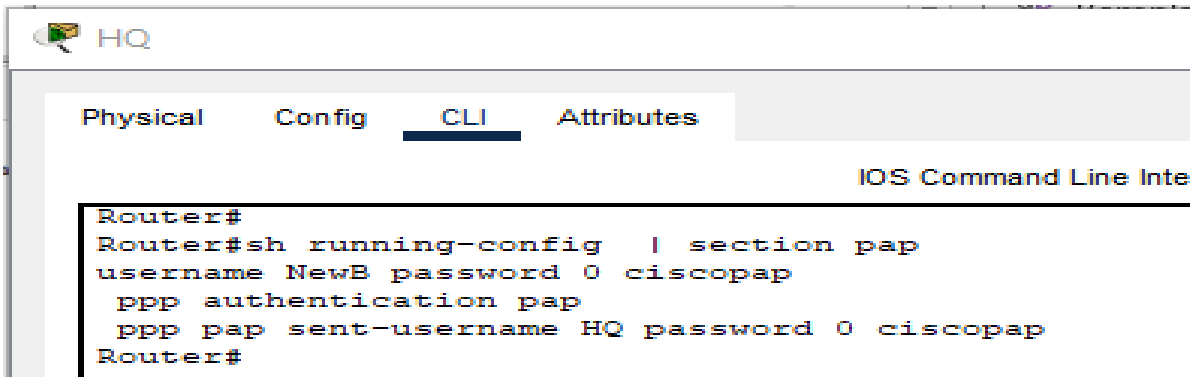
Tâche 2 : configuration de l'encapsulation PPP avec authentification CHAP et PAP

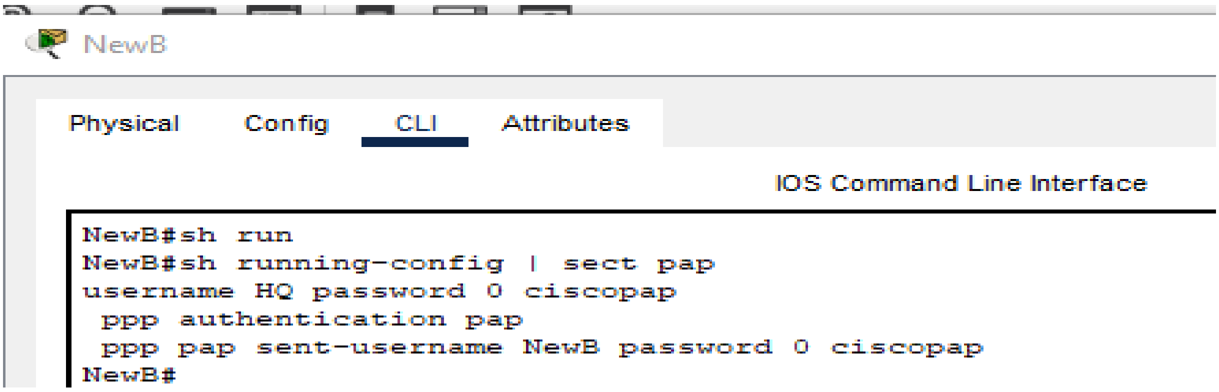
Étape 1 : activation de l'encapsulation PPP avec authentification CHAP sur la liaison WAN reliant HQ et ISP
Le mot de passe CHAP est ciscochap.



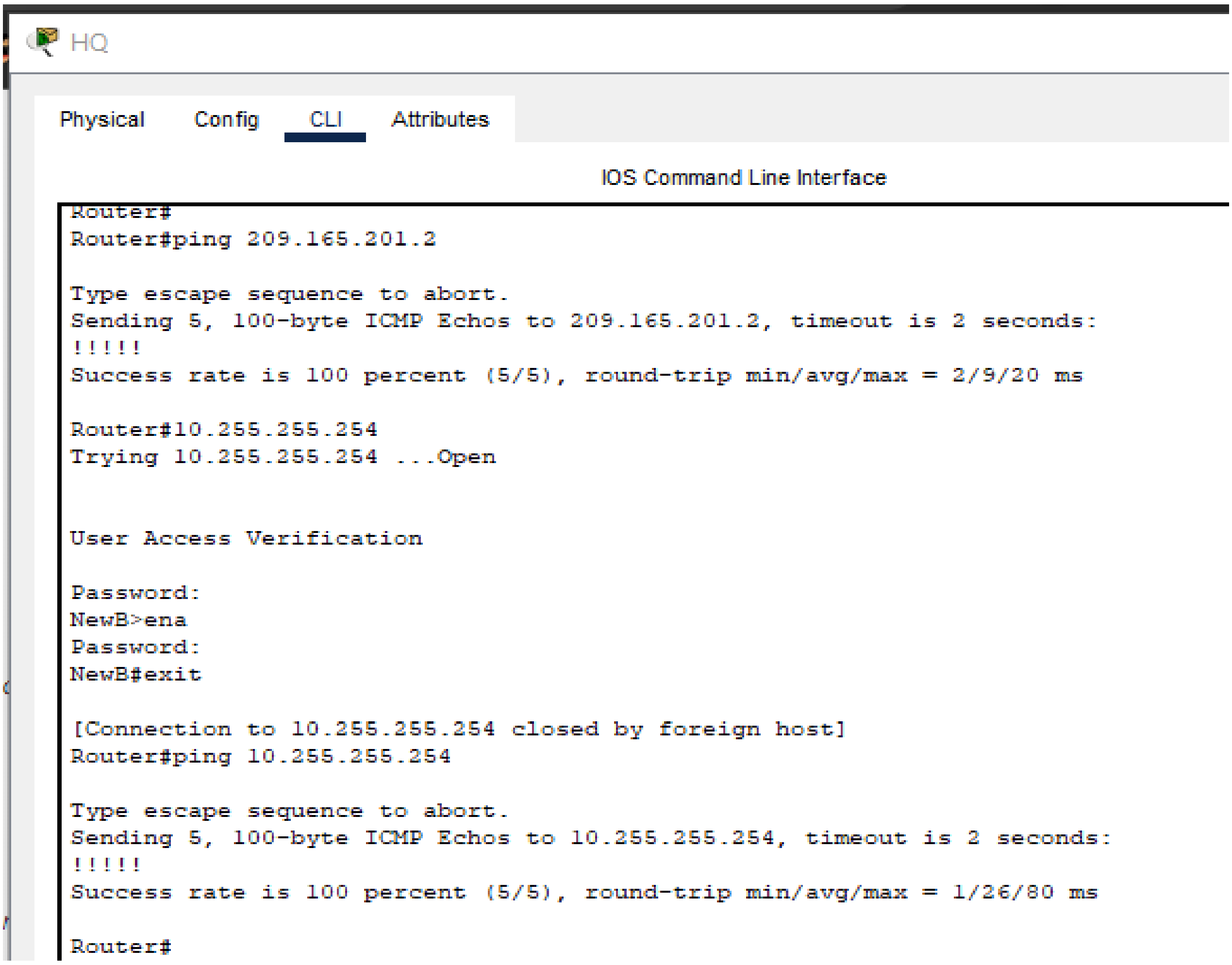
Étape 2 : activation de l'encapsulation PPP avec authentification PAP sur la liaison WAN reliant HQ et NewB

Vous devez connecter un câble aux interfaces adéquates. HQ est l'équipement de communication de données (DCE) de
la liaison. Choisissez la fréquence d'horloge. Le mot de passe PAP est ciscopap.





Étape 3 : vérification de l'aboutissement des envois de requêtes ping vers ISP et NewB à partir de HQ

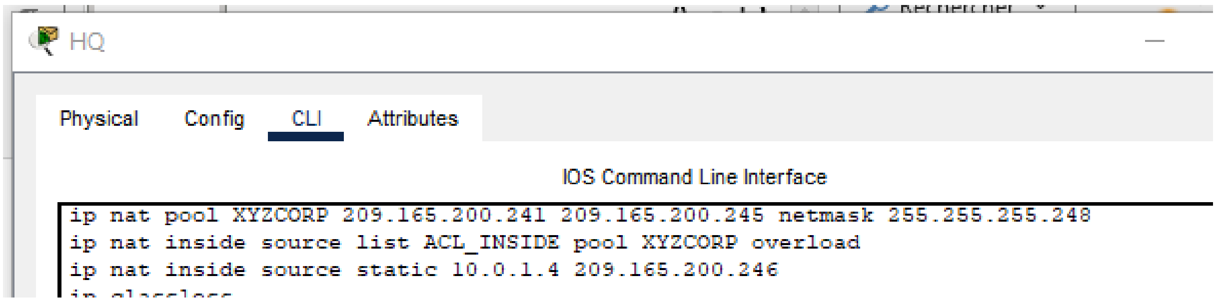


Tâche 3 : configuration de la traduction d’adresses de réseau statique et dynamique sur HQ

Étape 1 : configuration de la traduction d’adresses de réseau

Respectez les consignes suivantes :

- Autorisez la traduction de toutes les adresses de l’espace d’adressage 10.0.0.0/8.
- XYZ Corporation détient l’espace d’adressage 209.165.200.240/29. Le pool d’adresses XYZCORP utilise les adresses .241 à .245 avec un masque /29.
- Le site Web www.xyzcorp.com, dont l’adresse est 10.0.1.2, est inscrit auprès du système DNS public à l’adresse IP 209.165.200.246.



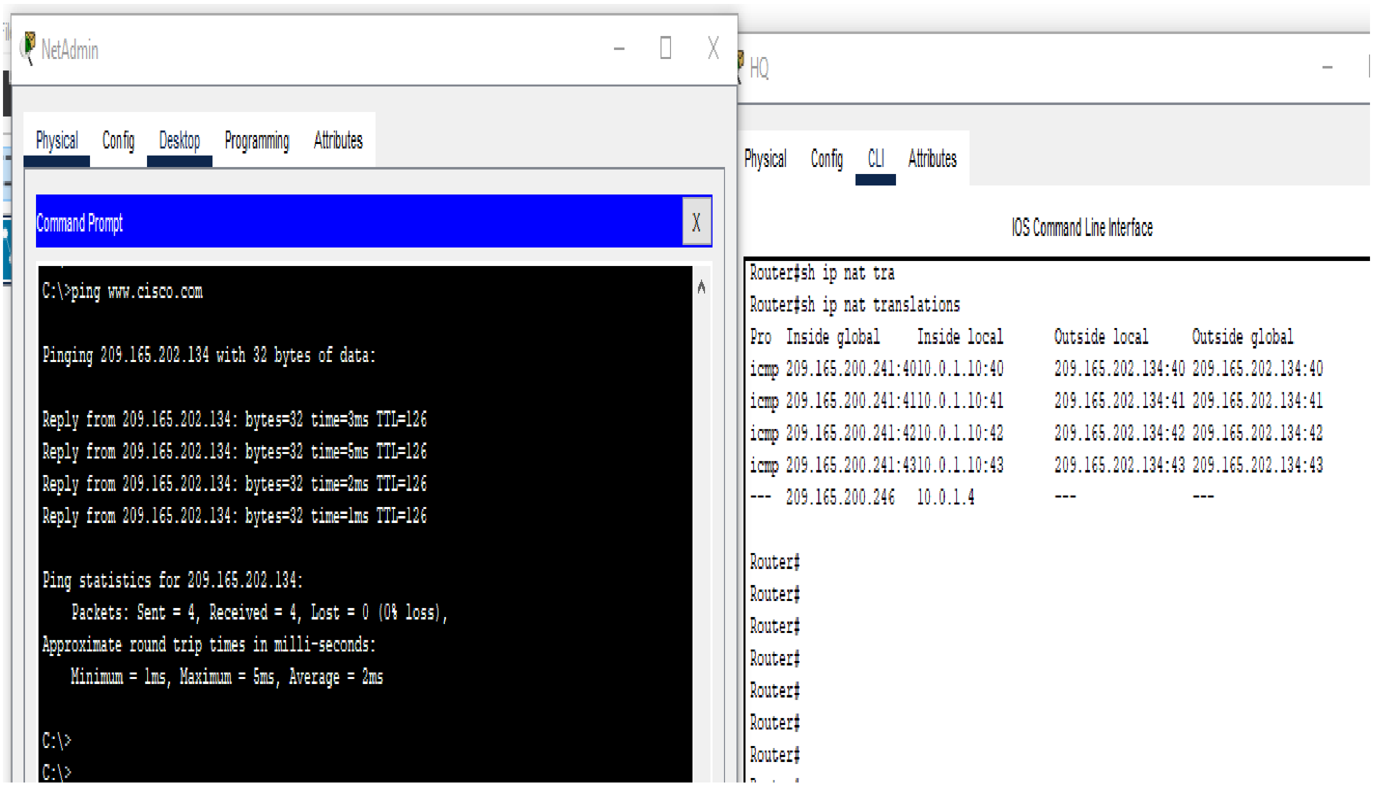
```
!
ip access-list standard ACL_INSIDE
 permit 10.0.0.0 0.255.255.255
```

Étape 2 : vérification du fonctionnement de la traduction d'adresses de réseau par l'envoi de requêtes ping étendues

À partir de HQ, envoyez des requêtes ping vers l'interface série 0/0/0 d'ISP en utilisant comme adresse source l'interface du réseau local HQ. En principe, cet envoi de requêtes ping doit aboutir.

À l'aide de la commande show ip nat translations, vérifiez que la traduction d'adresses de réseau a bien traduit l'envoi de requêtes ping.

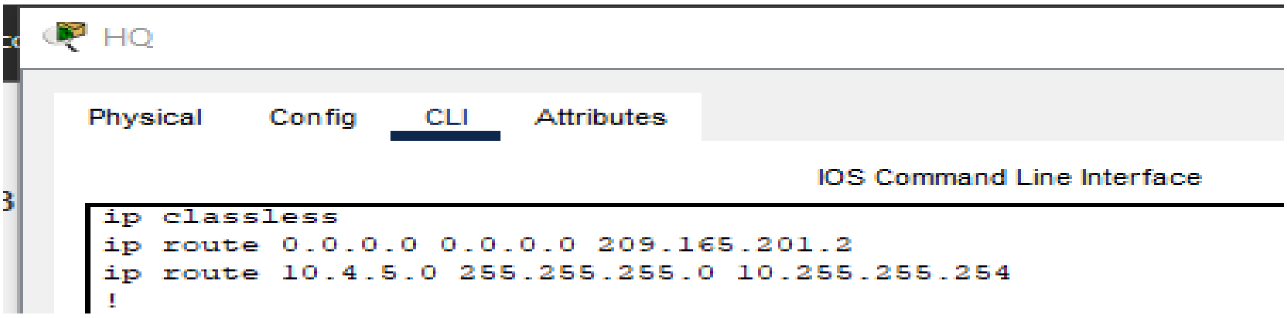
```
% Incomplete command.
Router#sh running-config | section  ip nat
 ip nat inside
 ip nat inside
 ip nat inside
 ip nat inside
 ip nat inside
 ip nat outside
ip nat pool XYZCORP 209.165.200.241 209.165.200.245 netmask 255.255.255.248
ip nat inside source list ACL_INSIDE pool XYZCORP overload
ip nat inside source static 10.0.1.4 209.165.200.246
Router#
```



Tâche 4 : configuration du routage statique et du routage par défaut

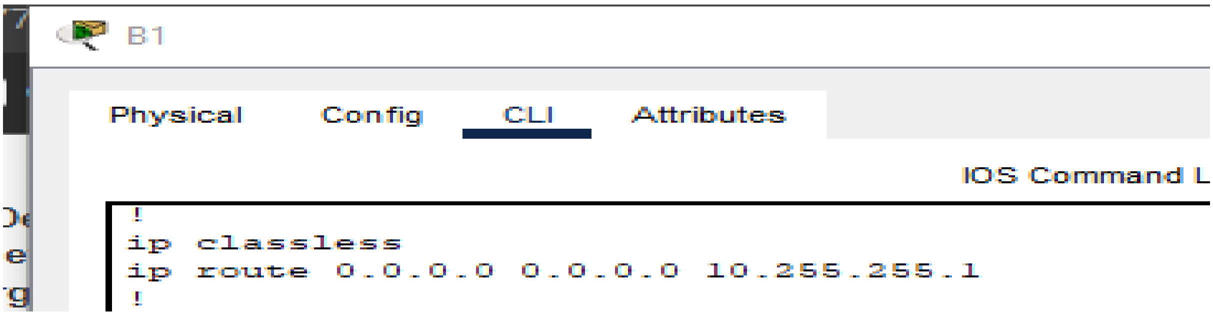
Étape 1 : configuration d’une route par défaut vers ISP et d’une route statique vers le réseau local NewB sur HQ

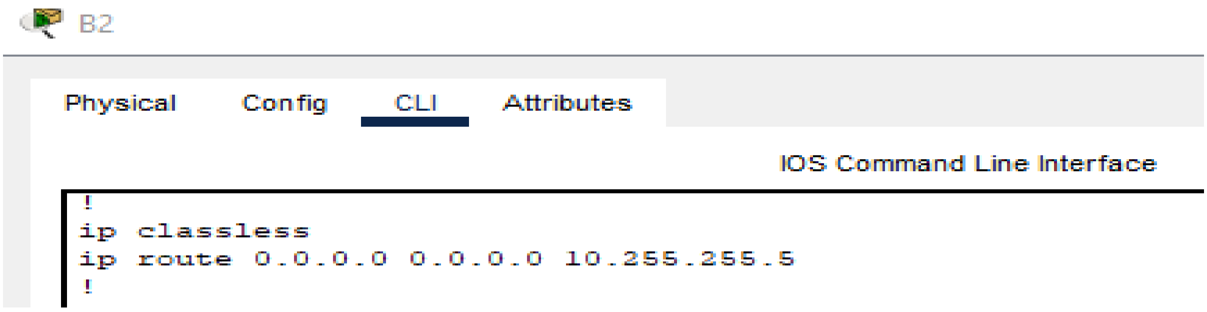
Utilisez l’argument exit interface.

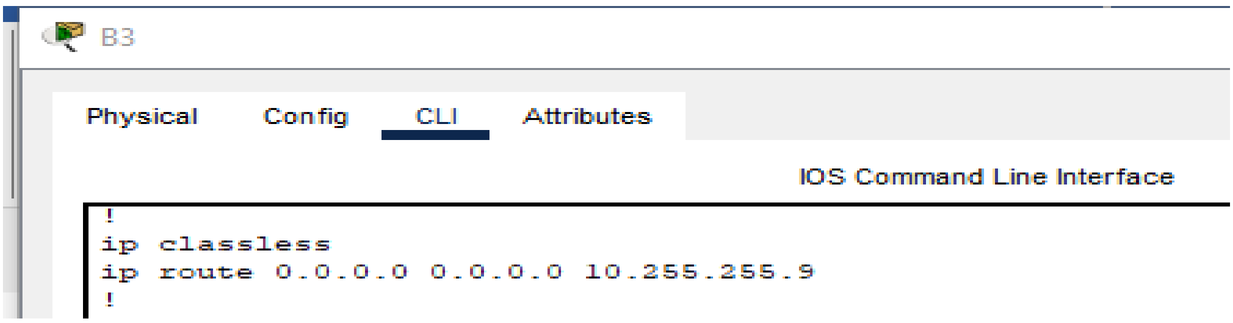


Étape 2 : configuration d’une route par défaut vers HQ sur les routeurs Branch

Utilisez l’adresse IP du tronçon suivant comme argument.

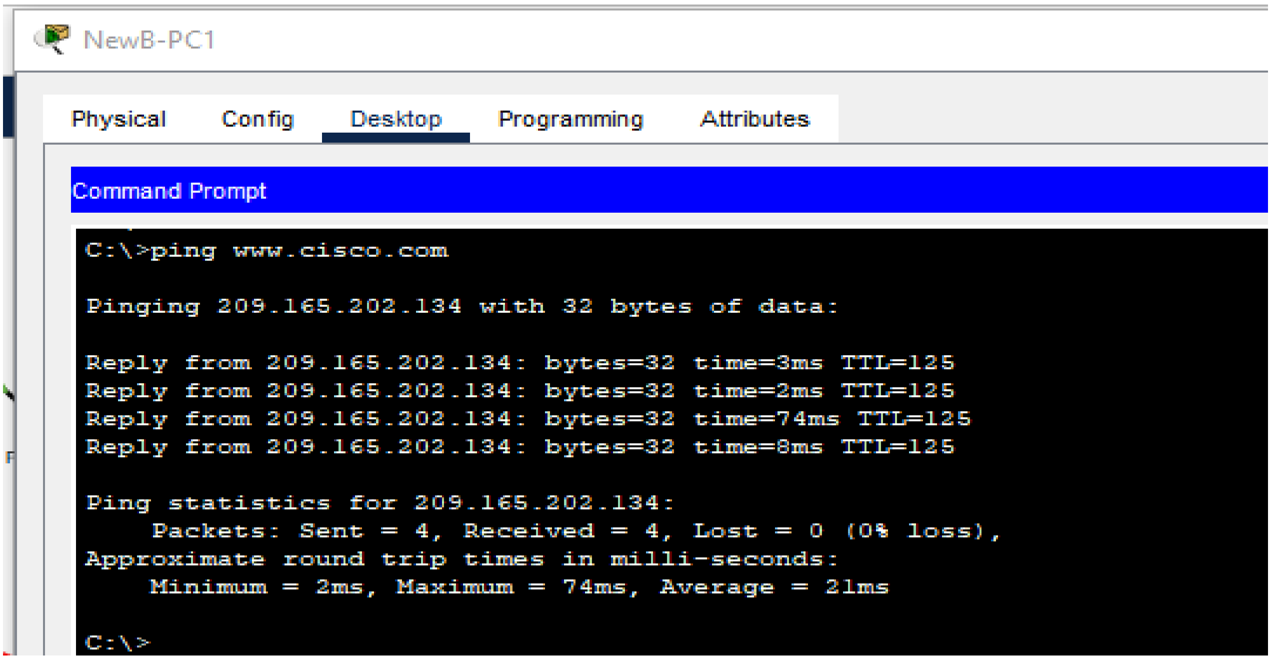


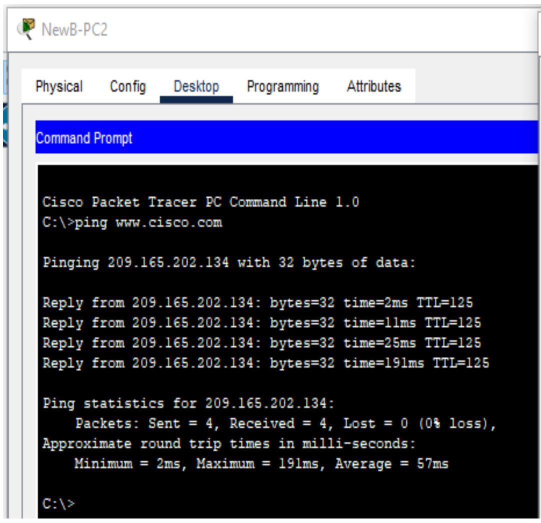


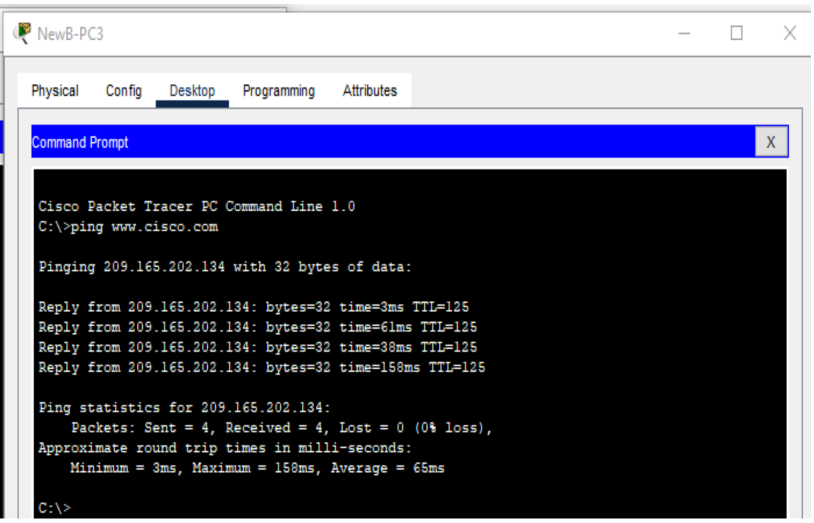


Étape 3 : vérification de la connectivité au-delà d'ISP

Les trois PC NewB et le PC NetAdmin doivent pouvoir envoyer des requêtes ping vers le serveur Web www.cisco.com.





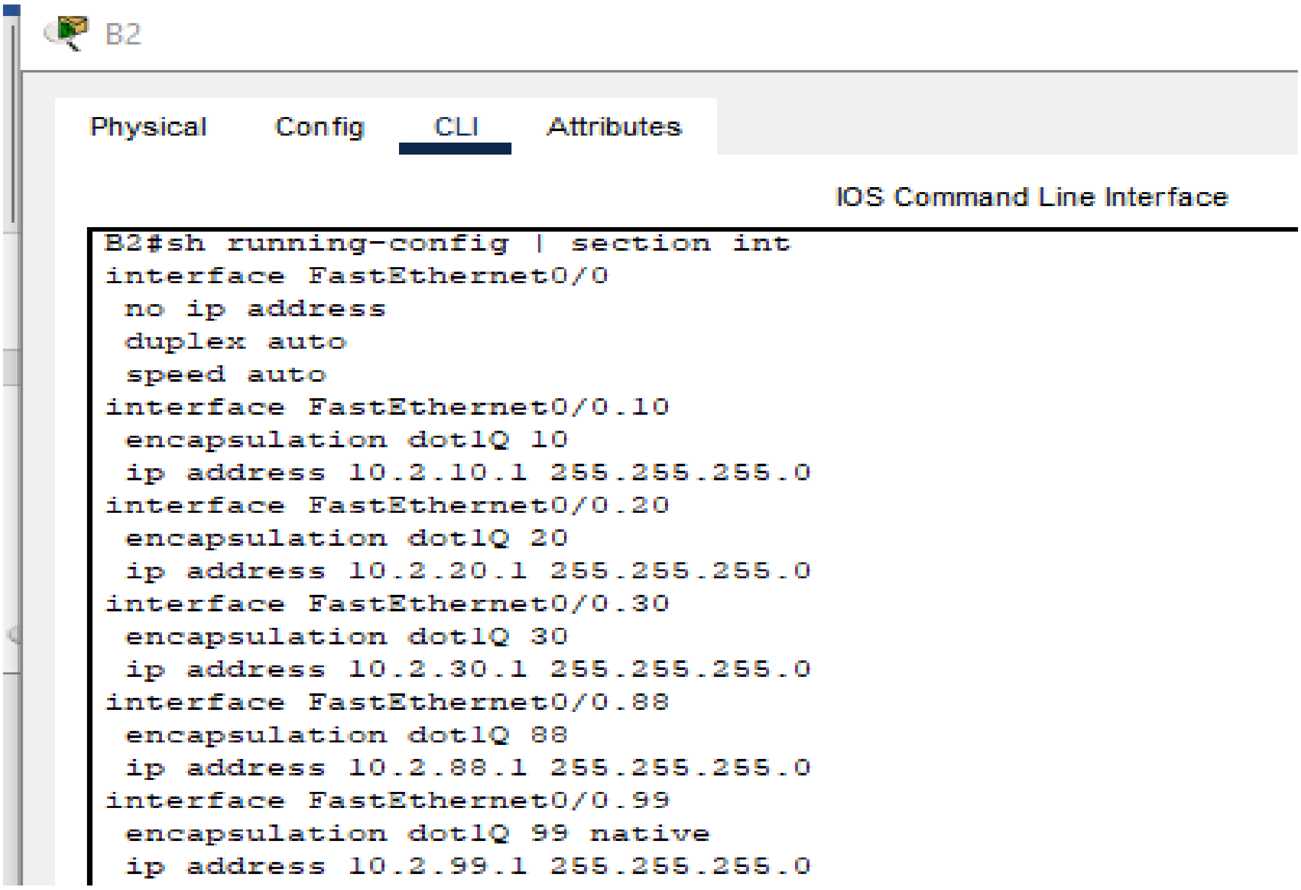


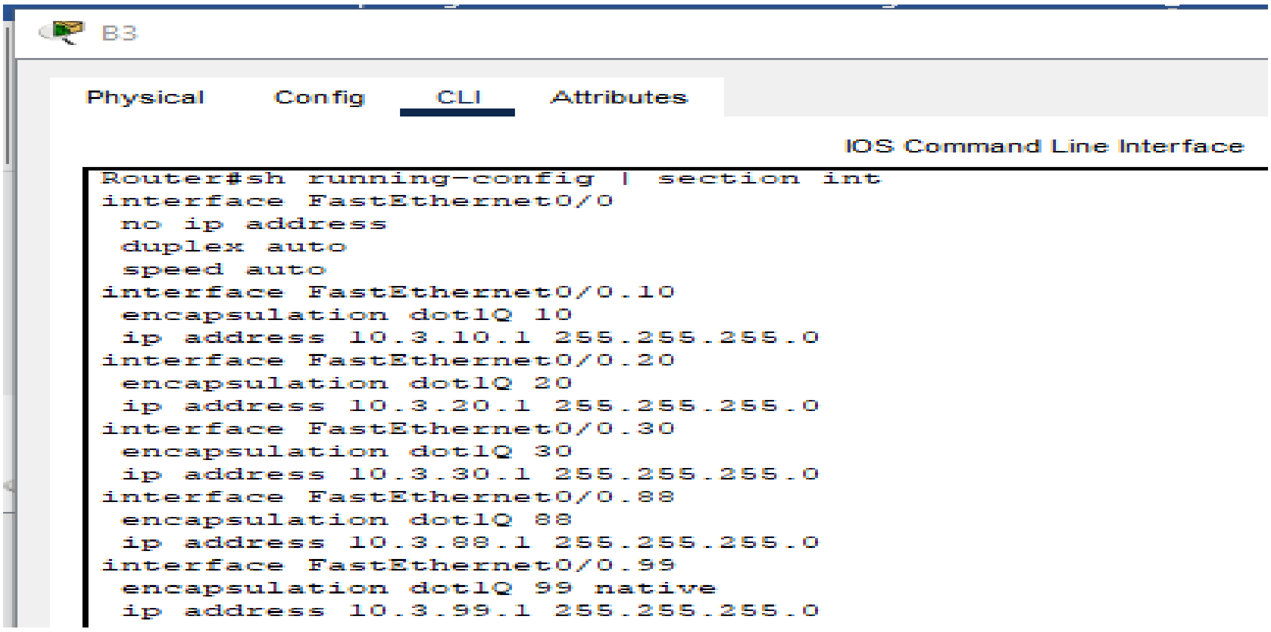
Tâche 5 : configuration du routage entre les réseaux locaux virtuels

Étape 1 : configuration du routage entre les réseaux locaux virtuels sur chaque routeur Branch

À l'aide de la table d'adressage des routeurs Branch, configurez et activez le routage entre les réseaux locaux virtuels sur l'interface de réseau local. Le réseau local virtuel VLAN 99 est le VLAN natif.

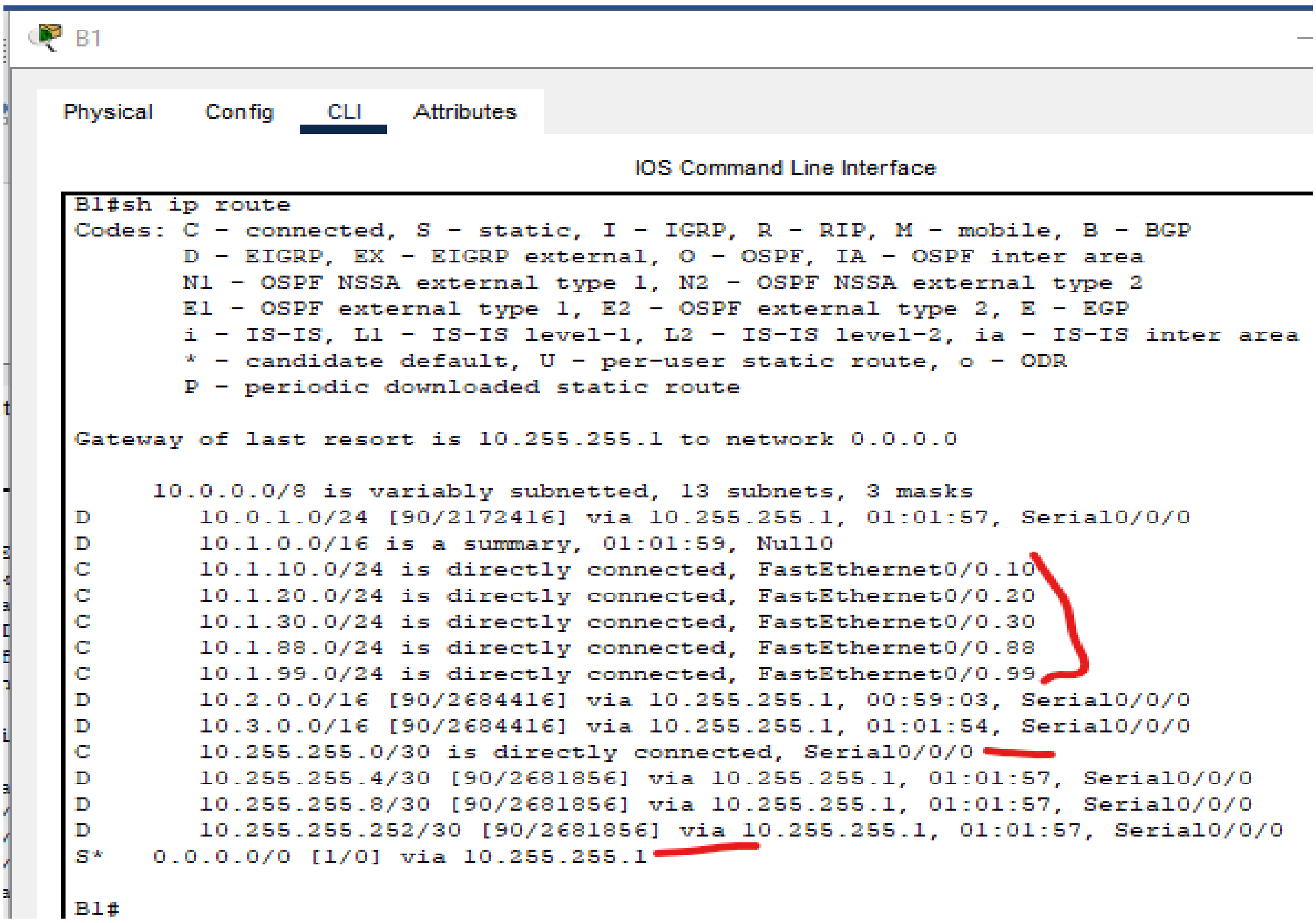

B1
Physical
Config
CLI
Attributes
IOS Command Line Interfa
B1#sh running-config | section int
B1#sh running-config | section int
interface FastEthernet0/0
no ip address
duplex auto
speed auto
interface FastEthernet0/0.10
encapsulation dot1Q 10
ip address 10.1.10.1 255.255.255.0
interface FastEthernet0/0.20
encapsulation dot1Q 20
ip address 10.1.20.1 255.255.255.0
interface FastEthernet0/0.30
encapsulation dot1Q 30
ip address 10.1.30.1 255.255.255.0
interface FastEthernet0/0.88
encapsulation dot1Q 88
ip address 10.1.88.1 255.255.255.0
interface FastEthernet0/0.99
encapsulation dot1Q 99 native
ip address 10.1.99.1 255.255.255.0


B2
Physical
Config
CLI
Attributes
IOS Command Line Interface
B2#sh running-config | section int
interface FastEthernet0/0
no ip address
duplex auto
speed auto
interface FastEthernet0/0.10
encapsulation dot1Q 10
ip address 10.2.10.1 255.255.255.0
interface FastEthernet0/0.20
encapsulation dot1Q 20
ip address 10.2.20.1 255.255.255.0
interface FastEthernet0/0.30
encapsulation dot1Q 30
ip address 10.2.30.1 255.255.255.0
interface FastEthernet0/0.88
encapsulation dot1Q 88
ip address 10.2.88.1 255.255.255.0
interface FastEthernet0/0.99
encapsulation dot1Q 99 native
ip address 10.2.99.1 255.255.255.0



Étape 2 : vérification des tables de routage

Chaque routeur Branch doit désormais disposer de six réseaux directement connectés et d'une route par défaut statique.



B2

Physical Config CLI Attributes

IOS Command Line Interface

```
B2#sh ip rou
B2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.255.5 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 13 subnets, 3 masks
D       10.0.1.0/24 [90/2172416] via 10.255.255.5, 00:57:52, Serial0/0/0
D       10.1.0.0/16 [90/2684416] via 10.255.255.5, 00:57:52, Serial0/0/0
D       10.2.0.0/16 is a summary, 01:00:48, Null0
C       10.2.10.0/24 is directly connected, FastEthernet0/0.10
C       10.2.20.0/24 is directly connected, FastEthernet0/0.20
C       10.2.30.0/24 is directly connected, FastEthernet0/0.30
C       10.2.88.0/24 is directly connected, FastEthernet0/0.88
C       10.2.99.0/24 is directly connected, FastEthernet0/0.99
D       10.3.0.0/16 [90/2684416] via 10.255.255.5, 00:57:52, Serial0/0/0
D       10.255.255.0/30 [90/2681856] via 10.255.255.5, 00:57:52, Serial0/0/0
C       10.255.255.4/30 is directly connected, Serial0/0/0
D       10.255.255.8/30 [90/2681856] via 10.255.255.5, 00:57:52, Serial0/0/0
D       10.255.255.252/30 [90/2681856] via 10.255.255.5, 00:57:52, Serial0/0/0
S*   0.0.0.0/0 [1/0] via 10.255.255.5

B2#
```

B3

Physical Config CLI Attributes

IOS Command Line Interface

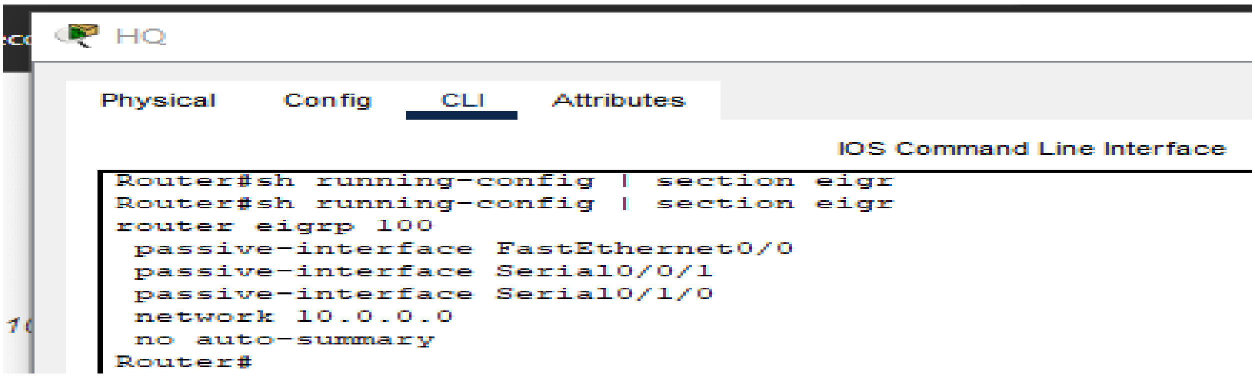
```
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.255.9 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 13 subnets, 3 masks
D       10.0.1.0/24 [90/2172416] via 10.255.255.9, 01:01:33, Serial0/0/0
D       10.1.0.0/16 [90/2684416] via 10.255.255.9, 01:01:30, Serial0/0/0
D       10.2.0.0/16 [90/2684416] via 10.255.255.9, 00:58:39, Serial0/0/0
D       10.3.0.0/16 is a summary, 01:01:35, Null0
C       10.3.10.0/24 is directly connected, FastEthernet0/0.10
C       10.3.20.0/24 is directly connected, FastEthernet0/0.20
C       10.3.30.0/24 is directly connected, FastEthernet0/0.30
C       10.3.88.0/24 is directly connected, FastEthernet0/0.88
C       10.3.99.0/24 is directly connected, FastEthernet0/0.99
D       10.255.255.0/30 [90/2681856] via 10.255.255.9, 01:01:33, Serial0/0/0
D       10.255.255.4/30 [90/2681856] via 10.255.255.9, 01:01:33, Serial0/0/0
C       10.255.255.8/30 is directly connected, Serial0/0/0
D       10.255.255.252/30 [90/2681856] via 10.255.255.9, 01:01:33, Serial0/0/0
S*   0.0.0.0/0 [1/0] via 10.255.255.9

Router#
```

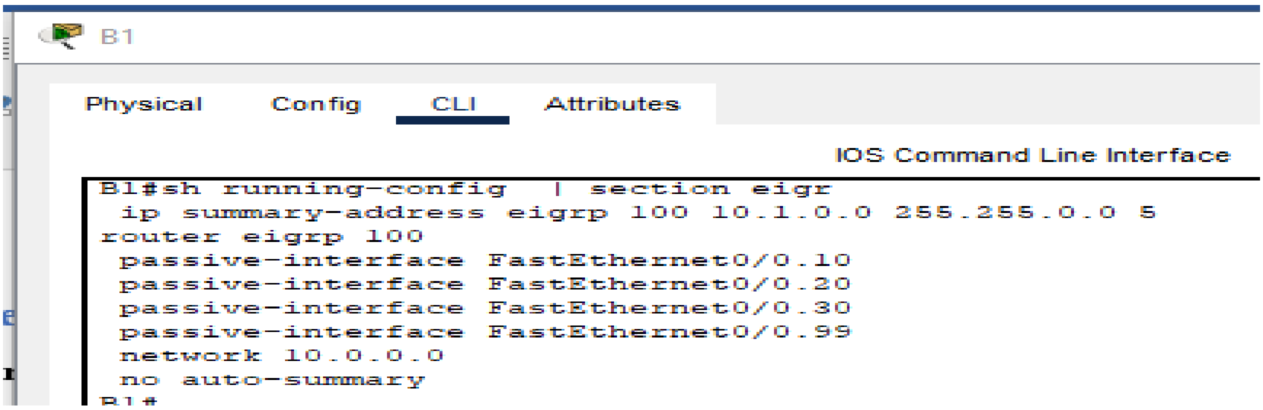
Tâche 6 : configuration et optimisation du routage EIGRP

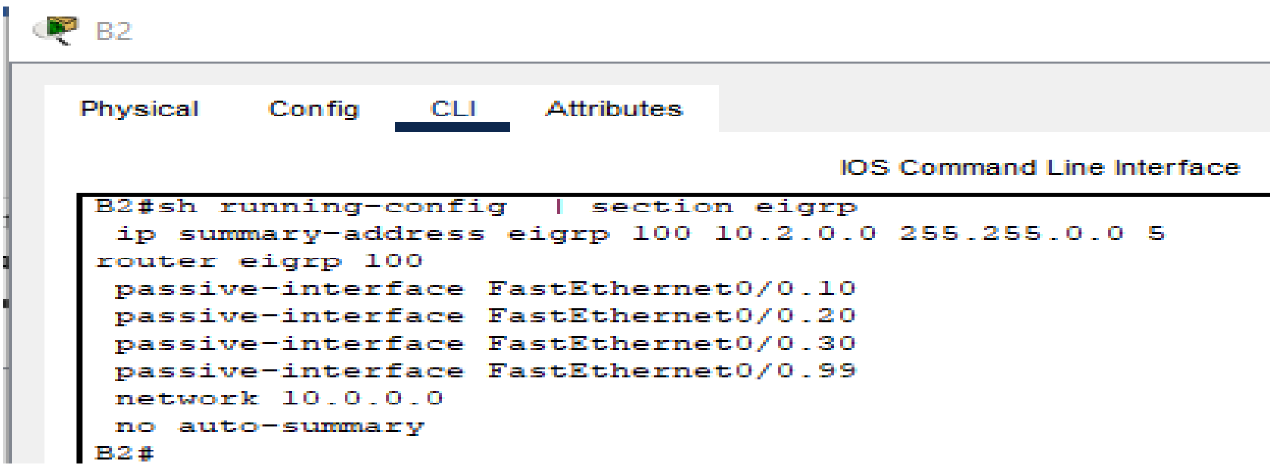
Étape 1 : configuration du protocole EIGRP sur HQ, B1, B2 et B3

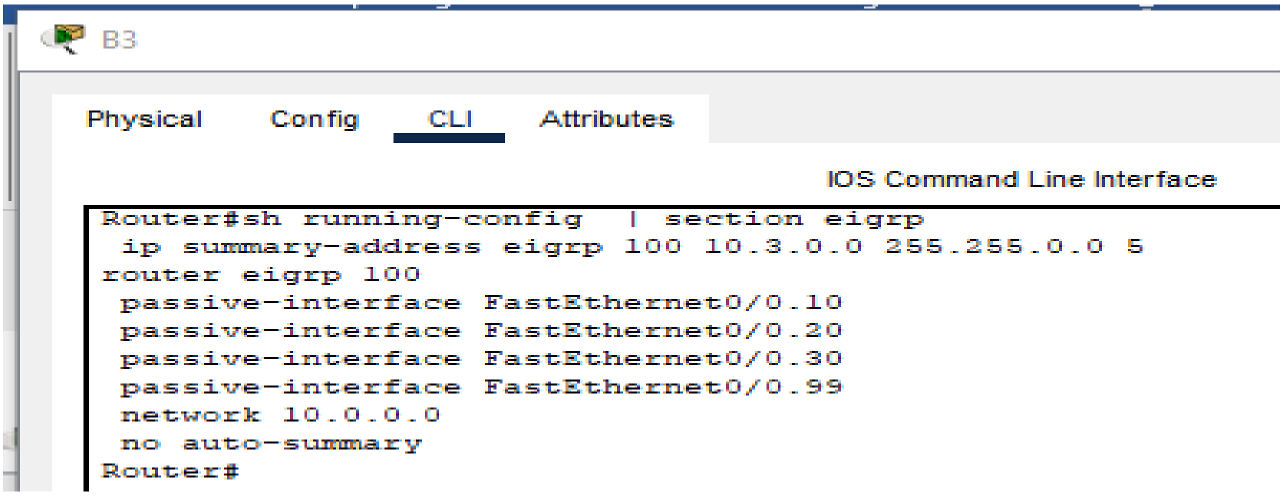
- Utilisez AS 100.
- Désactivez les mises à jour EIGRP sur les interfaces adéquates.
- Résumez manuellement le routage EIGRP de manière à ce que chaque routeur Branch annonce uniquement l'espace d'adressage 10.X.0.0/16 à HQ.

Remarque : Packet Tracer ne simule pas correctement les avantages liés aux résumés du routage EIGRP. Les tables de routage continueront à afficher tous les sous-réseaux, même si vous avez configuré correctement le résumé du routage.





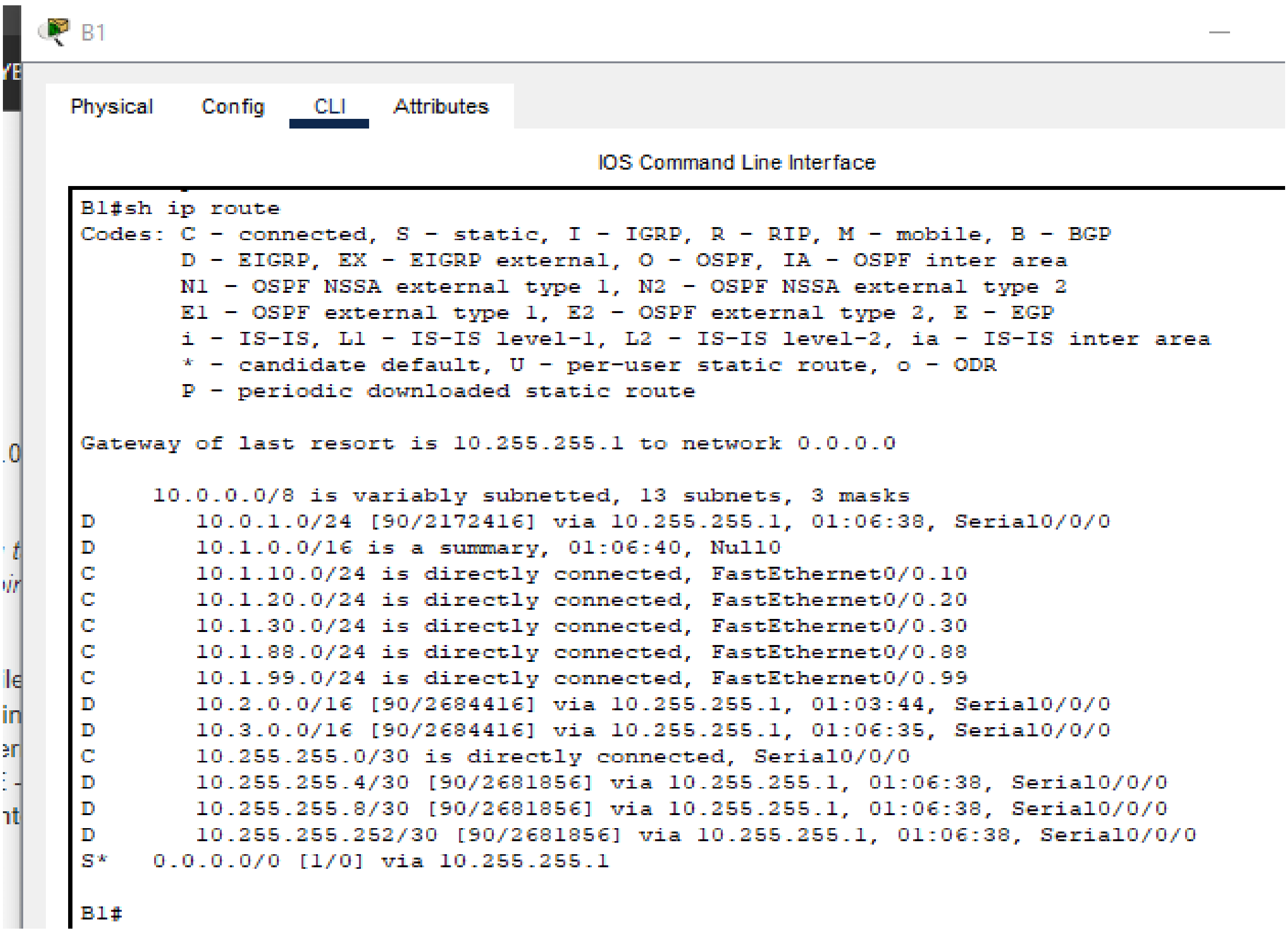




Étape 2 : vérification des tables de routage et de la connectivité

Les routeurs HQ et Branch doivent désormais disposer de tables de routage complètes.

Le PC NetAdmin doit désormais être capable d'envoyer des paquets ping vers chacune des sous-interfaces VLAN des routeurs Branch.



B2

Physical Config CLI Attributes

IOS Command Line Interface

```
B2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.255.5 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 13 subnets, 3 masks
D       10.0.1.0/24 [90/2172416] via 10.255.255.5, 01:04:08, Serial0/0/0
D       10.1.0.0/16 [90/2684416] via 10.255.255.5, 01:04:08, Serial0/0/0
D       10.2.0.0/16 is a summary, 01:07:04, Null0
C       10.2.10.0/24 is directly connected, FastEthernet0/0.10
C       10.2.20.0/24 is directly connected, FastEthernet0/0.20
C       10.2.30.0/24 is directly connected, FastEthernet0/0.30
C       10.2.88.0/24 is directly connected, FastEthernet0/0.88
C       10.2.99.0/24 is directly connected, FastEthernet0/0.99
D       10.3.0.0/16 [90/2684416] via 10.255.255.5, 01:04:08, Serial0/0/0
D       10.255.255.0/30 [90/2681856] via 10.255.255.5, 01:04:08, Serial0/0/0
C       10.255.255.4/30 is directly connected, Serial0/0/0
D       10.255.255.8/30 [90/2681856] via 10.255.255.5, 01:04:08, Serial0/0/0
D       10.255.255.252/30 [90/2681856] via 10.255.255.5, 01:04:08, Serial0/0/0
S*   0.0.0.0/0 [1/0] via 10.255.255.5

B2#
```

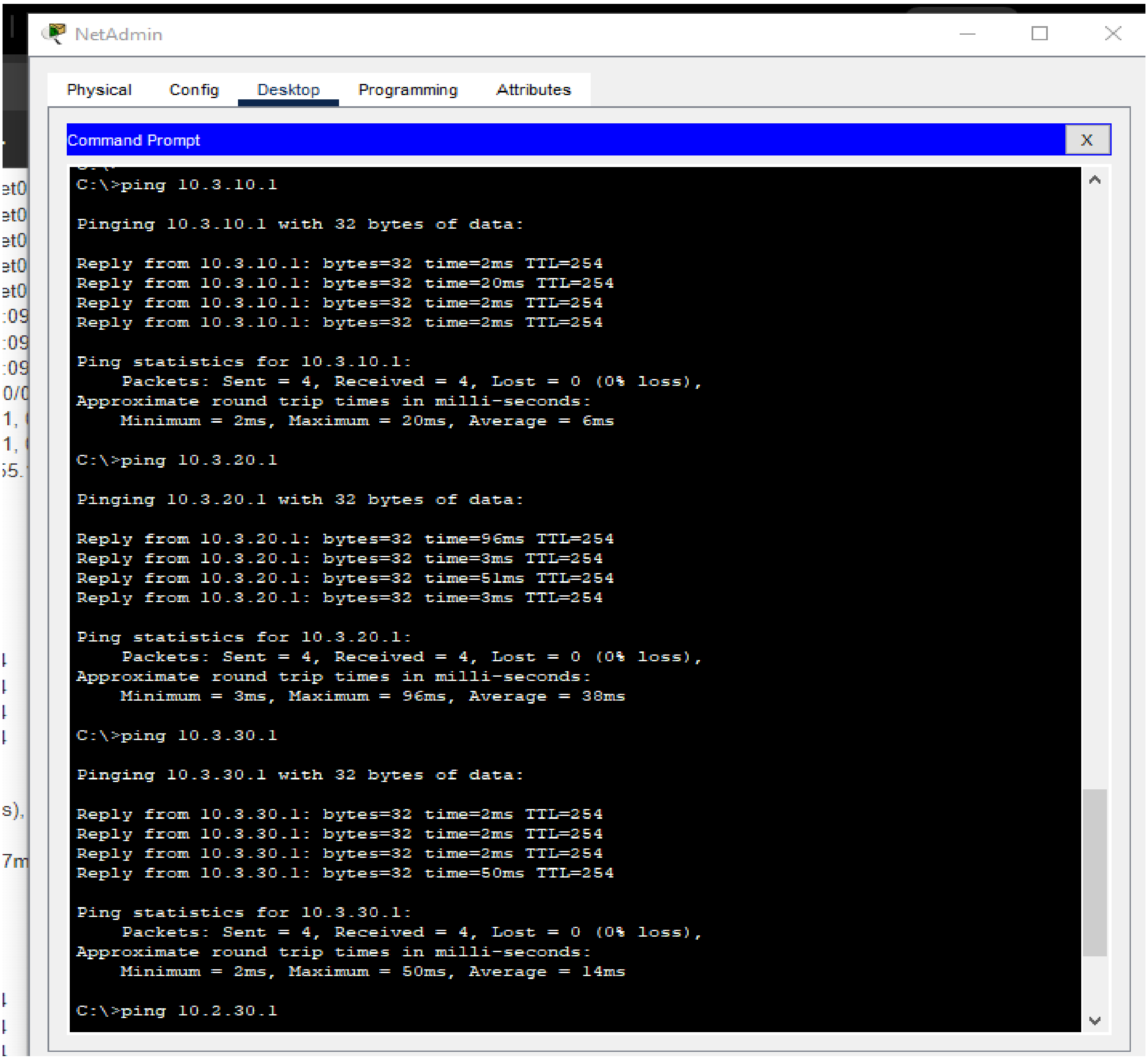
B3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#sh ip rou
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.255.9 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 13 subnets, 3 masks
D       10.0.1.0/24 [90/2172416] via 10.255.255.9, 01:07:26, Serial0/0/0
D       10.1.0.0/16 [90/2684416] via 10.255.255.9, 01:07:23, Serial0/0/0
D       10.2.0.0/16 [90/2684416] via 10.255.255.9, 01:04:32, Serial0/0/0
D       10.3.0.0/16 is a summary, 01:07:28, Null0
C       10.3.10.0/24 is directly connected, FastEthernet0/0.10
C       10.3.20.0/24 is directly connected, FastEthernet0/0.20
C       10.3.30.0/24 is directly connected, FastEthernet0/0.30
C       10.3.88.0/24 is directly connected, FastEthernet0/0.88
C       10.3.99.0/24 is directly connected, FastEthernet0/0.99
D       10.255.255.0/30 [90/2681856] via 10.255.255.9, 01:07:26, Serial0/0/0
D       10.255.255.4/30 [90/2681856] via 10.255.255.9, 01:07:26, Serial0/0/0
C       10.255.255.8/30 is directly connected, Serial0/0/0
D       10.255.255.252/30 [90/2681856] via 10.255.255.9, 01:07:26, Serial0/0/0
S*   0.0.0.0/0 [1/0] via 10.255.255.9

Router#
```

Test de connectivité



Tâche 7 : configuration du protocole VTP, de l'agrégation, de l'interface VLAN et des réseaux locaux virtuels

Les conditions suivantes s'appliquent aux trois branches. Configurez un jeu de trois commutateurs. Utilisez ensuite les scripts de ces commutateurs sur les deux autres jeux de commutateurs.

Étape 1 : configuration du protocole VTP sur les commutateurs Branch

BX-S1 est le serveur VTP. BX-S2 et BX-S3 sont des clients VTP.

Le nom de domaine est XYZCORP.

Le mot de passe est xyzvtp.

B1-S1

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#sh vtp status
VTP Version                     : 1
Configuration Revision          : 22
Maximum VLANs supported locally : 255
Number of existing VLANs        : 10
VTP Operating Mode              : Server
VTP Domain Name                 : XYZCORP
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Disabled
VTP Traps Generation            : Disabled
MD5 digest                      : 0xA7 0x51 0x3D 0x9F 0x09 0xE7 0xD3 0xA6
Configuration last modified by 10.1.99.21 at 3-1-93 02:02:16
Local updater ID is 10.1.99.21 on interface Vl99 (lowest numbered VLAN interface found)
Switch#
Switch#
```

B1-S2

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#sh vtp status
VTP Version                     : 1
Configuration Revision          : 22
Maximum VLANs supported locally : 255
Number of existing VLANs        : 10
VTP Operating Mode              : Client
VTP Domain Name                 : XYZCORP
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Disabled
VTP Traps Generation            : Disabled
MD5 digest                      : 0xA7 0x51 0x3D 0x9F 0x09 0xE7 0xD3 0xA6
Configuration last modified by 10.1.99.21 at 3-1-93 02:02:16
Switch#
```

B1-S3

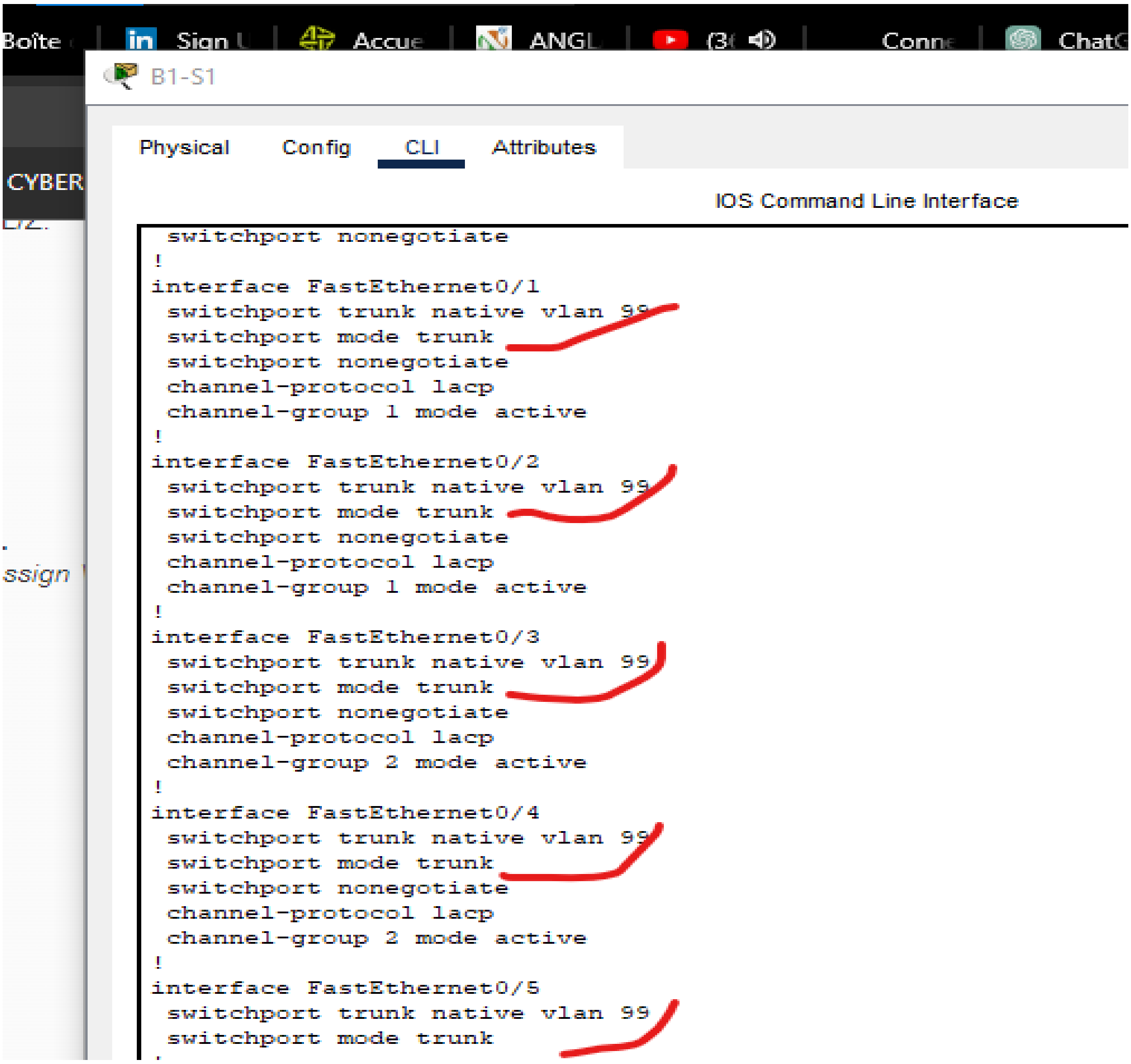
Physical Config CLI Attributes

IOS Command Line Interface

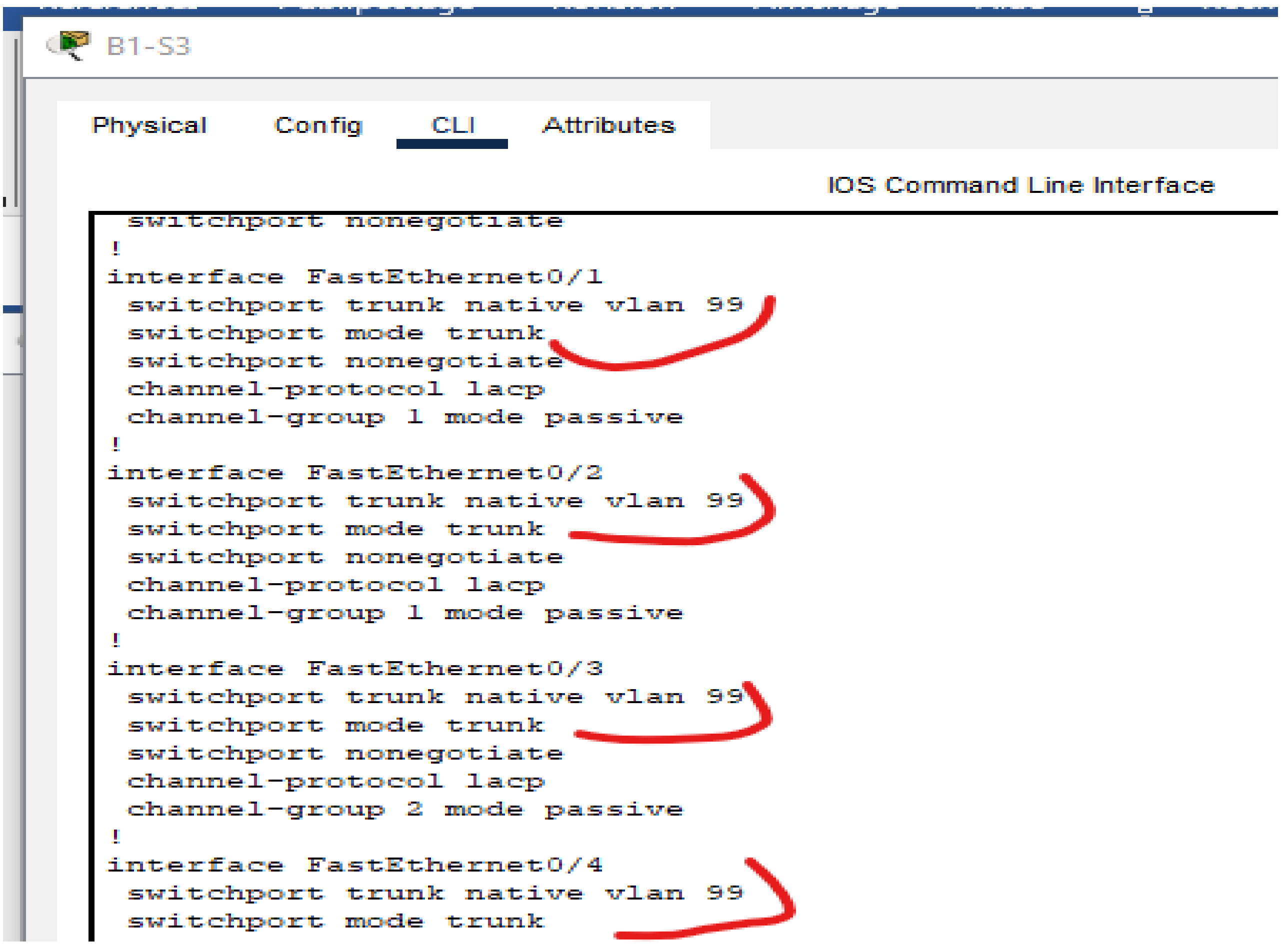
```
Switch#sh vtp status
VTP Version                     : 1
Configuration Revision          : 22
Maximum VLANs supported locally : 255
Number of existing VLANs        : 10
VTP Operating Mode              : Client
VTP Domain Name                 : XYZCORP
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Disabled
VTP Traps Generation            : Disabled
MD5 digest                      : 0xA7 0x51 0x3D 0x9F 0x09 0xE7 0xD3 0xA6
Configuration last modified by 10.1.99.21 at 3-1-93 02:02:16
Switch#
```

Étape 2 : configuration de l'agrégation sur BX-S1, BX-S2 et BX-S3

Configurez les interfaces adéquates en mode d'agrégation et désignez le réseau local virtuel VLAN 99 en tant que VLAN natif




B1-S2
Physical
Config
CLI
Attributes
IOS Command Line Interface
switchport nonegotiate
!
interface FastEthernet0/1
switchport trunk native vlan 99
switchport mode trunk
switchport nonegotiate
channel-protocol lacp
channel-group 2 mode active
!
interface FastEthernet0/2
switchport trunk native vlan 99
switchport mode trunk
switchport nonegotiate
channel-protocol lacp
channel-group 2 mode active
!
interface FastEthernet0/3
switchport trunk native vlan 99
switchport mode trunk
switchport nonegotiate
channel-protocol lacp
channel-group 1 mode passive
!
interface FastEthernet0/4
switchport trunk native vlan 99
switchport mode trunk
switchport nonegotiate
channel-protocol lacp
channel-group 1 mode passive

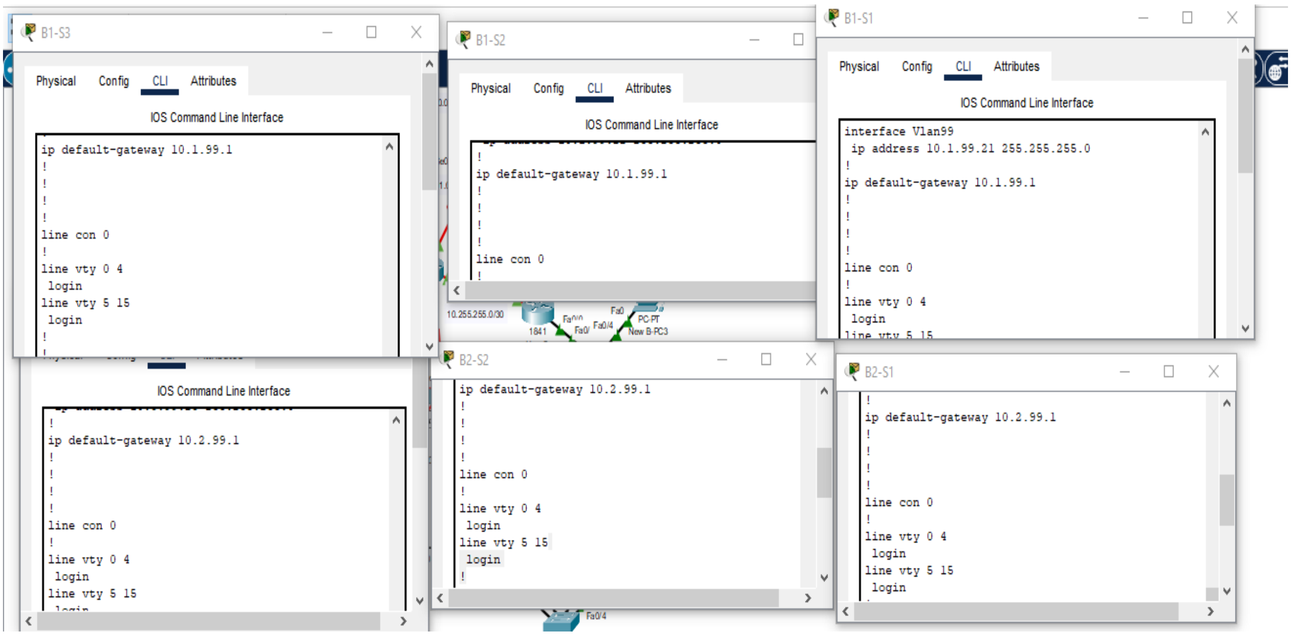

B1-S3
Physical
Config
CLI
Attributes
IOS Command Line Interface
switchport nonegotiate
!
interface FastEthernet0/1
switchport trunk native vlan 99
switchport mode trunk
switchport nonegotiate
channel-protocol lacp
channel-group 1 mode passive
!
interface FastEthernet0/2
switchport trunk native vlan 99
switchport mode trunk
switchport nonegotiate
channel-protocol lacp
channel-group 1 mode passive
!
interface FastEthernet0/3
switchport trunk native vlan 99
switchport mode trunk
switchport nonegotiate
channel-protocol lacp
channel-group 2 mode passive
!
interface FastEthernet0/4
switchport trunk native vlan 99
switchport mode trunk

Faire la même chose pour :

B2-S1, B2-S2, B2-S3

B3-S1, B3-S2, B3-S3

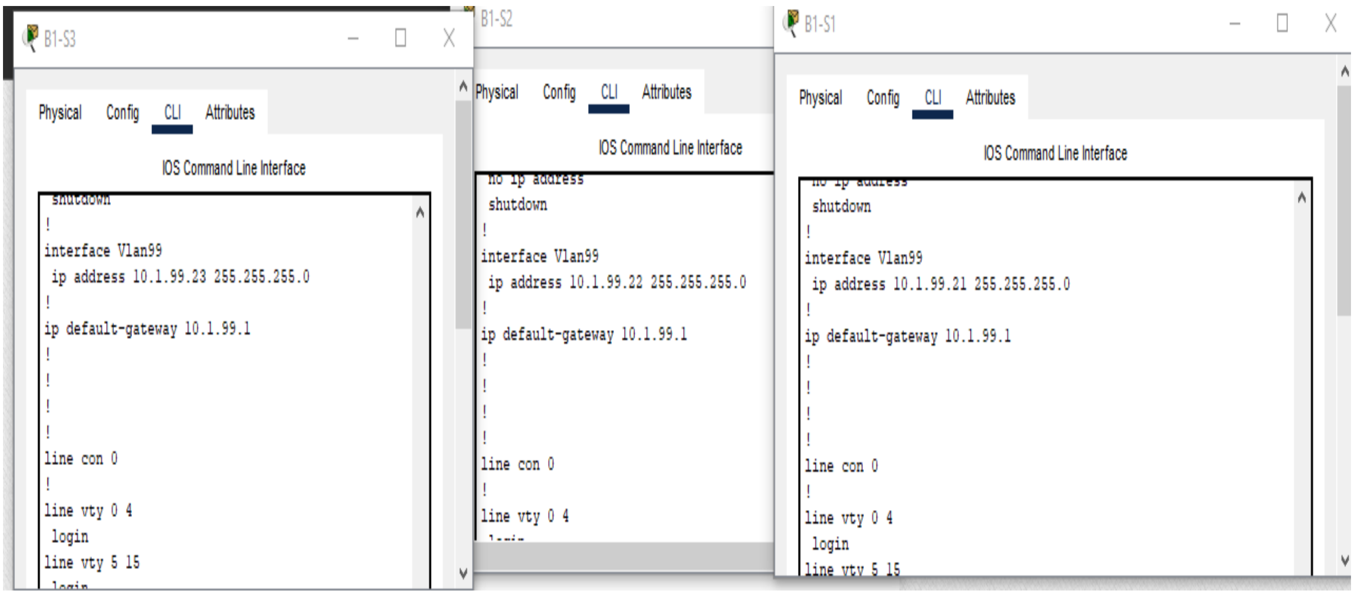
Étape 3 : configuration de l'interface VLAN et de la passerelle par défaut sur BX-S1, BX-S2 et BX-S3



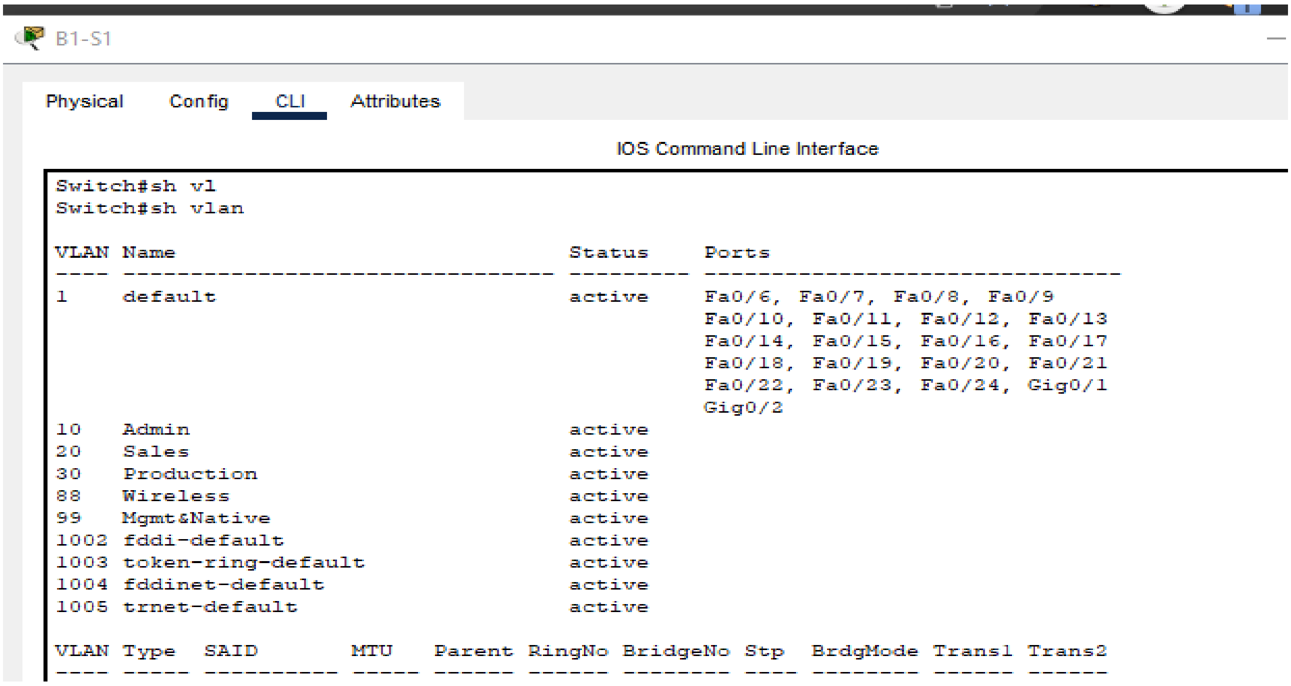
B3-S1 :10.3.99.1

B3-S2 :10.3.99.1

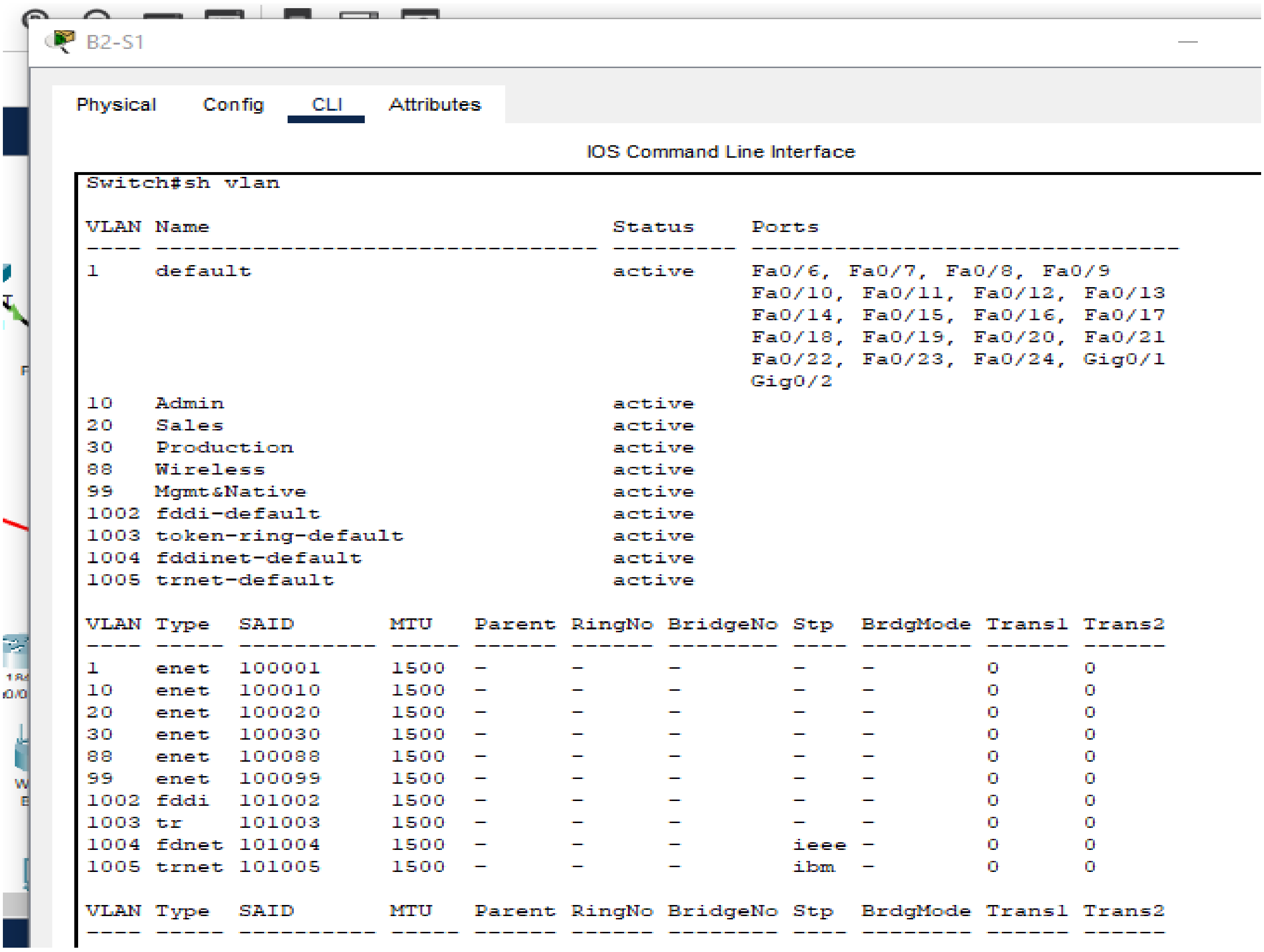
B3-S2 :10.3.99.1



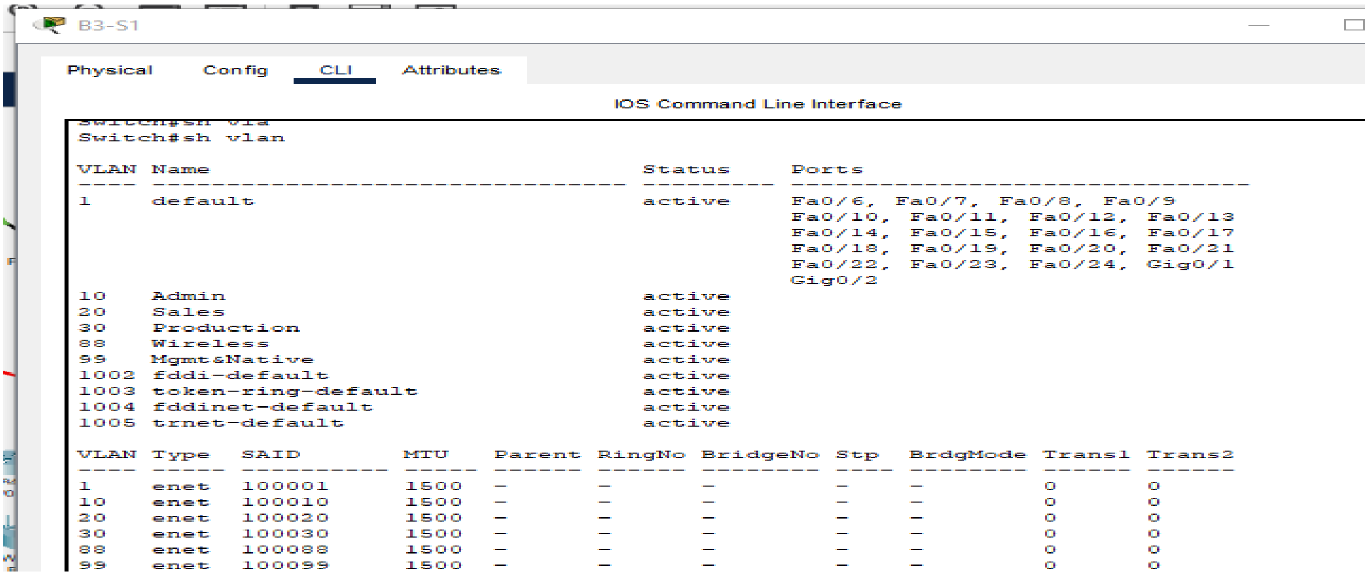
Étape 4 : création des réseaux locaux virtuels (VLAN) sur BX-S1




```
Switch#sh vl
Switch#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   Admin                            active
20   Sales                            active
30   Production                       active
88   Wireless                         active
99   Mgmt&Native                      active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
```



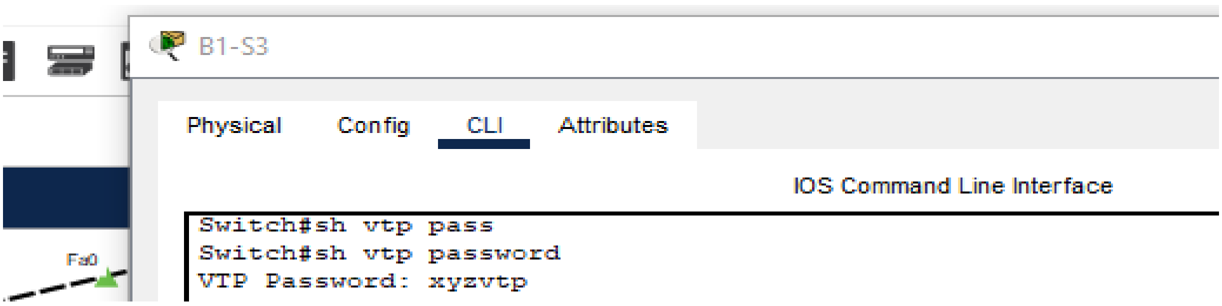

```
Switch#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   Admin                            active
20   Sales                            active
30   Production                       active
88   Wireless                         active
99   Mgmt&Native                      active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
10   enet  100010     1500  -      -      -        -    -        0      0
20   enet  100020     1500  -      -      -        -    -        0      0
30   enet  100030     1500  -      -      -        -    -        0      0
88   enet  100088     1500  -      -      -        -    -        0      0
99   enet  100099     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
```

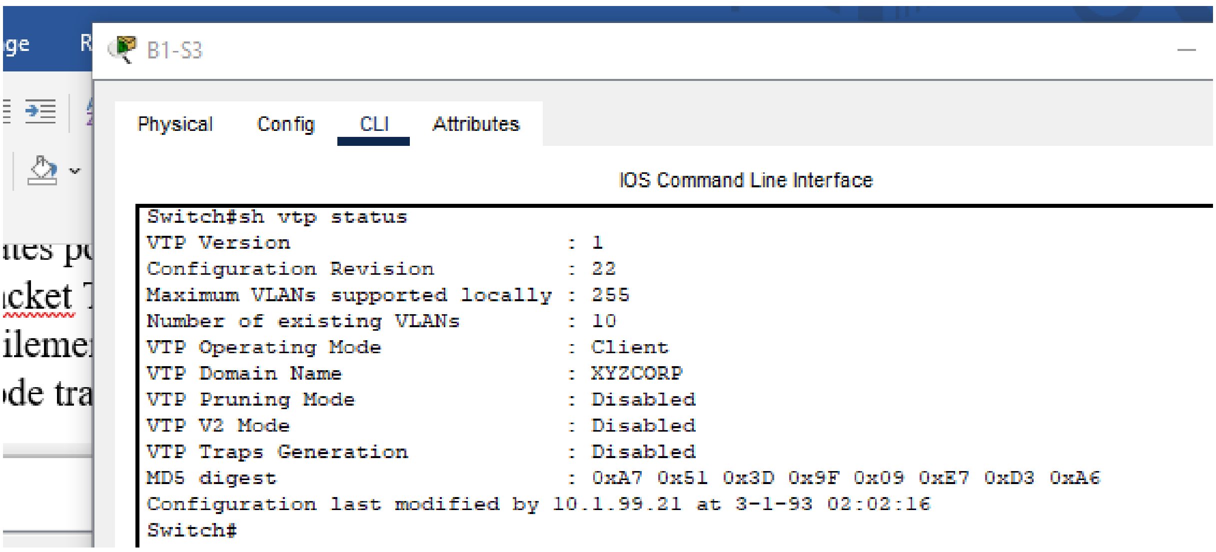


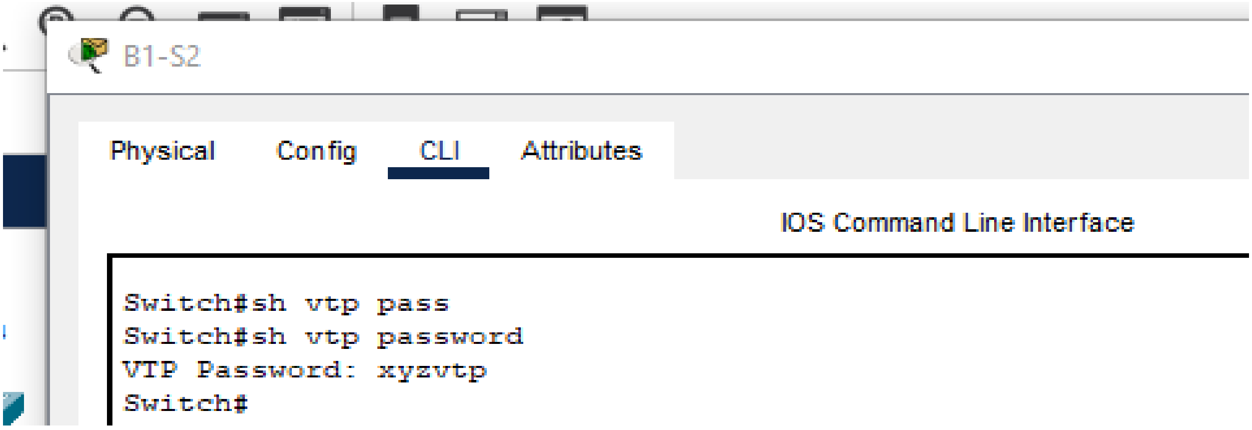
Sur BX-S1 uniquement, créez et nommez les réseaux locaux virtuels (VLAN) répertoriés dans le tableau Configuration VLAN et mappages de ports. VTP annonce les nouveaux réseaux locaux virtuels à BX-S1 et BX-S2.

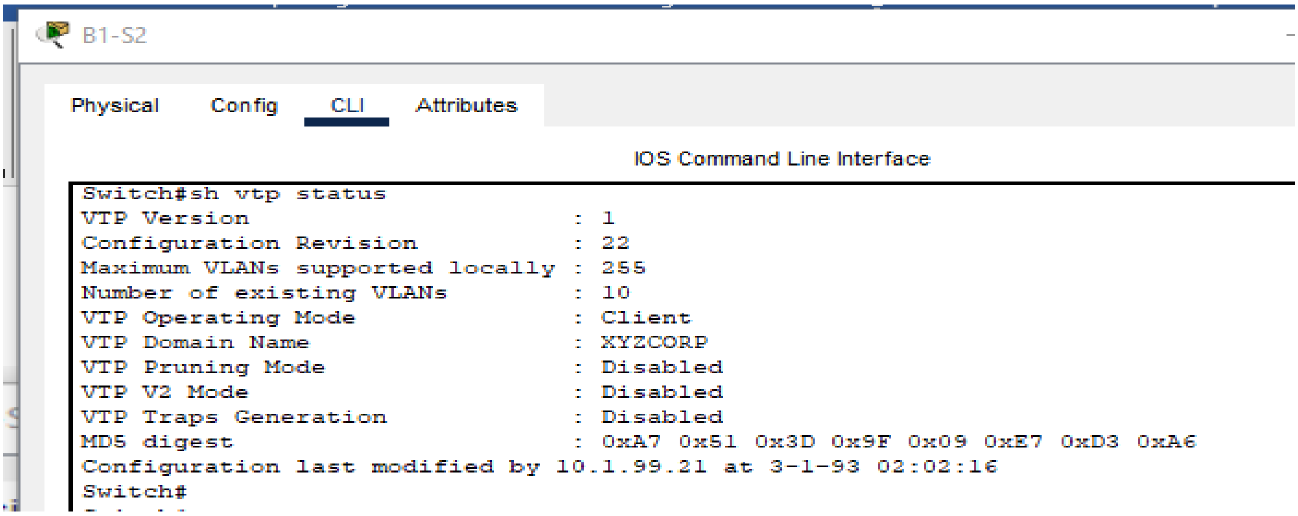
Étape 5 : vérification de l'envoi des réseaux locaux virtuels à BX-S2 et BX-S3

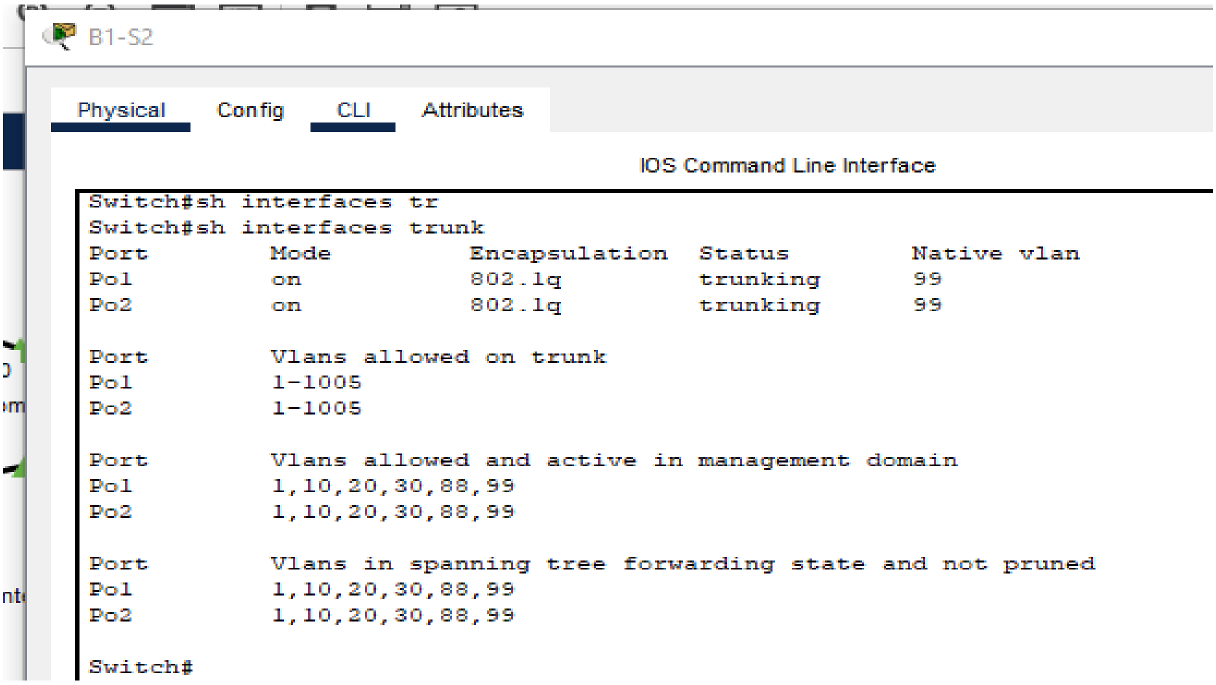
Utilisez les commandes adéquates pour vérifier que S2 et S3 disposent désormais des réseaux locaux virtuels créés sur S1. Packet Tracer peut mettre plusieurs minutes à simuler les annonces VTP. Pour forcer facilement l'envoi des annonces VTP, vous pouvez basculer un des commutateurs client du mode transparent au mode client et inversement.






B1-S2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch#sh vtp pass
Switch#sh vtp password
VTP Password: xyzvtp
Switch#


B1-S2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch#sh vtp status
VTP Version : 1
Configuration Revision : 22
Maximum VLANs supported locally : 255
Number of existing VLANs : 10
VTP Operating Mode : Client
VTP Domain Name : XYZCORP
VTP Pruning Mode : Disabled
VTP V2 Mode : Disabled
VTP Traps Generation : Disabled
MD5 digest : 0xA7 0x51 0x3D 0x9F 0x09 0xE7 0xD3 0xA6
Configuration last modified by 10.1.99.21 at 3-1-93 02:02:16
Switch#

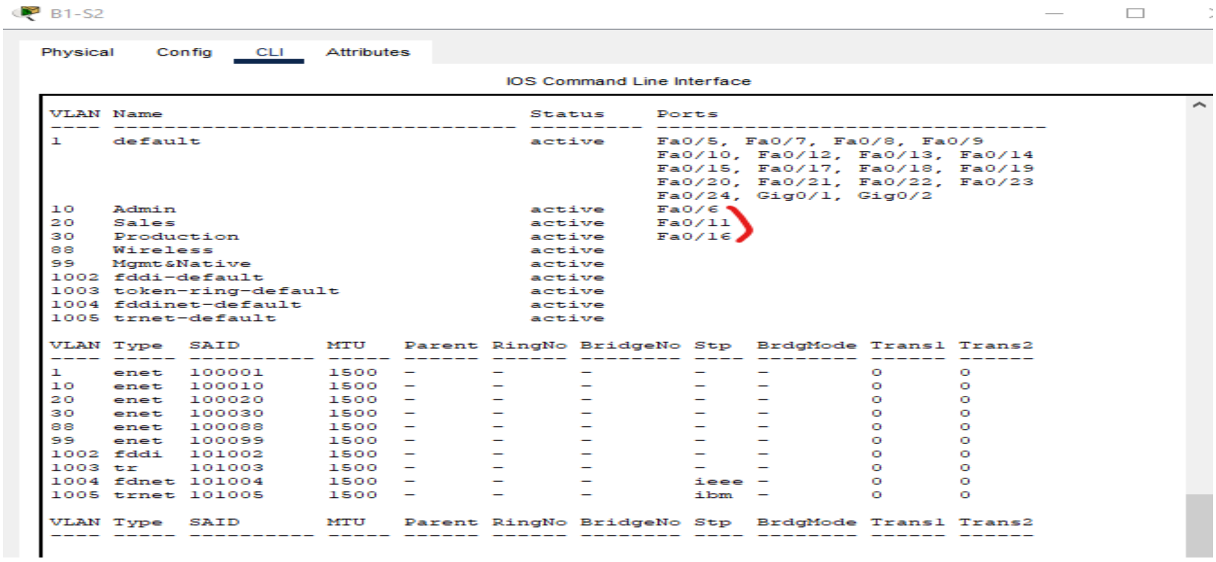

B1-S2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch#sh interfaces tr
Switch#sh interfaces trunk
Port Mode Encapsulation Status Native vlan
Po1 on 802.1q trunking 99
Po2 on 802.1q trunking 99
Port Vlans allowed on trunk
Po1 1-1005
Po2 1-1005
Port Vlans allowed and active in management domain
Po1 1,10,20,30,88,99
Po2 1,10,20,30,88,99
Port Vlans in spanning tree forwarding state and not pruned
Po1 1,10,20,30,88,99
Po2 1,10,20,30,88,99
Switch#

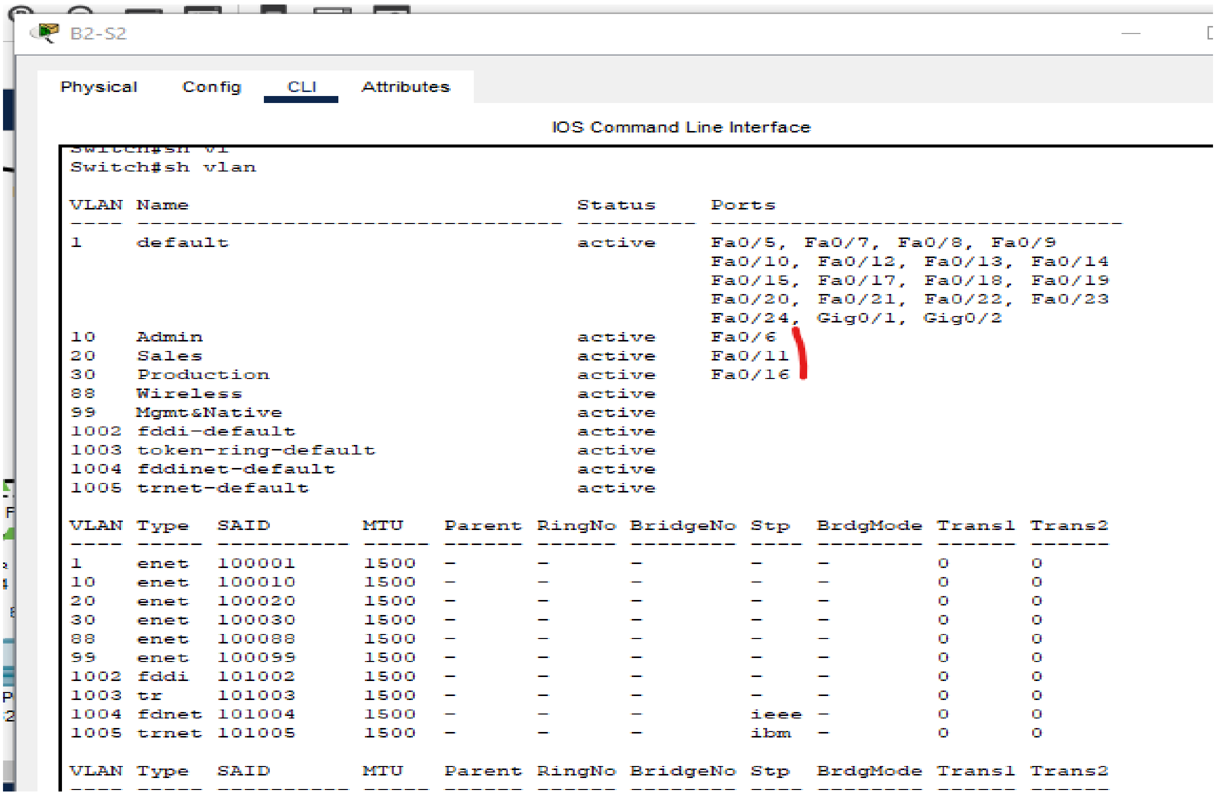
Tâche 8 : affectation des réseaux locaux virtuels et configuration de la sécurité des ports

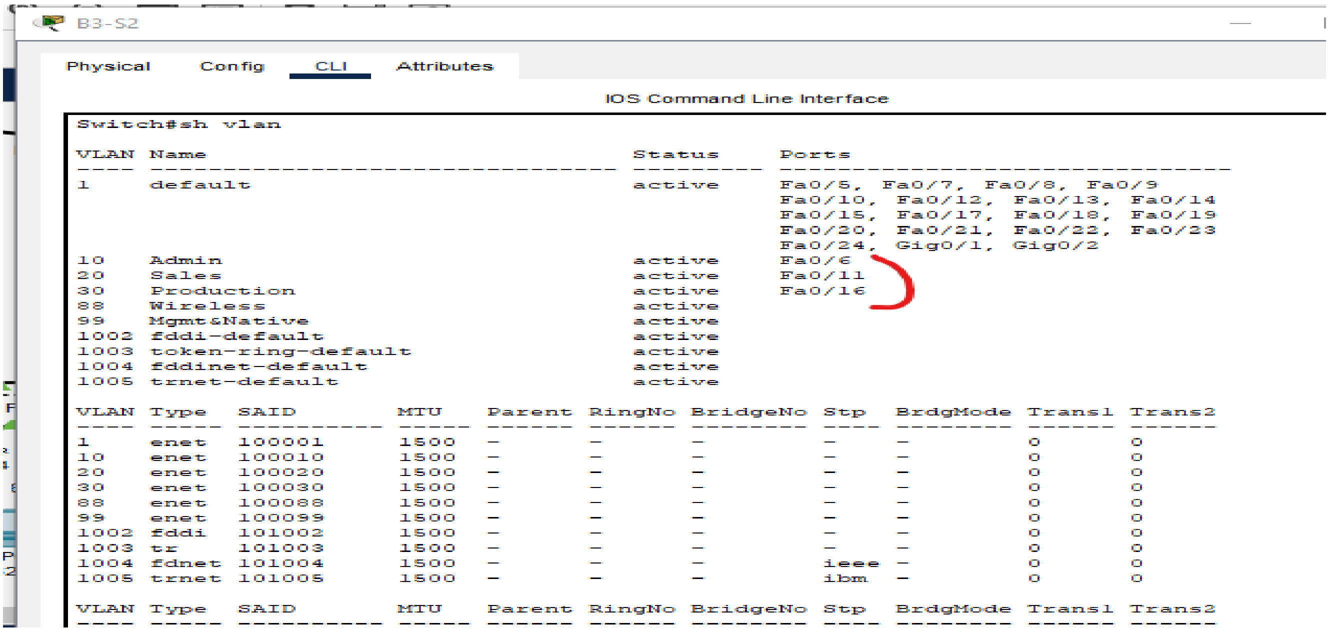
Étape 1 : affectation des réseaux locaux virtuels aux ports d'accès

Utilisez le tableau Configuration VLAN et mappages de ports pour réaliser les opérations suivantes :

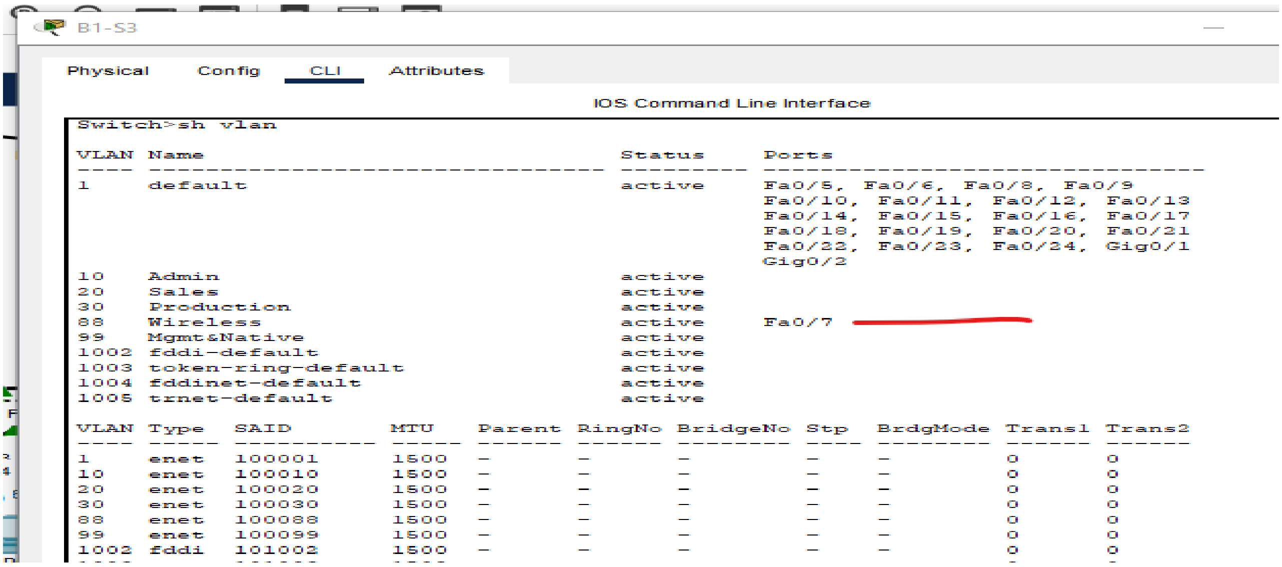
- Configurer les ports d'accès
- Affecter des réseaux locaux virtuels aux ports d'accès



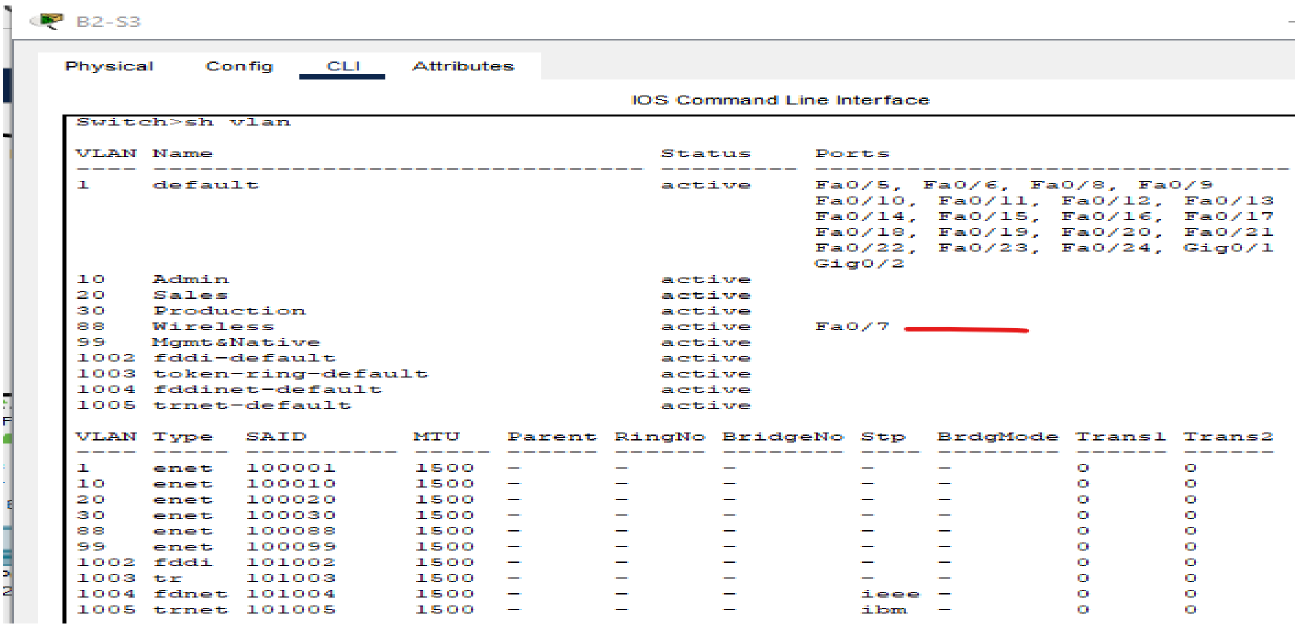




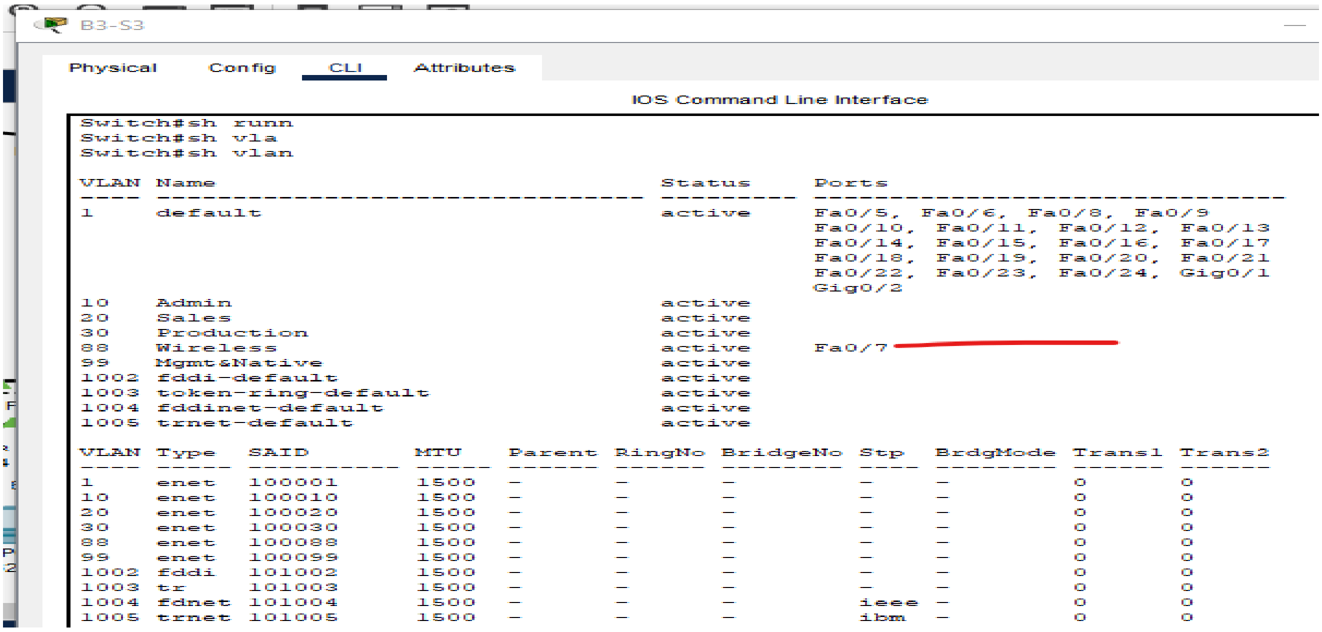

```
Switch#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   Admin                            active    Fa0/6
20   Sales                            active    Fa0/11
30   Production                       active    Fa0/16
88   Wireless                         active
99   Mgmt&Native                      active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
10   enet  100010     1500  -      -      -        -    -        0      0
20   enet  100020     1500  -      -      -        -    -        0      0
30   enet  100030     1500  -      -      -        -    -        0      0
88   enet  100088     1500  -      -      -        -    -        0      0
99   enet  100099     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
```




```
Switch>sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   Admin                            active
20   Sales                            active
30   Production                       active
88   Wireless                         active    Fa0/7
99   Mgmt&Native                      active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
10   enet  100010     1500  -      -      -        -    -        0      0
20   enet  100020     1500  -      -      -        -    -        0      0
30   enet  100030     1500  -      -      -        -    -        0      0
88   enet  100088     1500  -      -      -        -    -        0      0
99   enet  100099     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
```




```
Switch>sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   Admin                            active
20   Sales                            active
30   Production                       active
88   Wireless                         active    Fa0/7
99   Mgmt&Native                      active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
10   enet  100010     1500  -      -      -        -    -        0      0
20   enet  100020     1500  -      -      -        -    -        0      0
30   enet  100030     1500  -      -      -        -    -        0      0
88   enet  100088     1500  -      -      -        -    -        0      0
99   enet  100099     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0
```

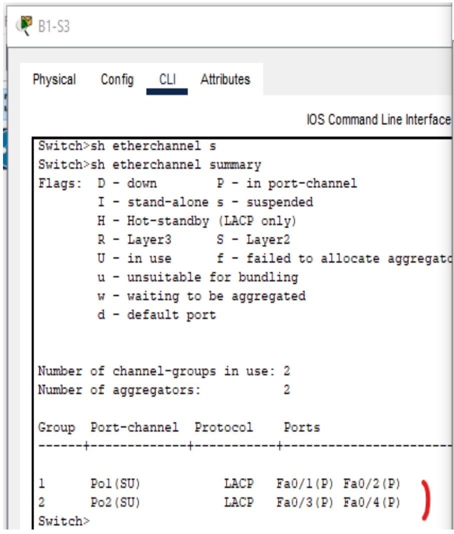


B3-S3

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#sh runn
Switch#sh vla
Switch#sh vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   Admin                            active
20   Sales                            active
30   Production                       active
88   Wireless                         active    Fa0/7
99   Mgmt&Native                      active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
10   enet  100010     1500  -      -      -        -    -        0      0
20   enet  100020     1500  -      -      -        -    -        0      0
30   enet  100030     1500  -      -      -        -    -        0      0
88   enet  100088     1500  -      -      -        -    -        0      0
99   enet  100099     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0
```

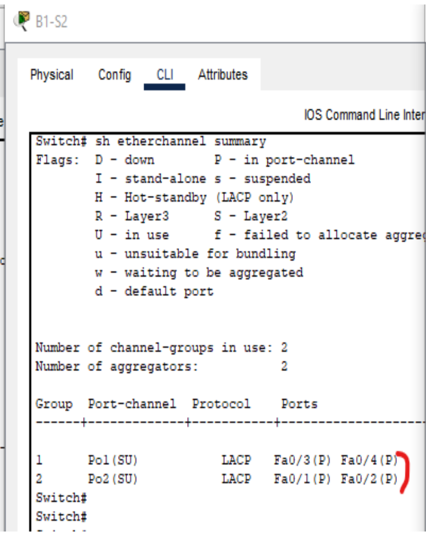


B1-S3

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>sh etherchannel s
Switch>sh etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregato
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------

1      Po1(SU)           LACP   Fa0/1(P) Fa0/2(P)
2      Po2(SU)           LACP   Fa0/3(P) Fa0/4(P)
Switch>
```

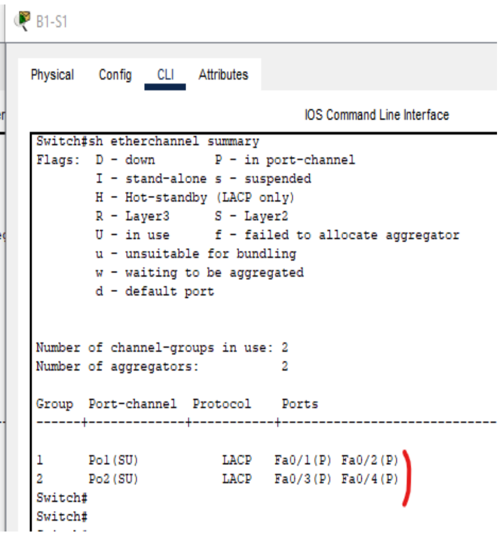


B1-S2

Physical Config CLI Attributes

IOS Command Line Inter

```
Switch# sh etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggreg
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------

1      Po1(SU)           LACP   Fa0/3(P) Fa0/4(P)
2      Po2(SU)           LACP   Fa0/1(P) Fa0/2(P)
Switch#
Switch#
```



B1-S1

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#sh etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------

1      Po1(SU)           LACP   Fa0/1(P) Fa0/2(P)
2      Po2(SU)           LACP   Fa0/3(P) Fa0/4(P)
Switch#
Switch#
```

Étape 2 : configuration de la sécurité des ports

Utilisez la stratégie suivante pour activer la sécurité des ports sur les ports d’accès BX-S2 :

- Autoriser une seule adresse MAC
- Forcer la première adresse MAC apprise à s’adapter à la configuration
- Forcer la désactivation du port en cas de violation de la sécurité

B1-S2

Physical Config CLI Attributes

IOS Command Line Interface

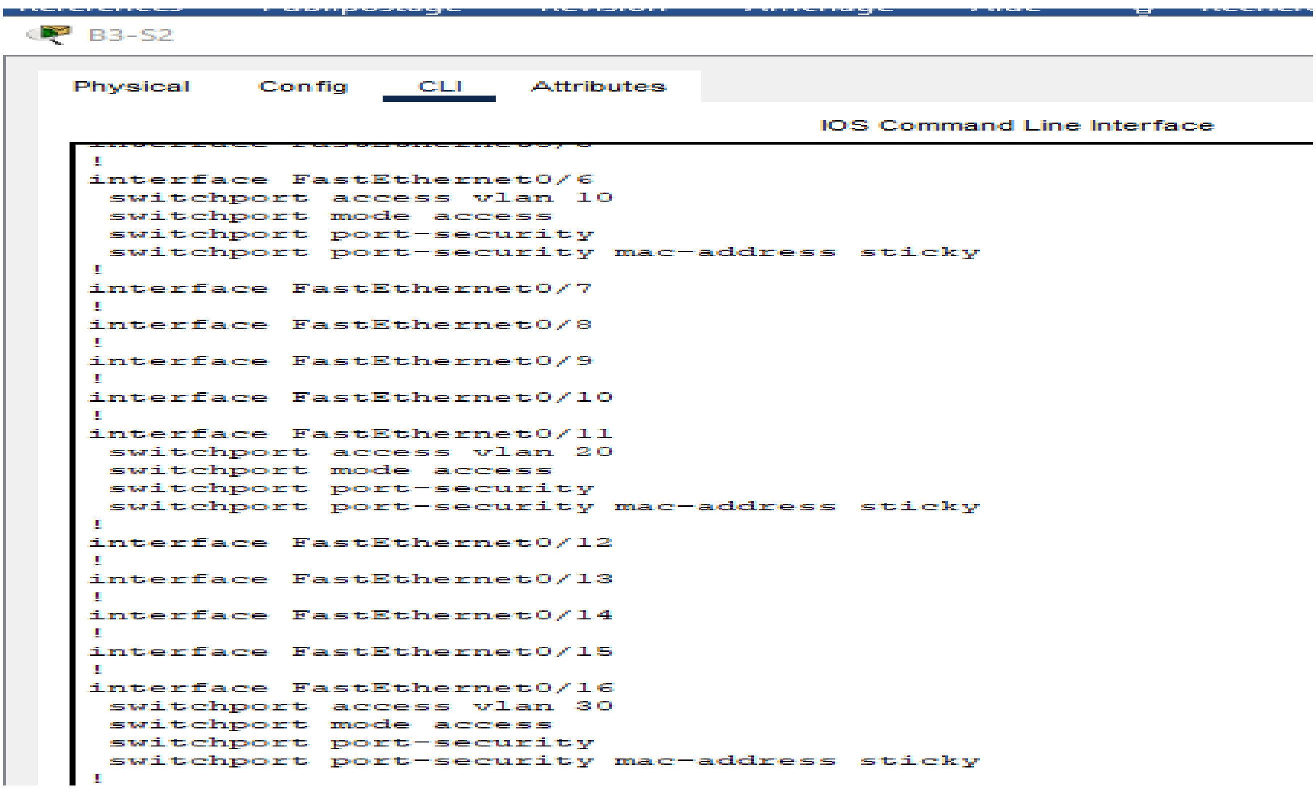
```
 switchport mode access
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky 00D0.BCCA.9C3A
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky 0001.9747.CE95
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
 switchport access vlan 30
 switchport mode access
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky 0001.4339.0C91
!
```

B2-S2

Physical Config CLI Attributes

IOS Command Line Interface

```
!
interface FastEthernet0/6
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky 000C.CFD8.751C
!
interface FastEthernet0/7
!
interface FastEthernet0/8
!
interface FastEthernet0/9
!
interface FastEthernet0/10
!
interface FastEthernet0/11
 switchport access vlan 20
 switchport mode access
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky 0001.64D9.7C5C
!
interface FastEthernet0/12
!
interface FastEthernet0/13
!
interface FastEthernet0/14
!
interface FastEthernet0/15
!
interface FastEthernet0/16
 switchport access vlan 30
 switchport mode access
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky 0002.1664.72A0
```



Étape 3 : vérification des affectations VLAN et de la sécurité des ports

Utilisez les commandes adéquates pour vérifier que les réseaux locaux virtuels (VLAN) d'accès sont correctement affectés et que la stratégie de sécurité des ports a été activée.

Tâche 9 : configuration du protocole STP

Étape 1 : configuration de BX-S1 comme pont racine

Définissez le niveau de priorité sur 4096 pour BX-S1, de manière à ce que ces commutateurs restent en permanence le pont racine de tous les réseaux locaux virtuels.

B1-S1(config)#spanning-tree vlan 1-1001 priority 4096

B2-S1(config)#spanning-tree vlan 1-1001 priority 4096

B3-S1(config)#spanning-tree vlan 1-1001 priority 4096

Étape 2 : configuration de BX-S3 comme pont racine de secours

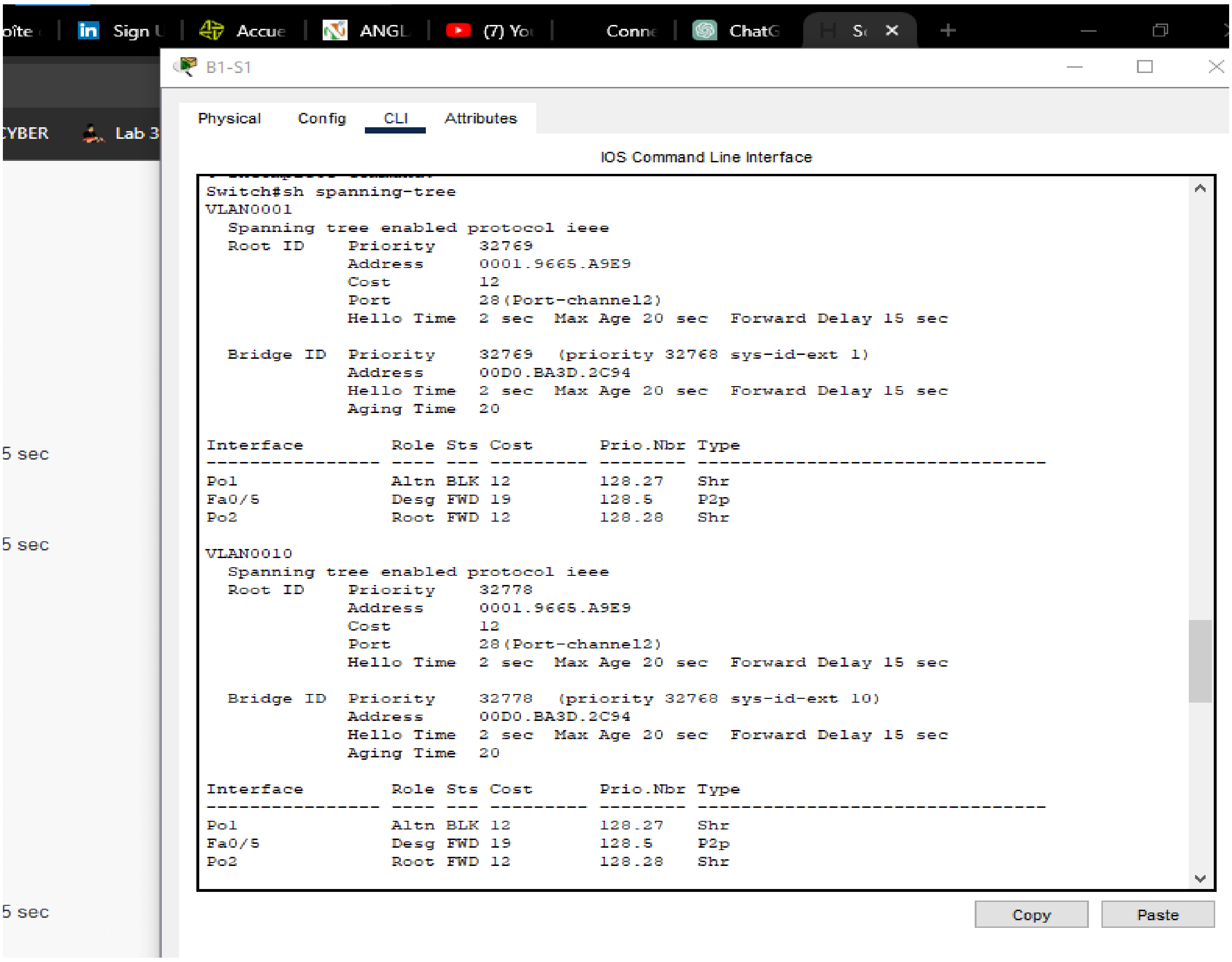
Définissez le niveau de priorité sur 8192 pour BX-S3, de manière à ce que ces commutateurs restent en permanence le pont racine de secours de tous les réseaux locaux virtuels.

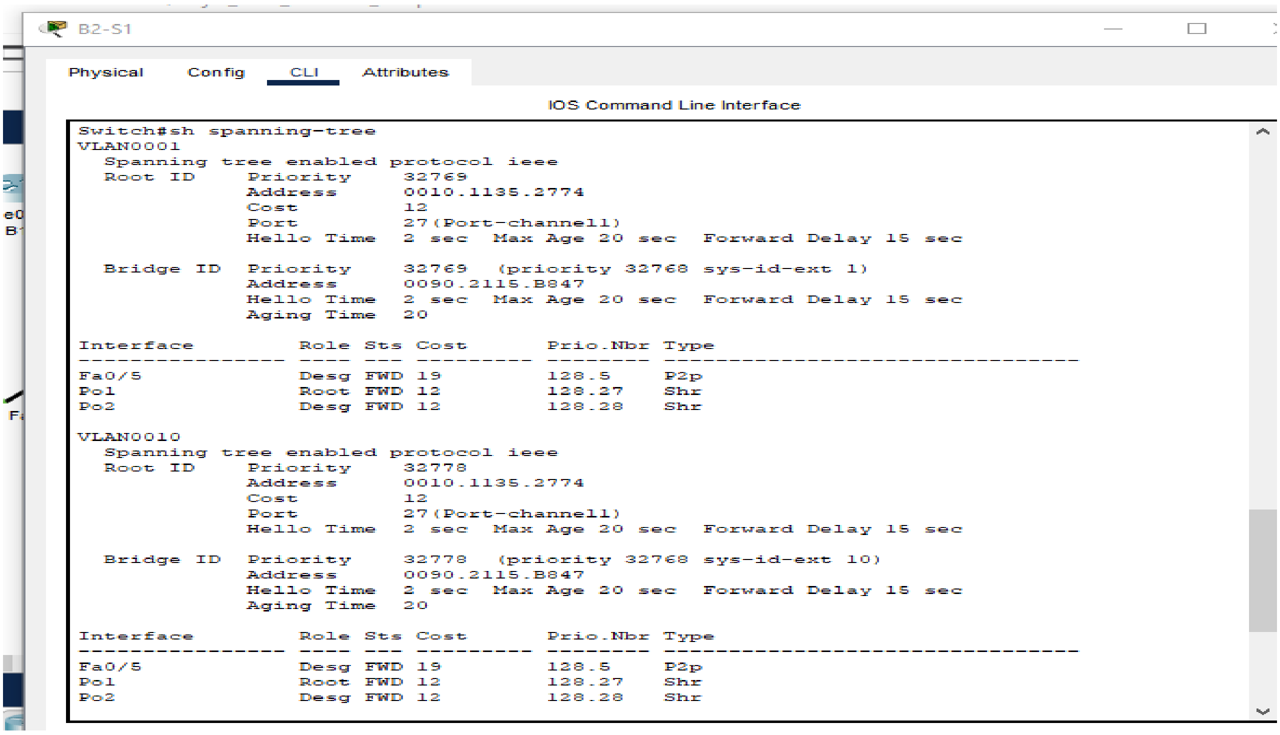
B1-S3(config)#spanning-tree vlan 1-1001 priority 8192

B2-S3(config)#spanning-tree vlan 1-1001 priority 8192

B3-S3(config)#spanning-tree vlan 1-1001 priority 8192

Étape 3 : vérification que BX-S1 est le pont racine

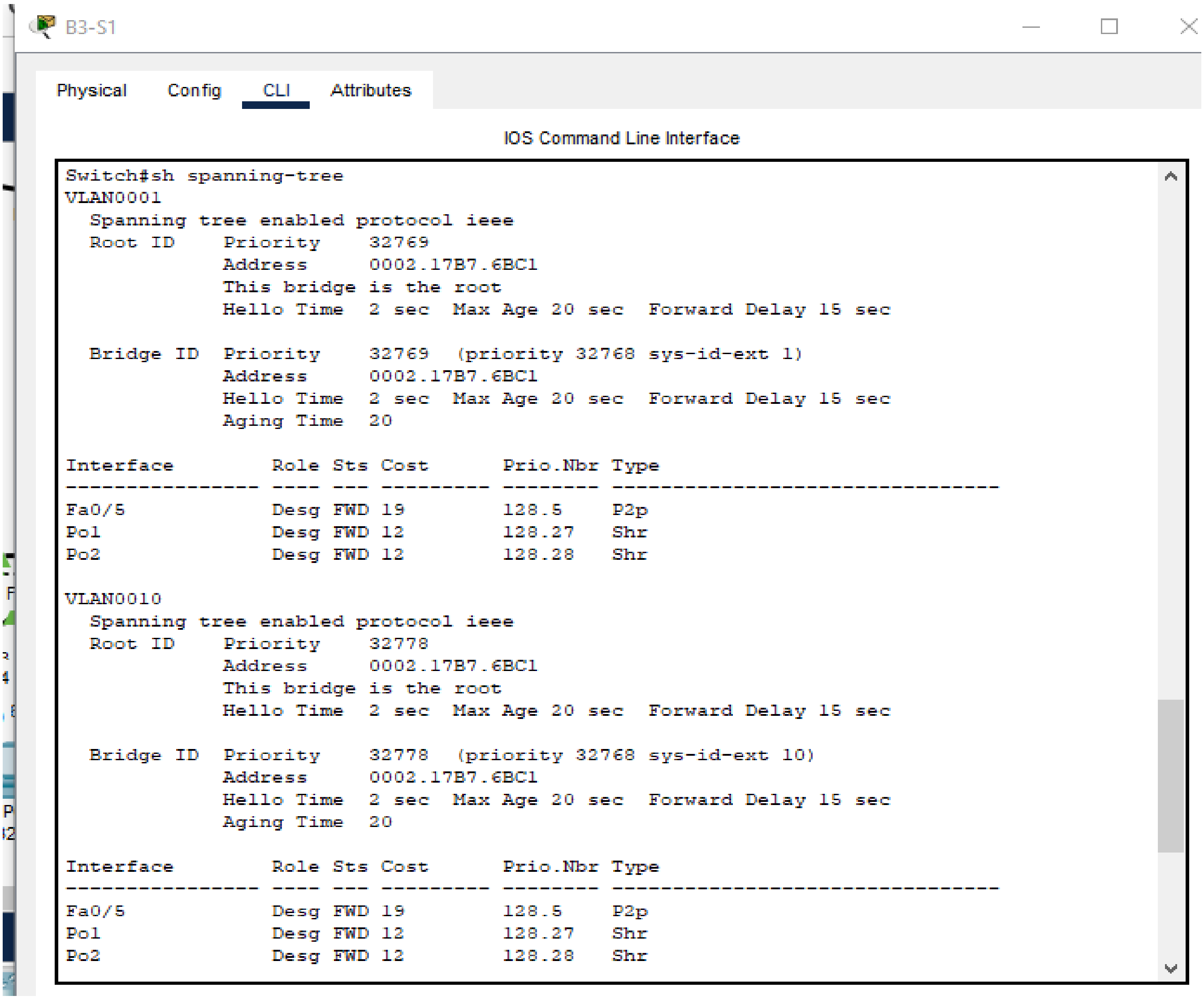



B2-S1

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0010.1135.2774
             Cost        12
             Port        27(Port-channel1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0090.2115.B847
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/5            Desg FWD 19        128.5    P2p
Po1              Root FWD 12        128.27   Shr
Po2              Desg FWD 12        128.28   Shr

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
             Address     0010.1135.2774
             Cost        12
             Port        27(Port-channel1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     0090.2115.B847
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/5            Desg FWD 19        128.5    P2p
Po1              Root FWD 12        128.27   Shr
Po2              Desg FWD 12        128.28   Shr
```


B3-S1

Physical Config CLI Attributes

IOS Command Line Interface

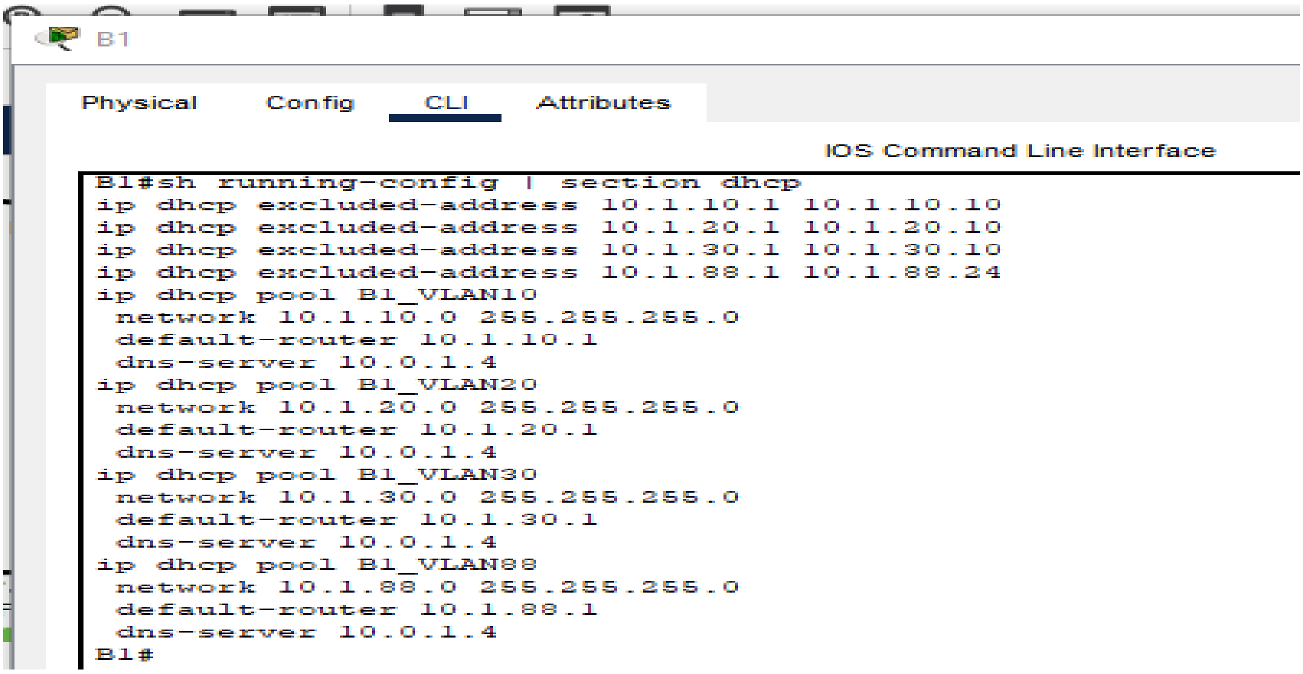
```
Switch#sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0002.17B7.6BC1
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0002.17B7.6BC1
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/5            Desg FWD 19        128.5    P2p
Po1              Desg FWD 12        128.27   Shr
Po2              Desg FWD 12        128.28   Shr

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
             Address     0002.17B7.6BC1
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     0002.17B7.6BC1
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/5            Desg FWD 19        128.5    P2p
Po1              Desg FWD 12        128.27   Shr
Po2              Desg FWD 12        128.28   Shr
```

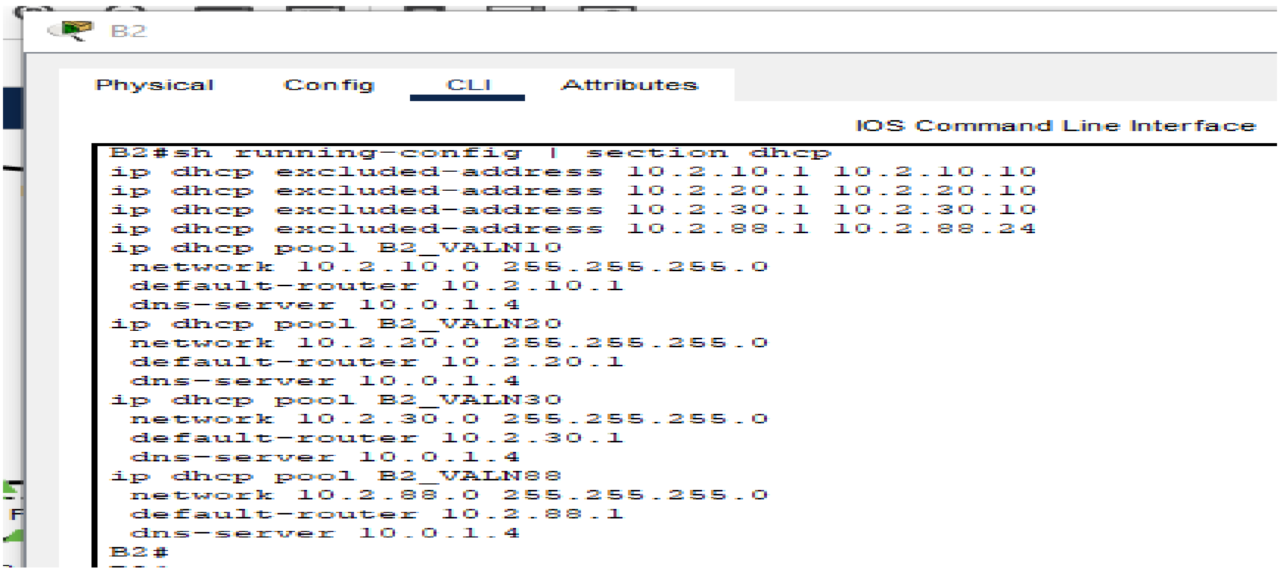
Tâche 10 : configuration du service DHCP

Étape 1 : configuration de pools DHCP pour chaque réseau local virtuel

Sur les routeurs Branch, configurez des pools DHCP pour chaque réseau local virtuel, en respectant les consignes suivantes :

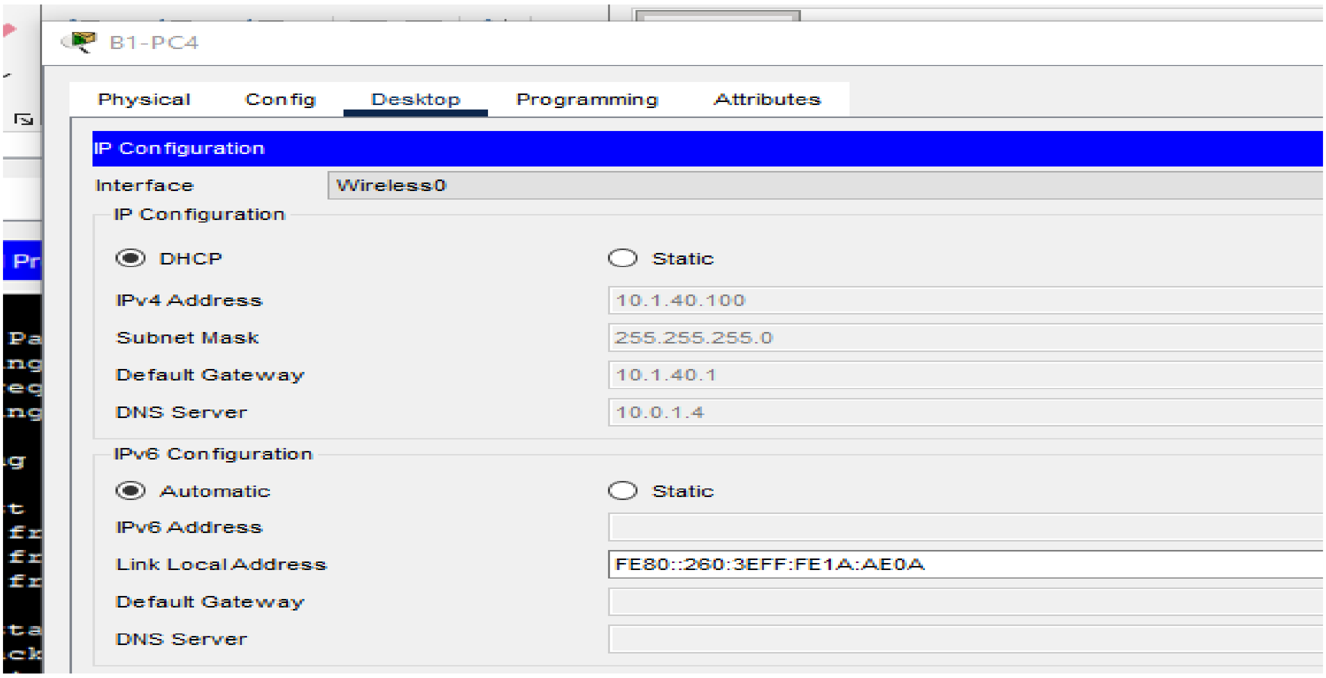
- Excluez les dix premières adresses IP de chaque pool pour les réserver aux réseaux locaux.
- Excluez les 24 premières adresses IP de chaque pool pour les réserver aux réseaux locaux sans fil.
- Le nom du pool est BX_VLAN##, où X est le numéro du routeur et ## est le numéro du réseau local virtuel.
  Dans le cadre de votre configuration DHCP, incluez le serveur DNS connecté à la batterie de serveurs HQ.

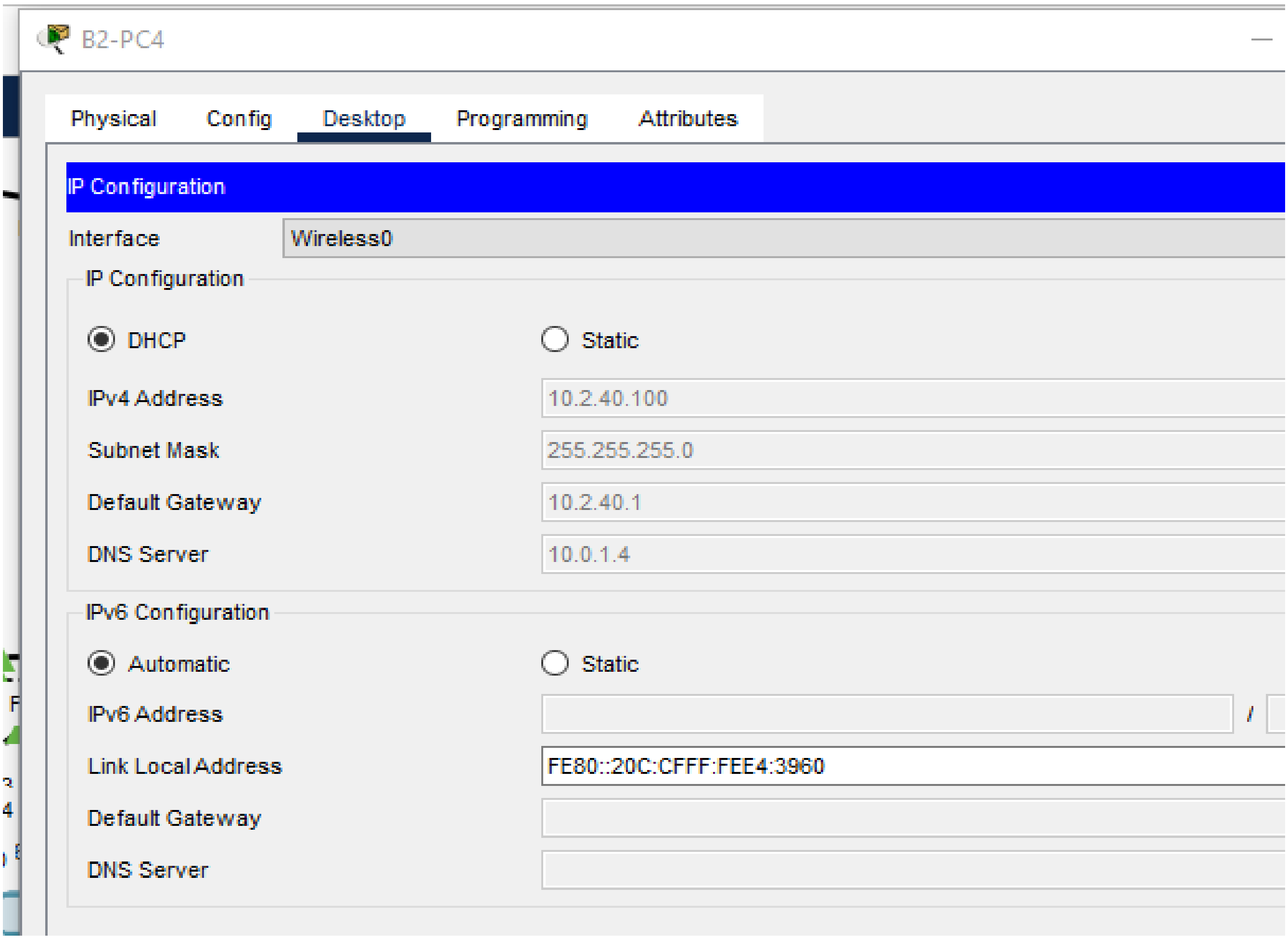


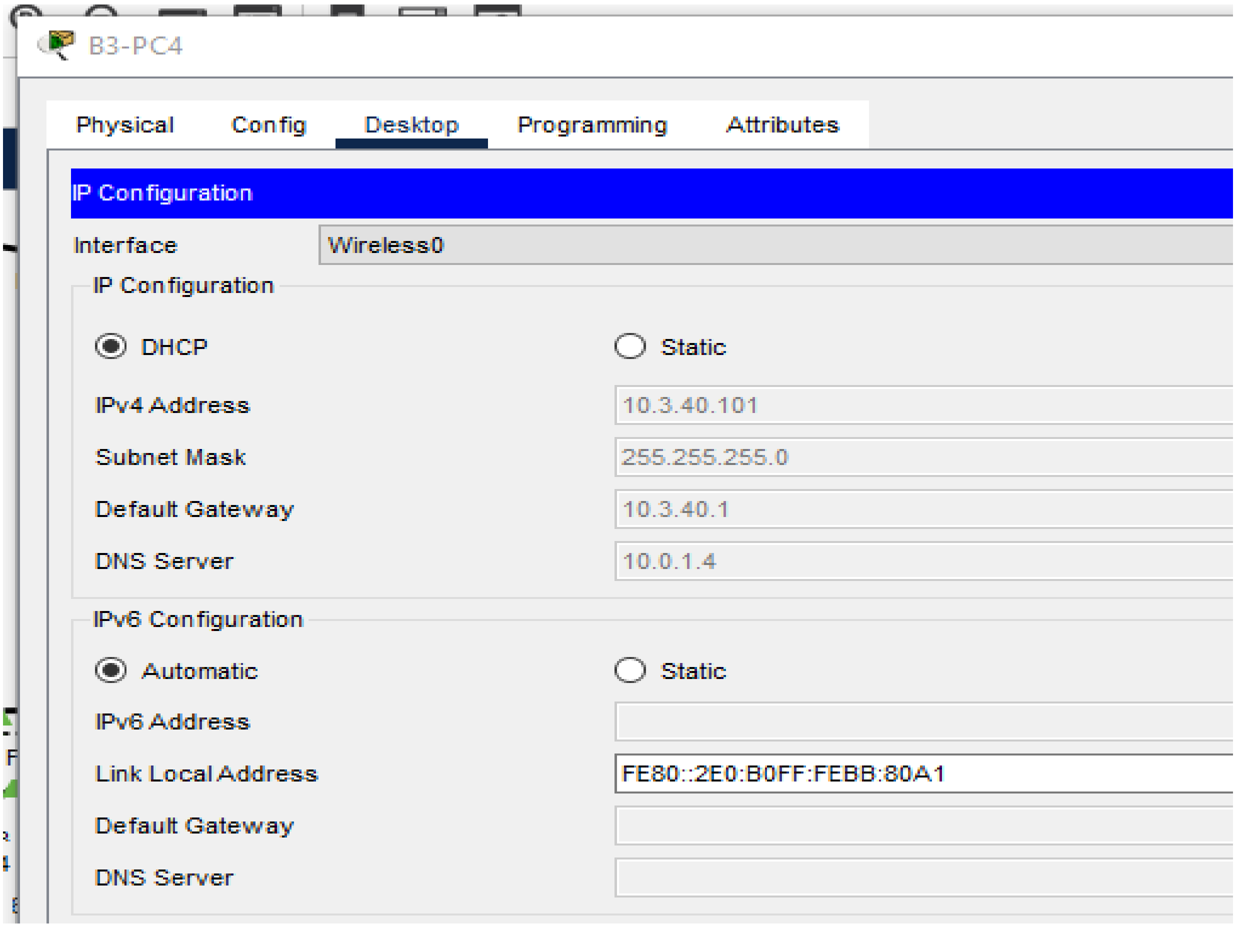


Étape 2 : activation de DHCP sur les PC

Actuellement, les ordinateurs sont configurés pour utiliser des adresses IP statiques. Modifiez cette configuration de sorte qu'ils utilisent le service DHCP

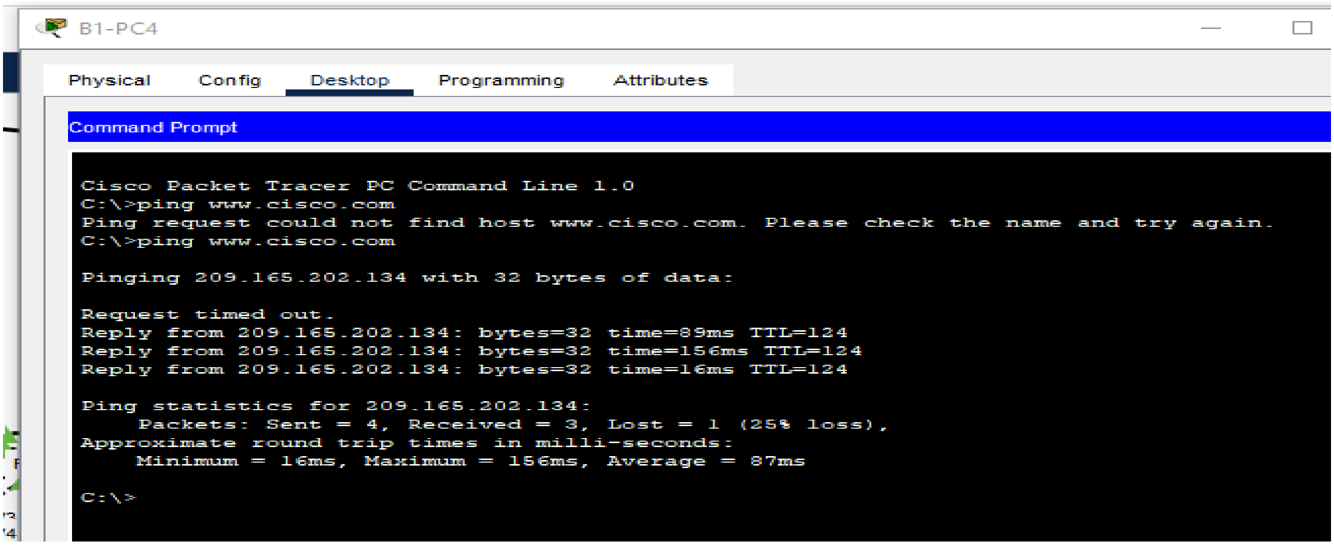


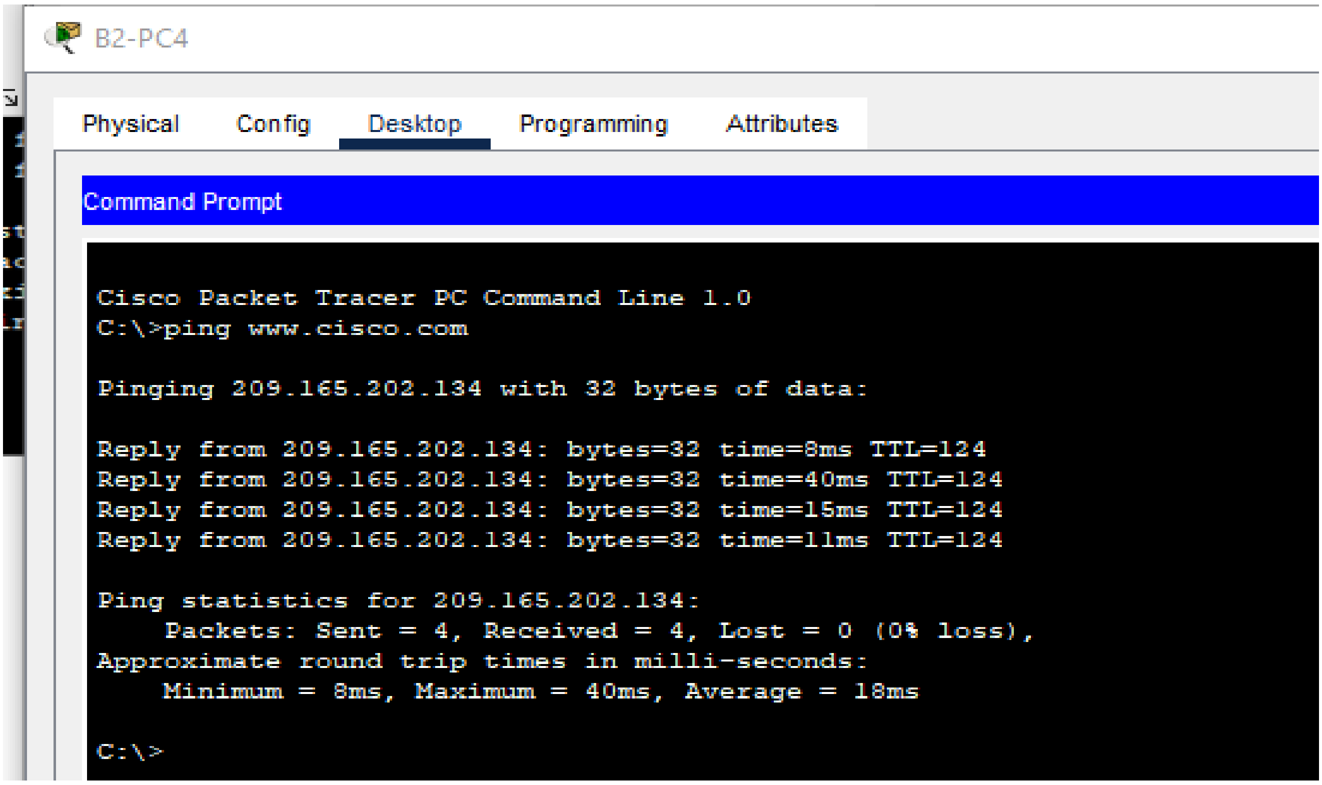


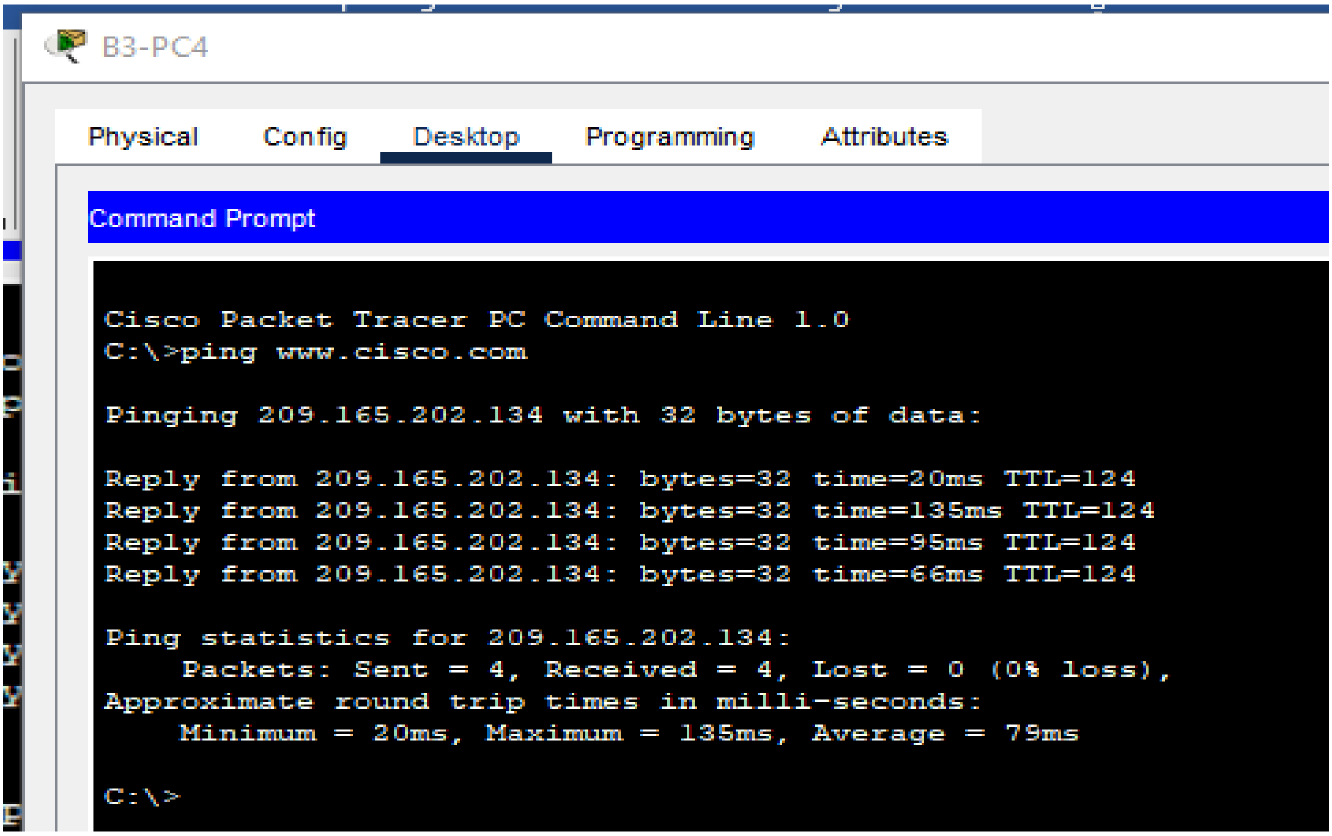


Étape 4 : vérification de la connectivité

Tous les PC physiquement connectés au réseau doivent pouvoir envoyer des requêtes ping vers le serveur Web
www.cisco.com.



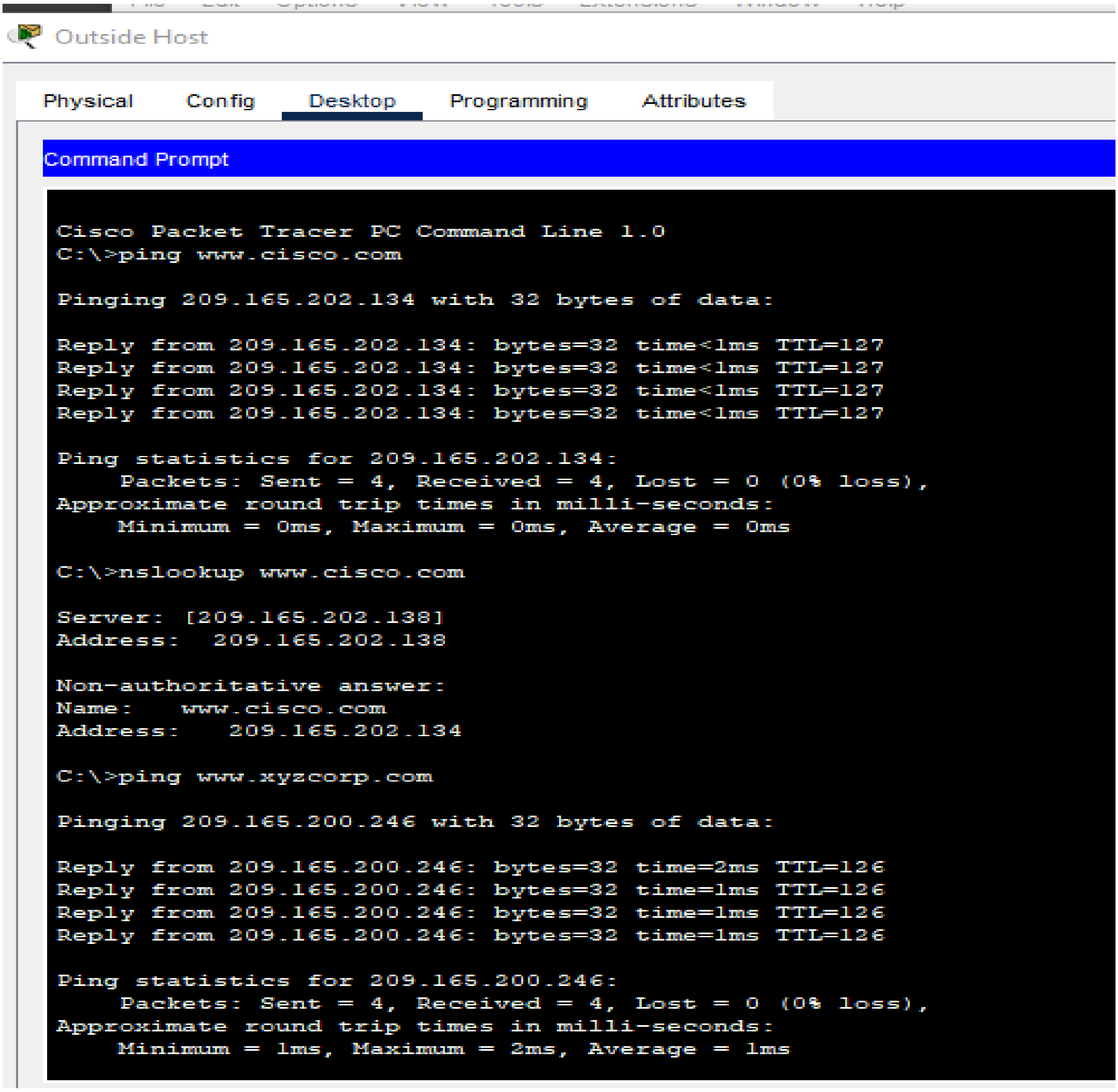

B2-PC4
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.cisco.com
Pinging 209.165.202.134 with 32 bytes of data:
Reply from 209.165.202.134: bytes=32 time=8ms TTL=124
Reply from 209.165.202.134: bytes=32 time=40ms TTL=124
Reply from 209.165.202.134: bytes=32 time=15ms TTL=124
Reply from 209.165.202.134: bytes=32 time=11ms TTL=124
Ping statistics for 209.165.202.134:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 8ms, Maximum = 40ms, Average = 18ms
C:\>


B3-PC4
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.cisco.com
Pinging 209.165.202.134 with 32 bytes of data:
Reply from 209.165.202.134: bytes=32 time=20ms TTL=124
Reply from 209.165.202.134: bytes=32 time=135ms TTL=124
Reply from 209.165.202.134: bytes=32 time=95ms TTL=124
Reply from 209.165.202.134: bytes=32 time=66ms TTL=124
Ping statistics for 209.165.202.134:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 20ms, Maximum = 135ms, Average = 79ms
C:\>

Tâche 11 : configuration d'une liste de contrôle d'accès pare-feu

Étape 1 : vérification de la connectivité à partir d'Outside Host

Le PC Outside Host doit être capable d'envoyer des requêtes ping au serveur à l'adresse www.xyzcorp.com.




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.cisco.com

Pinging 209.165.202.134 with 32 bytes of data:

Reply from 209.165.202.134: bytes=32 time<1ms TTL=127
Reply from 209.165.202.134: bytes=32 time<1ms TTL=127
Reply from 209.165.202.134: bytes=32 time<1ms TTL=127
Reply from 209.165.202.134: bytes=32 time<1ms TTL=127

Ping statistics for 209.165.202.134:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>nslookup www.cisco.com

Server: [209.165.202.138]
Address:  209.165.202.138

Non-authoritative answer:
Name:    www.cisco.com
Address:   209.165.202.134

C:\>ping www.xyzcorp.com

Pinging 209.165.200.246 with 32 bytes of data:

Reply from 209.165.200.246: bytes=32 time=2ms TTL=126
Reply from 209.165.200.246: bytes=32 time=1ms TTL=126
Reply from 209.165.200.246: bytes=32 time=1ms TTL=126
Reply from 209.165.200.246: bytes=32 time=1ms TTL=126

Ping statistics for 209.165.200.246:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

Étape 2 : mise en œuvre d'une liste de contrôle d'accès pare-feu rudimentaire

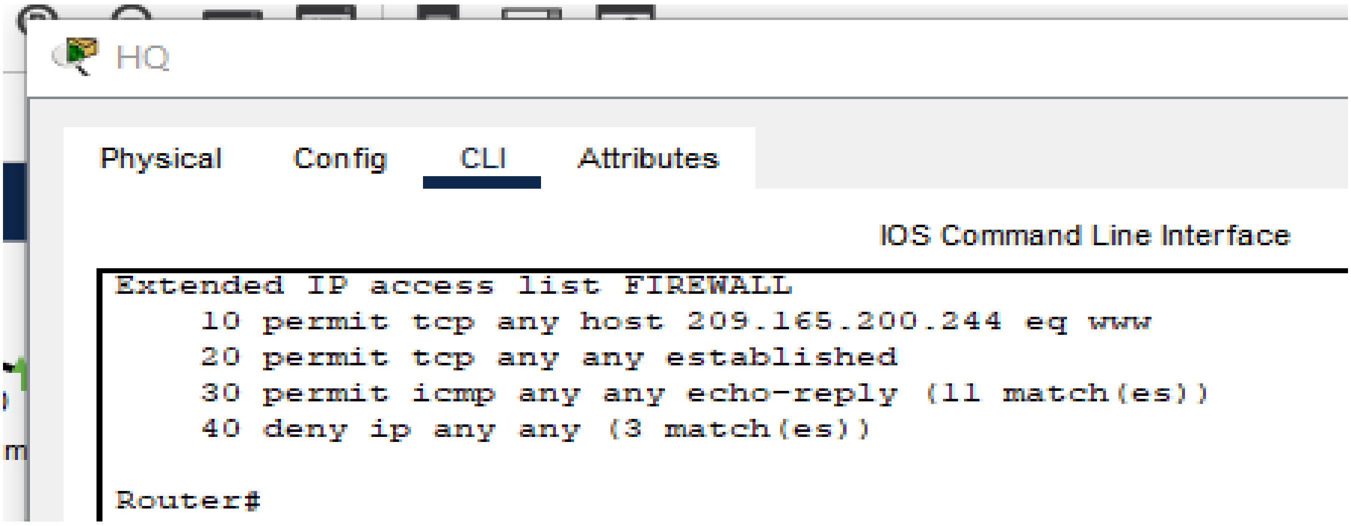
ISP permettant la connexion à Internet, configurez une liste de contrôle d'accès nommée, appelée FIREWALL, en ajoutant les instructions dans l'ordre suivant :

1. Autorisez les requêtes HTTP entrantes vers le serveur www.xyzcorp.com.

2. Autorisez uniquement les sessions TCP établies provenant d'ISP et des sources situées au-delà d'ISP.

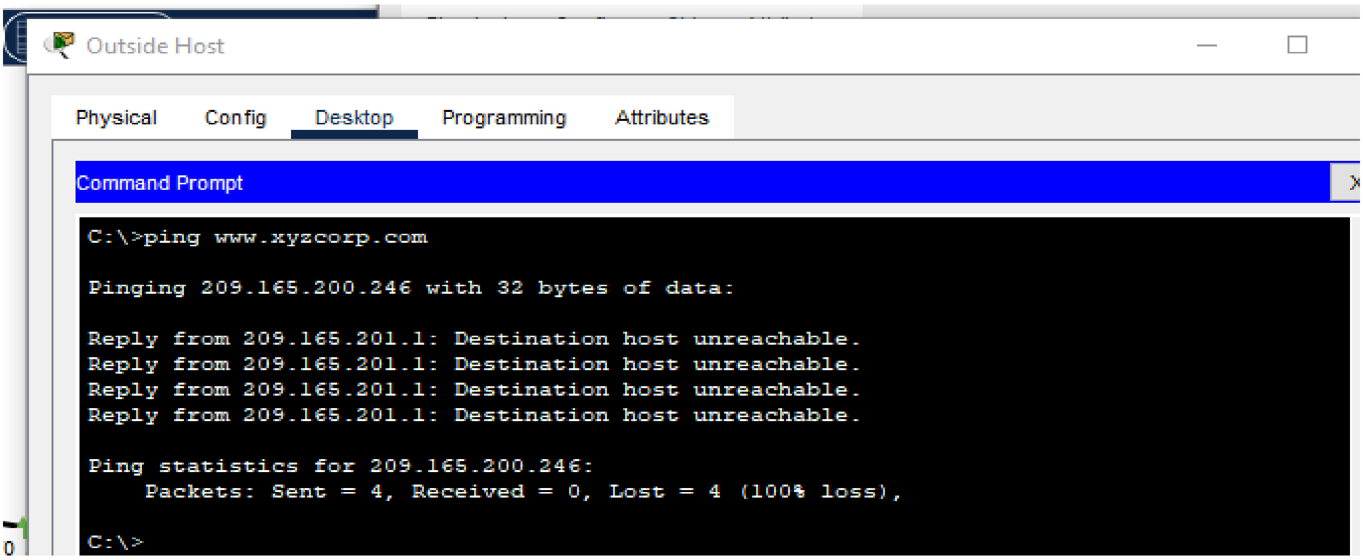
3. Autorisez uniquement les réponses ping entrantes provenant d'ISP et des sources situées au-delà d'ISP.

4. Bloquez de manière explicite tous les autres accès entrants provenant d'ISP et des sources situées au-delà d'ISP



Étape 3 : vérification de la connectivité à partir d'Outside Host

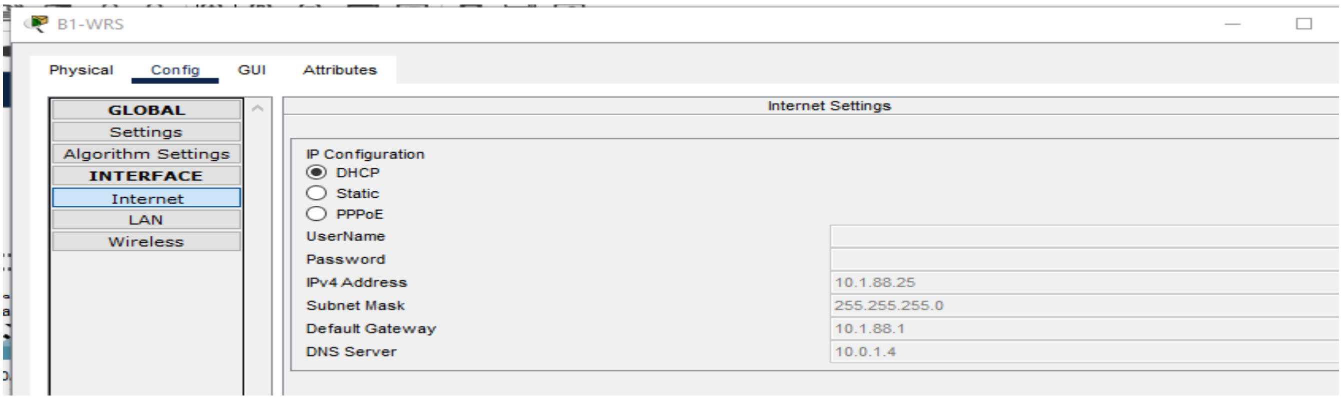
Le PC Outside Host ne doit pas être en mesure d'envoyer des requêtes ping vers le serveur à l'adresse www.xyzcorp.com. En revanche, le PC Outside Host doit pouvoir envoyer des requêtes d'affichage de pages Web.

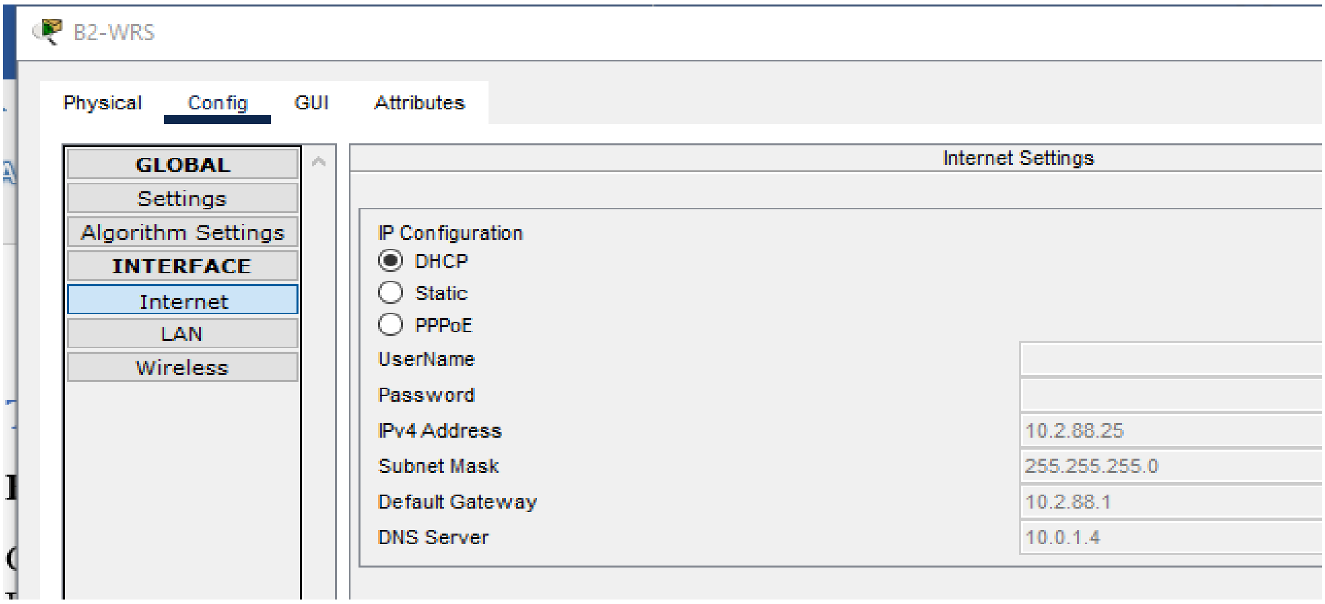


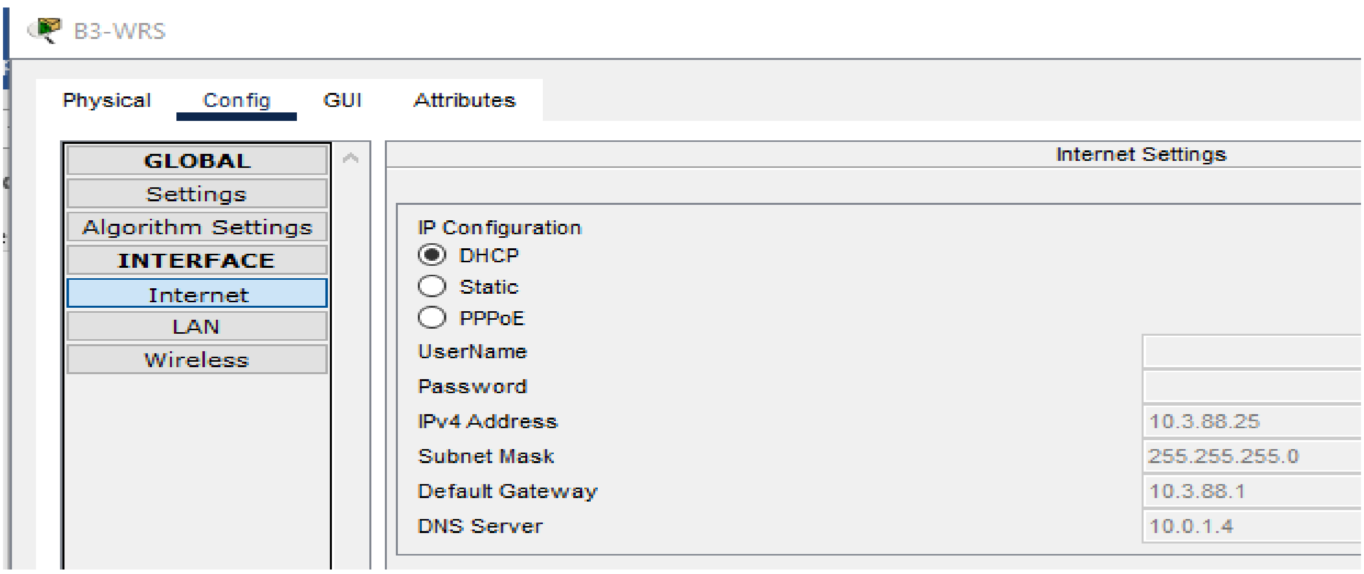
Tâche 12 : configuration de la connectivité sans fil

Étape 1 : vérification de la configuration DHCP

Chaque routeur BX-WRS doit déjà disposer d'un adressage IP obtenu auprès du service DHCP du routeur BX pour le réseau local virtuel VLAN 88.

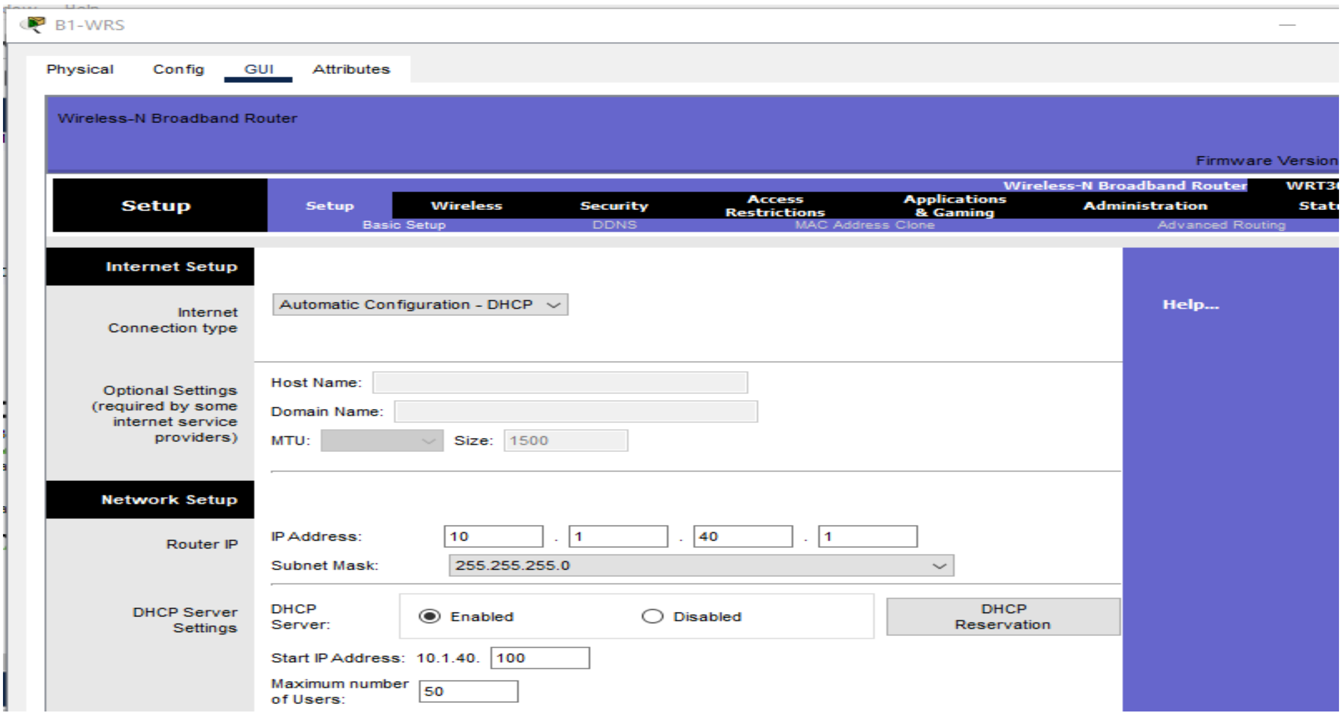


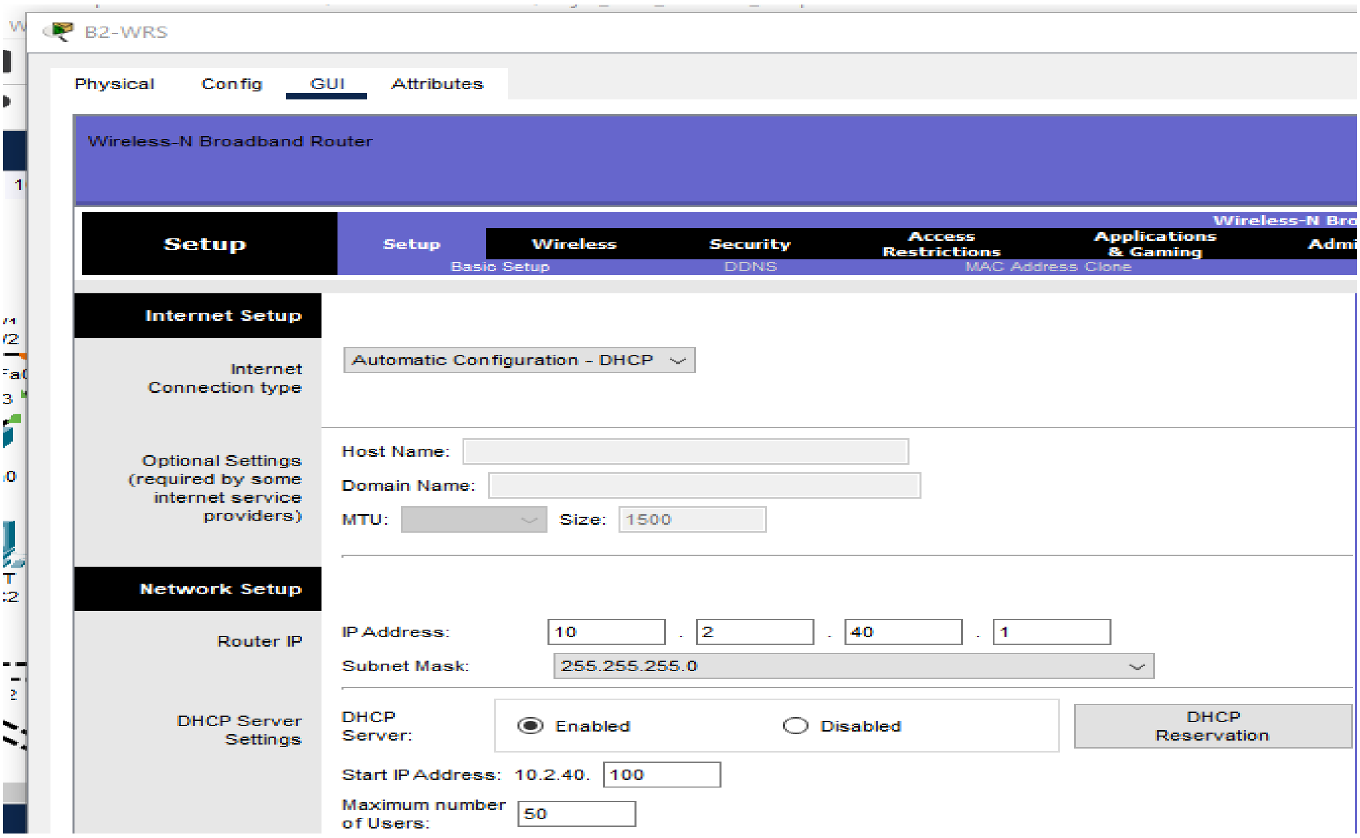


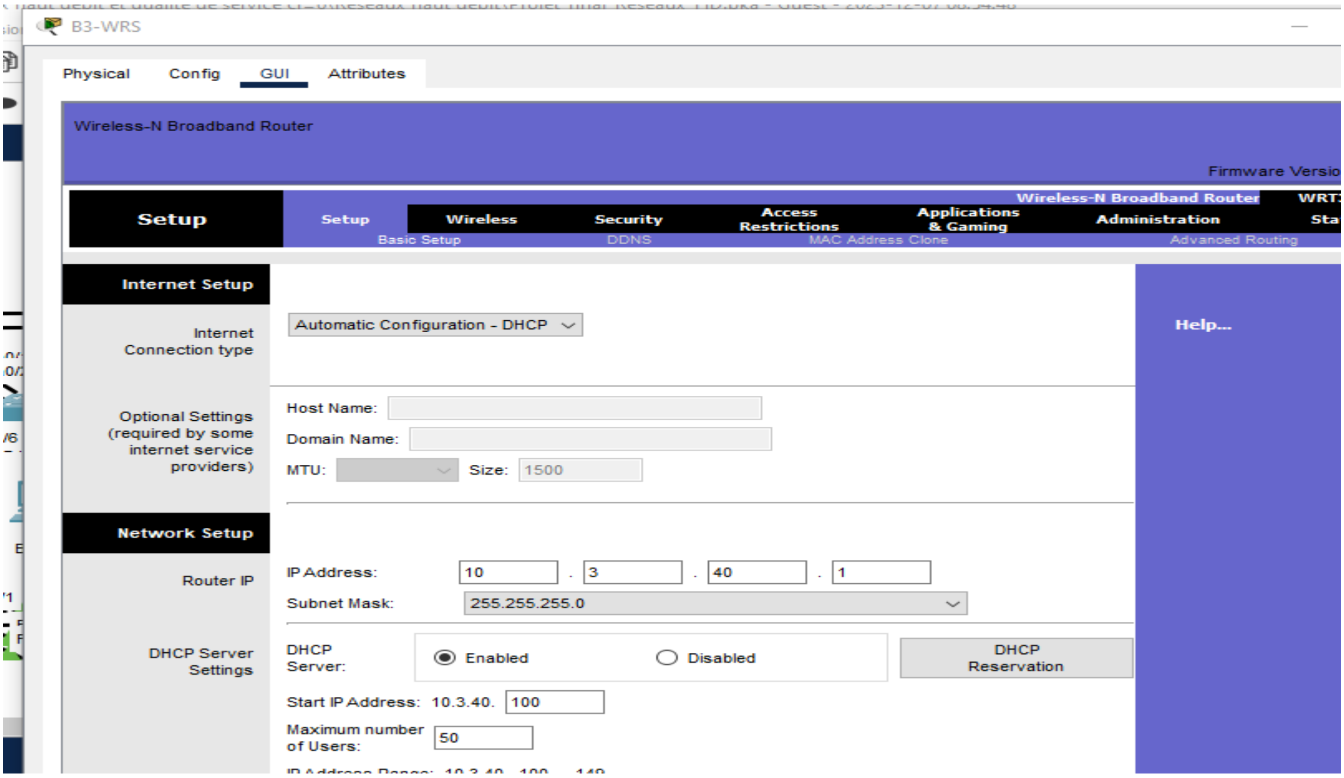


Étape 2 : configuration des paramètres de configuration réseau et de réseau local

Sous l'onglet GUI, le champ « Router IP » de la page Status doit correspondre à la première adresse IP du sous-réseau 10.X.40.0 /24. Conservez les valeurs par défaut des autres paramètres.

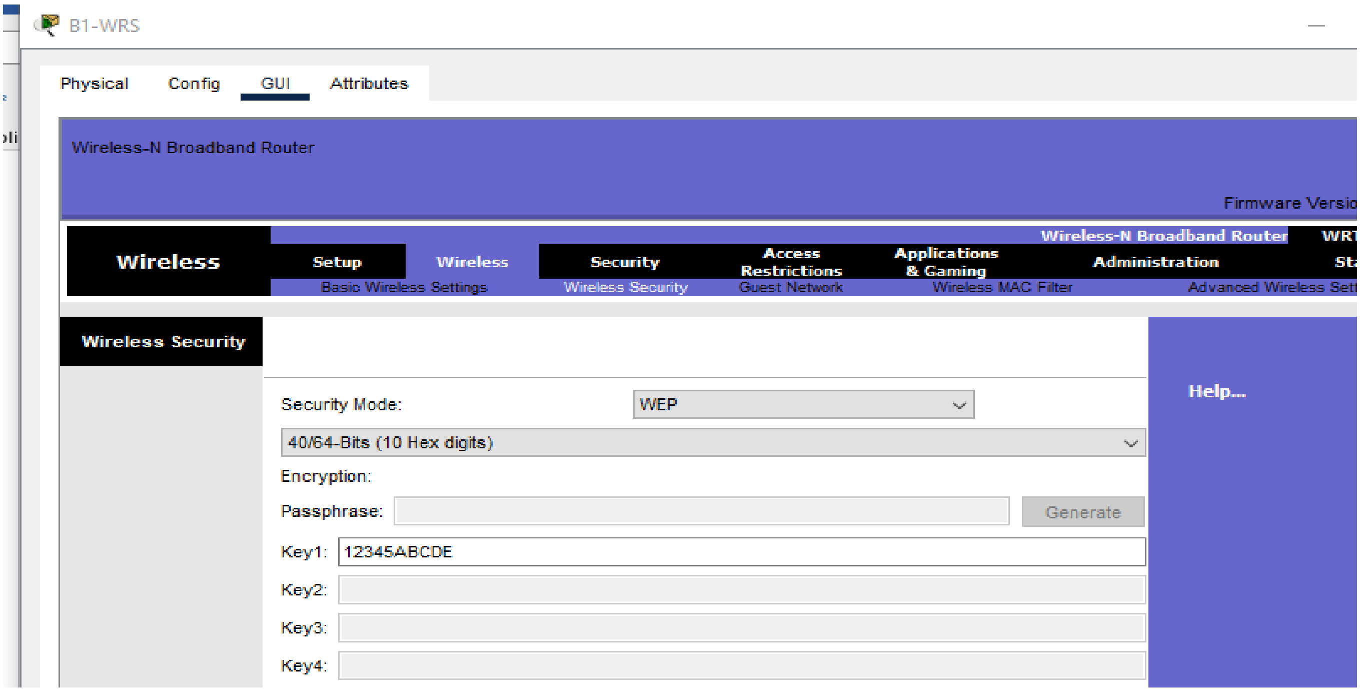






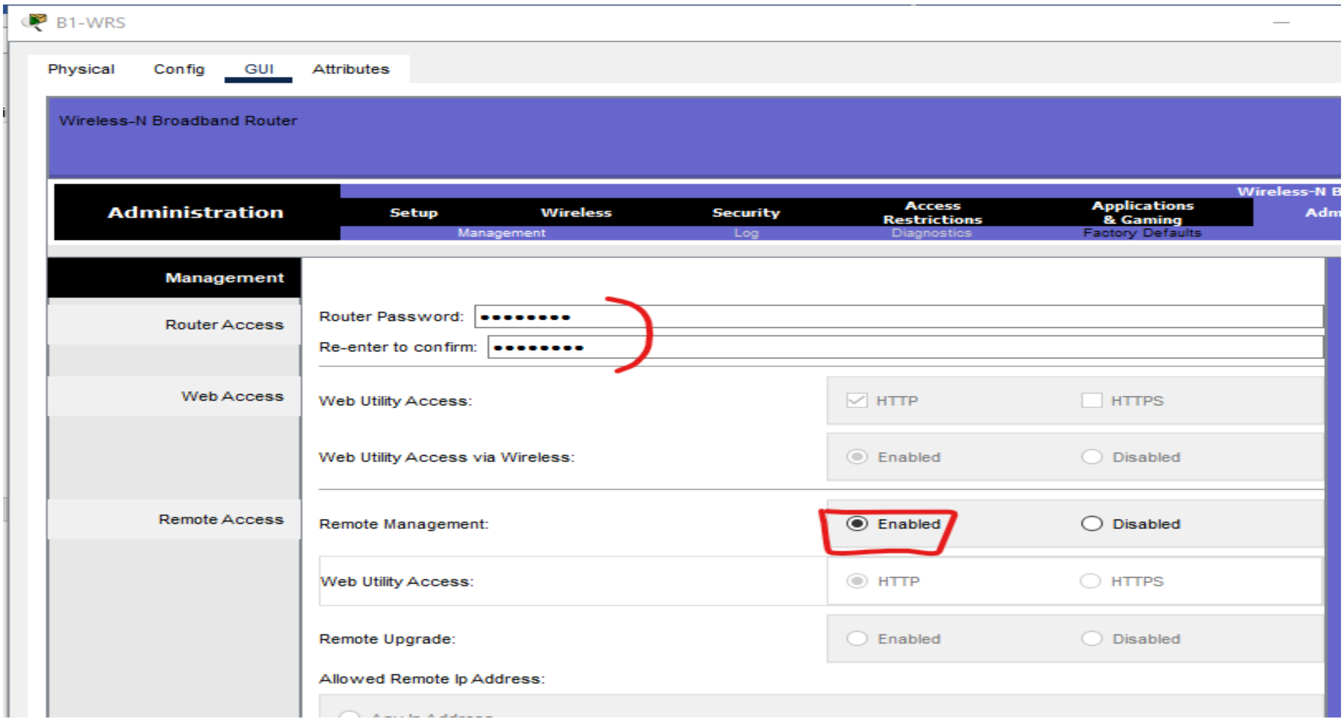
Étape 3 : configuration des paramètres de configuration de réseau sans fil

Les SSID des routeurs sont BX-WRS_LAN, où X correspond au numéro du routeur Branch. La clé WEP est 12345ABCDE.

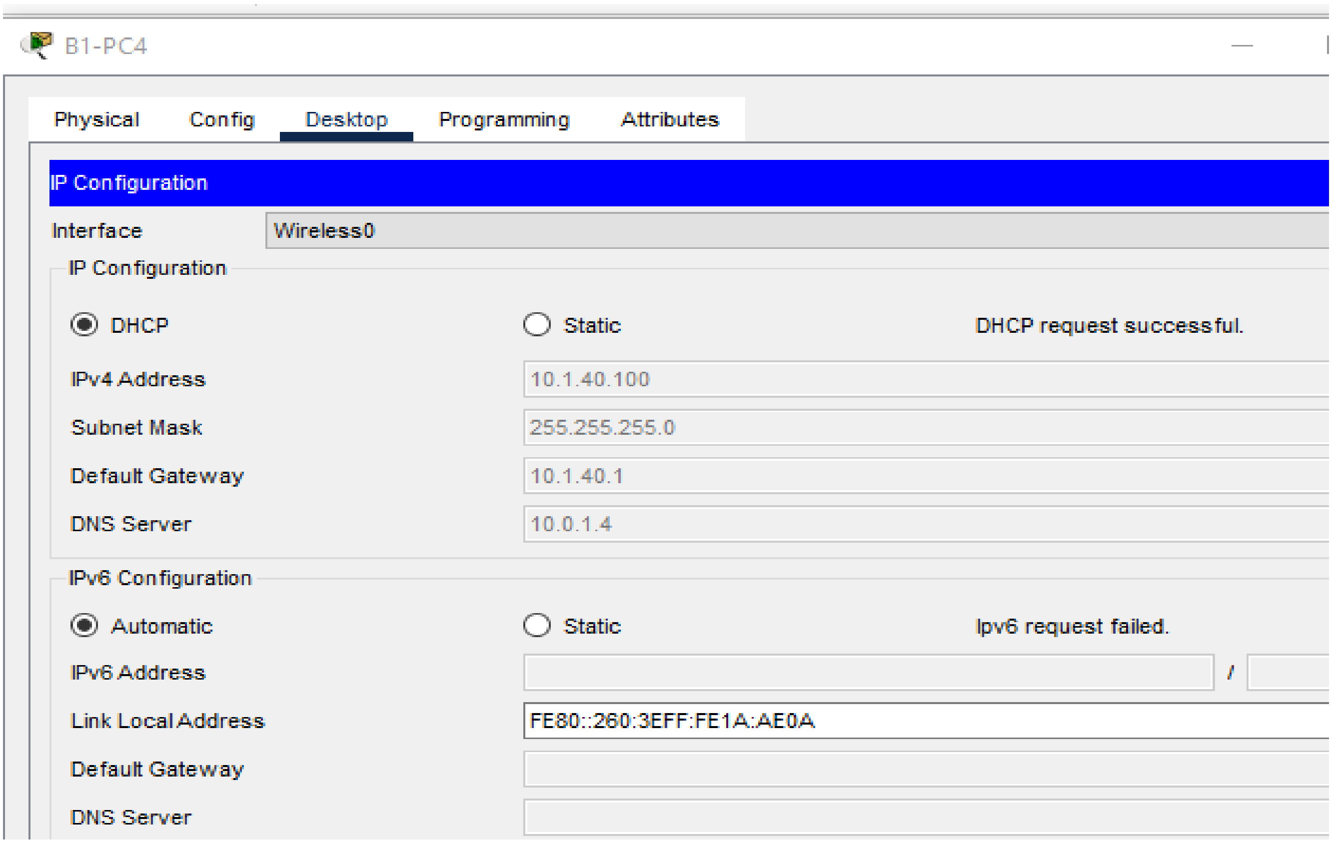


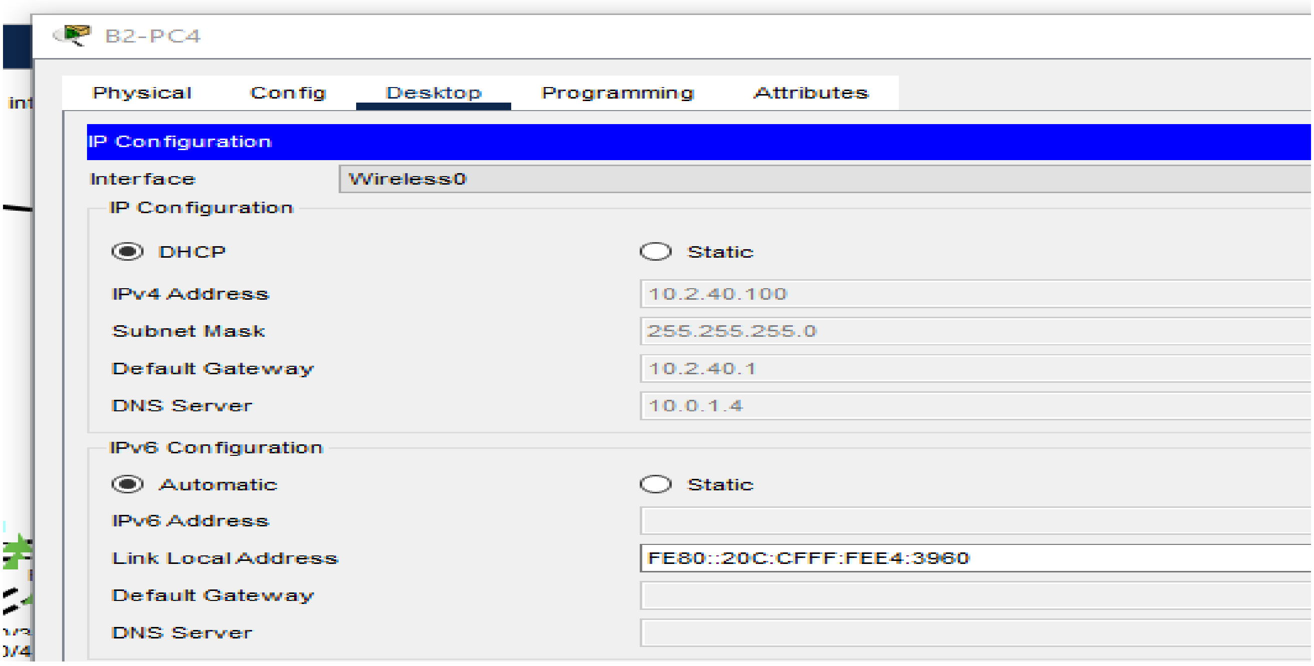
Étape 4 : configurez l'accès distant sur les routeurs sans fil.

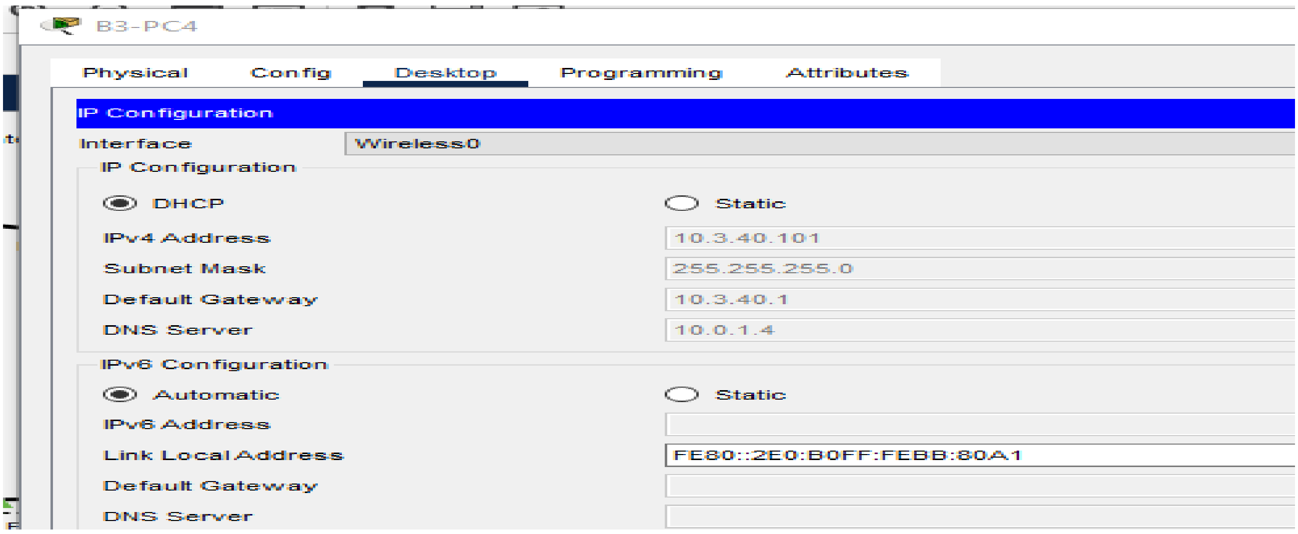
Définissez cisco123 comme mot de passe d'administration et activez l'administration à distance.



Étape 5 : configuration des PC BX-PC4 pour qu'ils accèdent au réseau sans fil par le biais du service DHCP

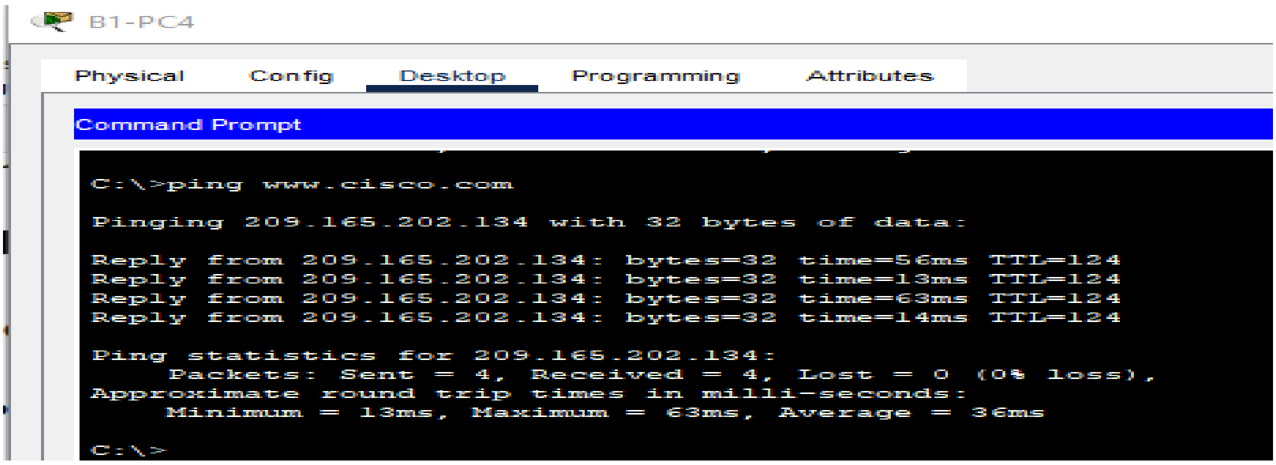


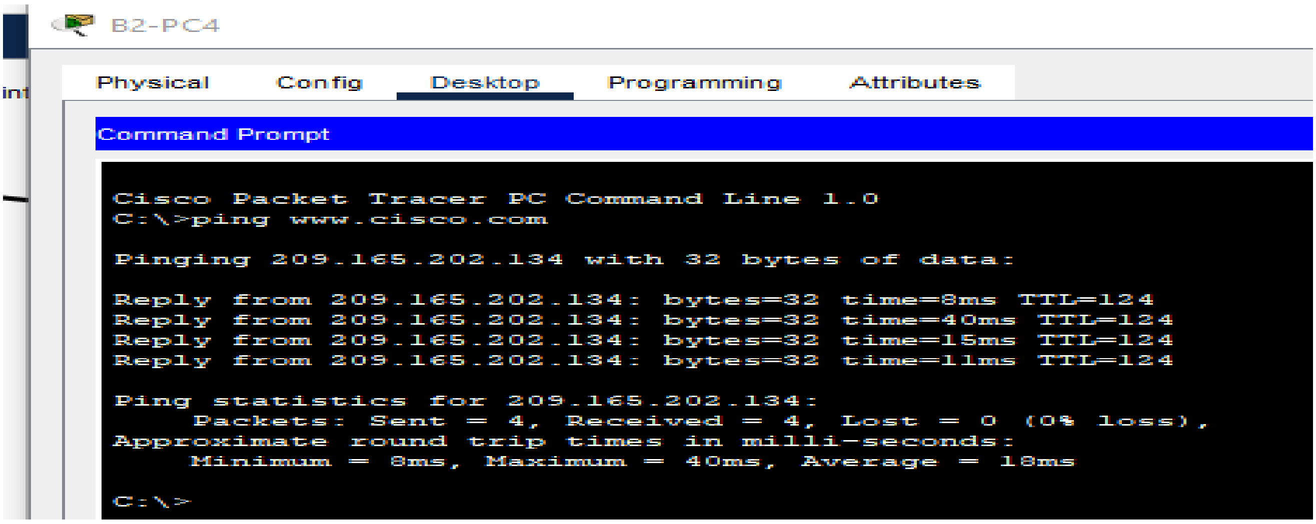


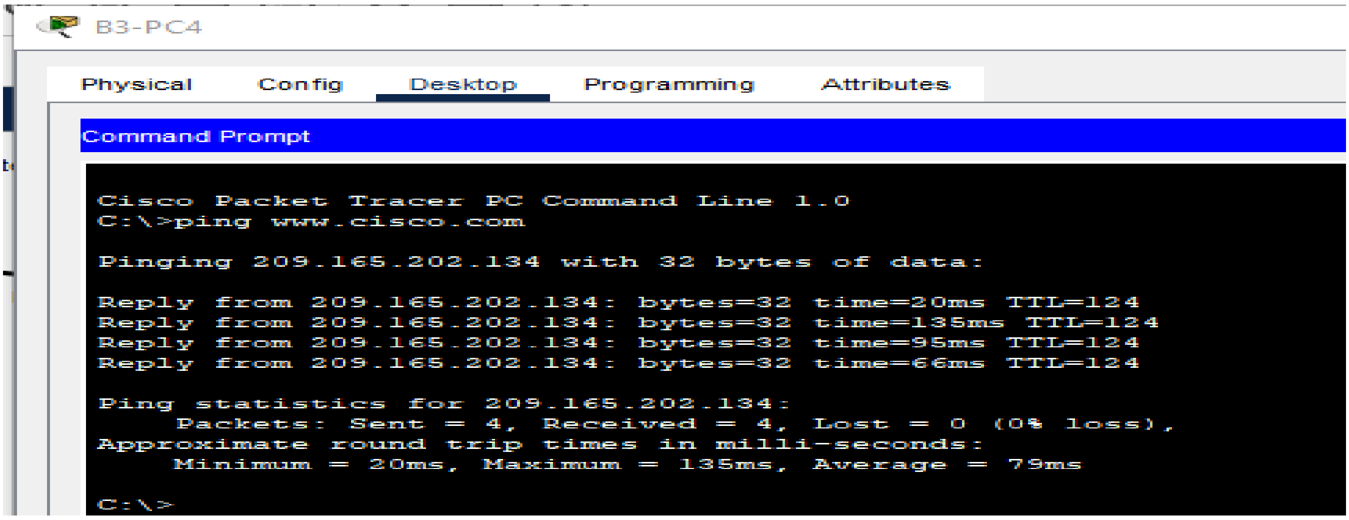


Etape 6 : vérification de la connectivité et du fonctionnement de l'administration à distance

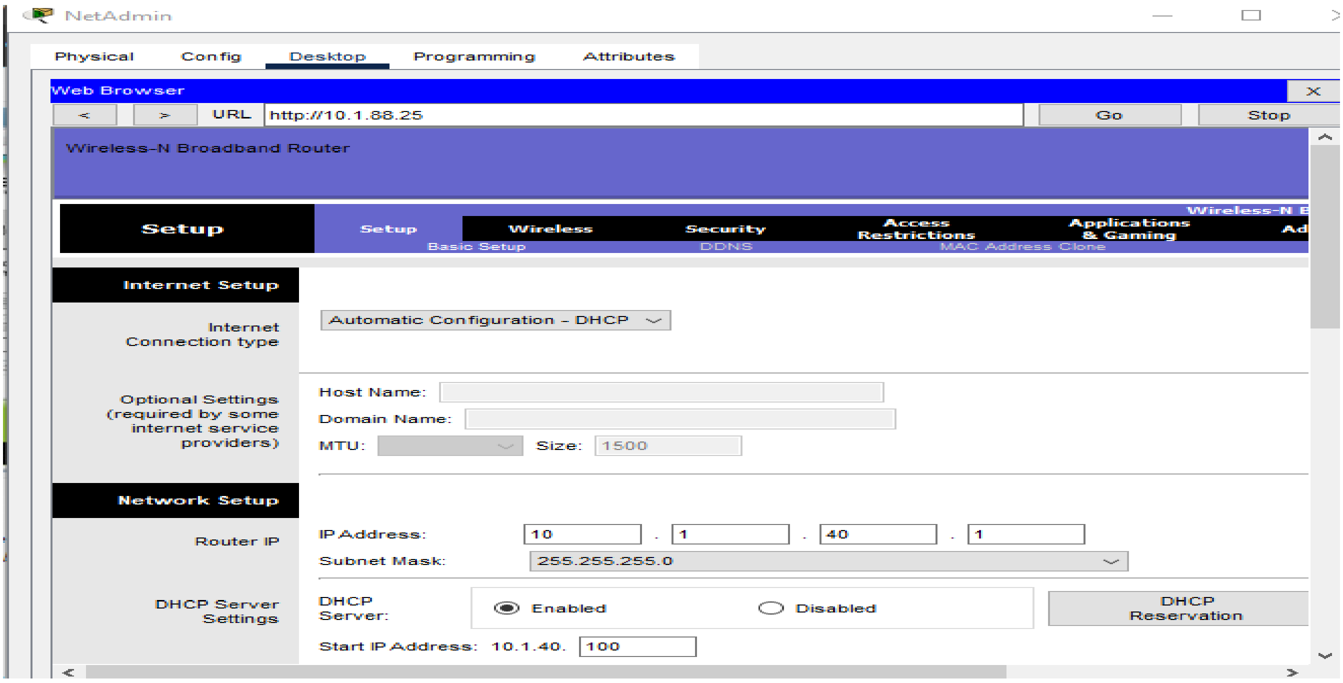
Tous les PC sans fil doivent pouvoir accéder au serveur Web www.cisco.com.

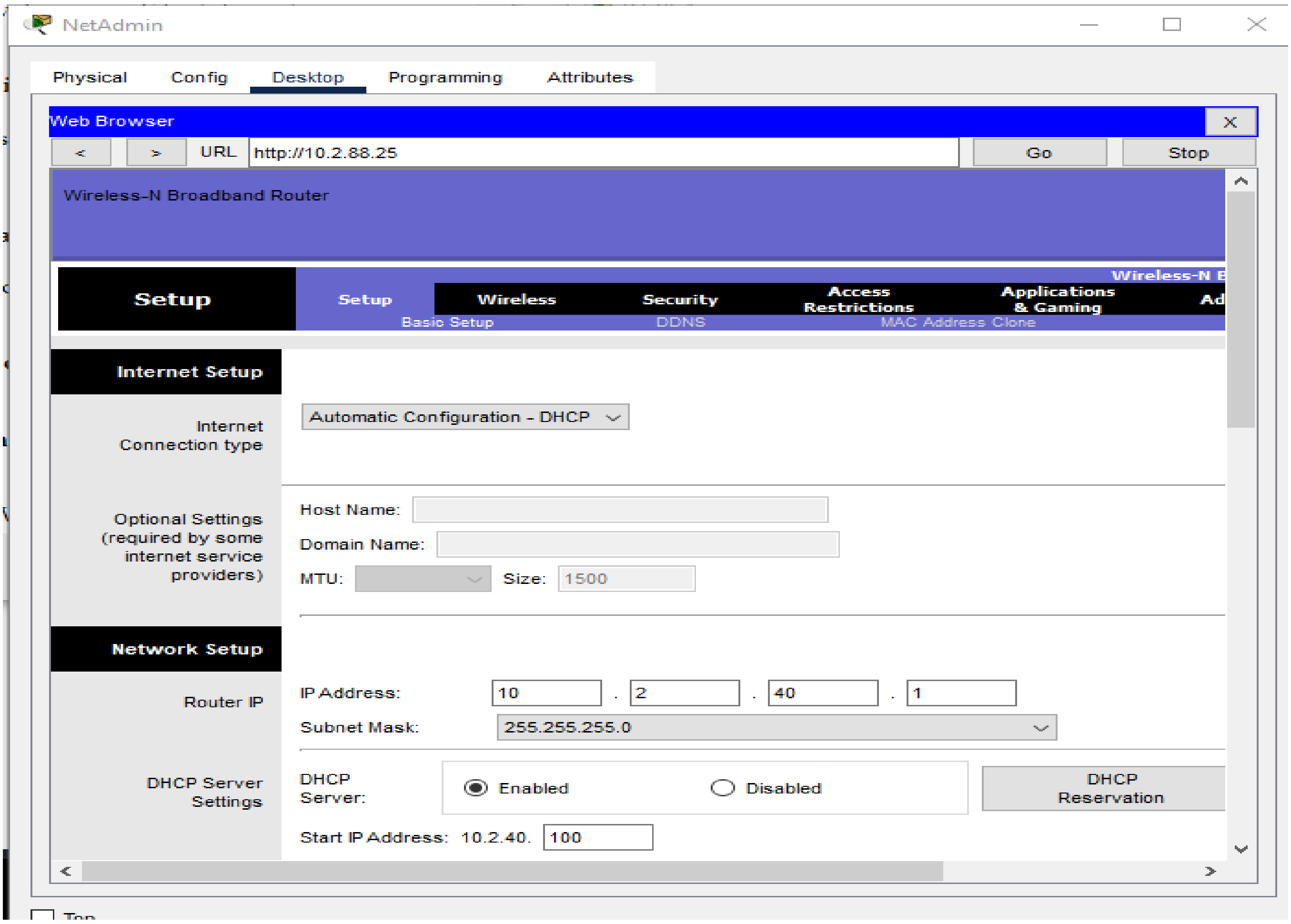


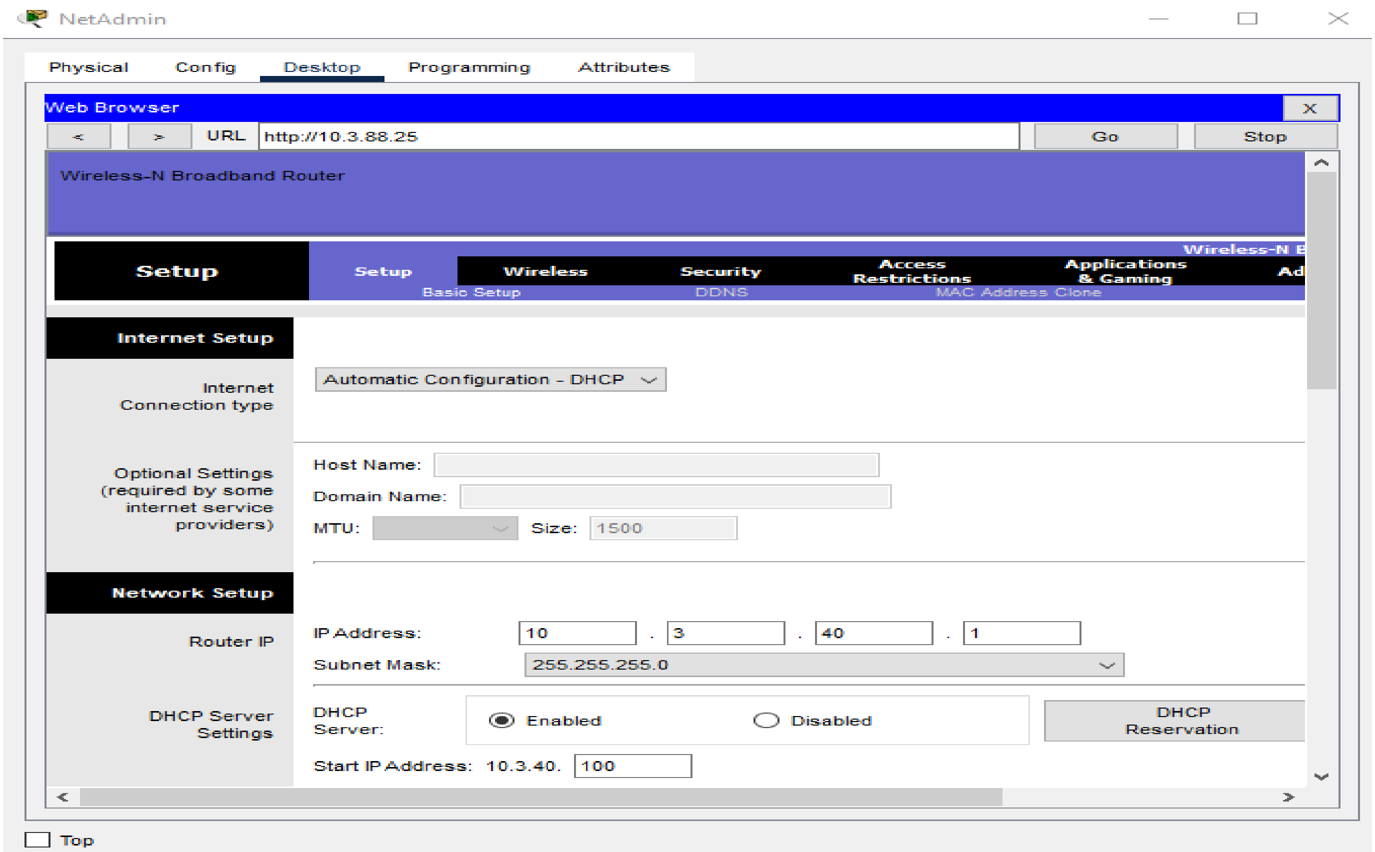


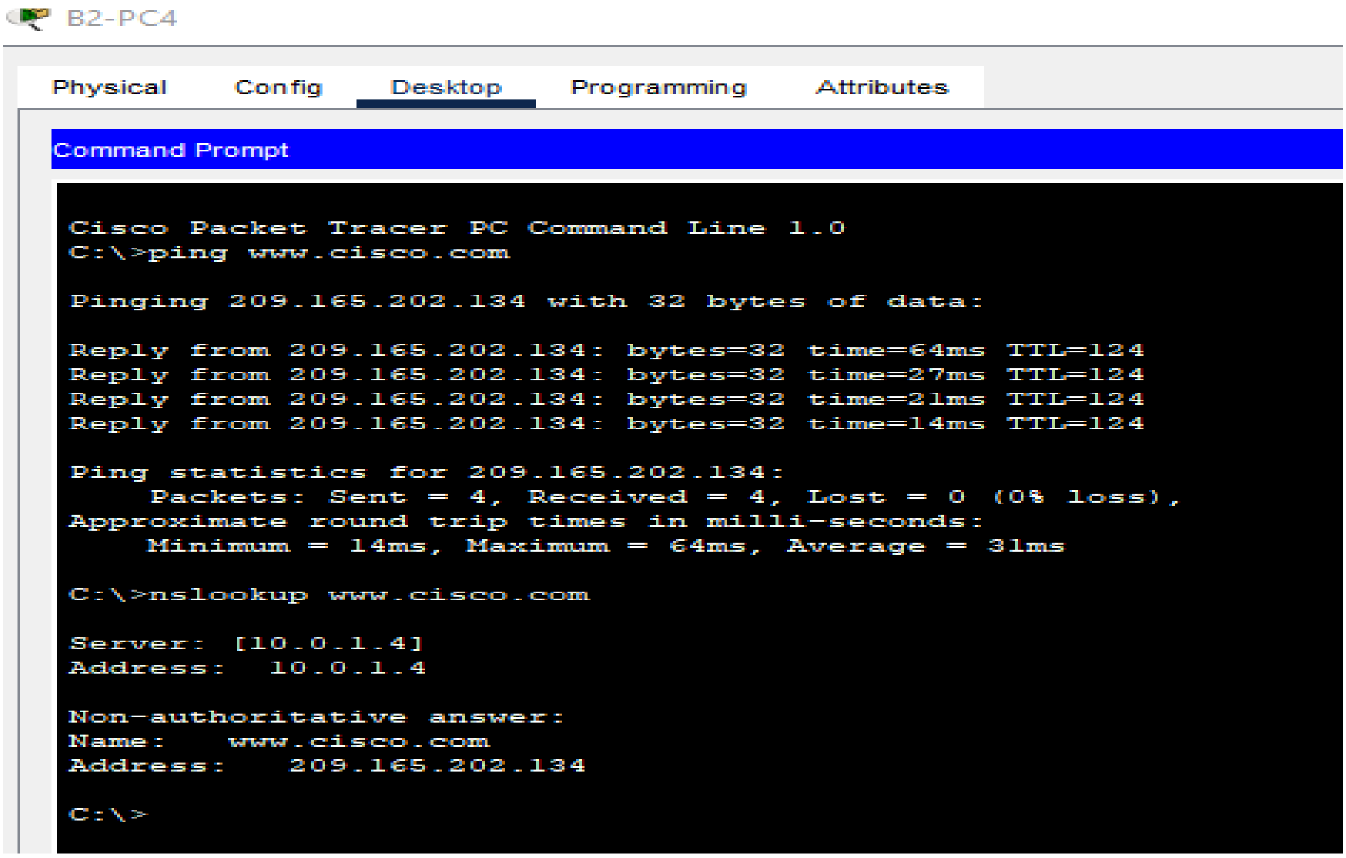


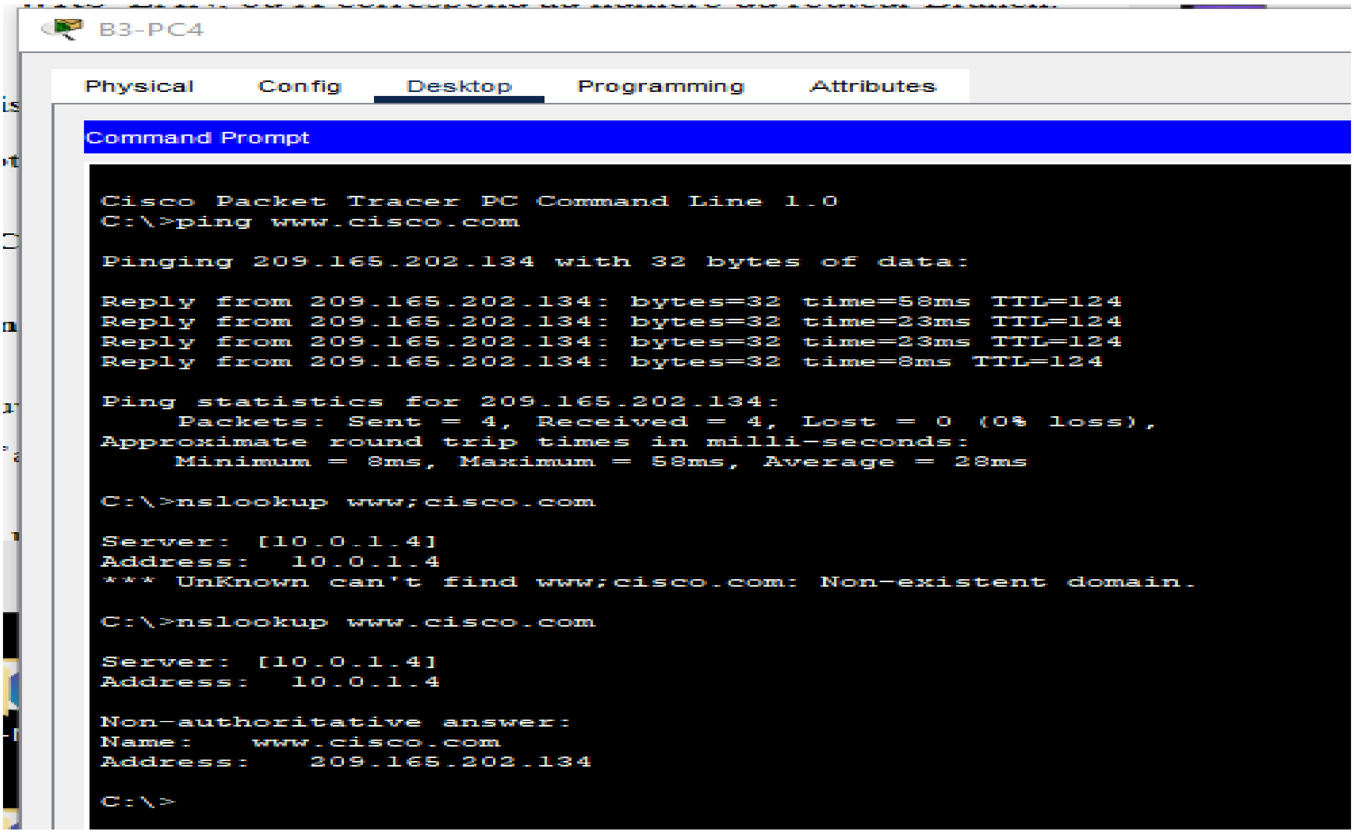
Vérifiez le fonctionnement de l'administration à distance en accédant au routeur sans fil par le biais du navigateur Web.




NetAdmin
Physical
Config
Desktop
Programming
Attributes
Web Browser
URL
http://10.2.88.25
Go
Stop
Wireless-N Broadband Router
Setup
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Basic Setup
DDNS
MAC Address Clone
Internet Setup
Internet Connection type
Automatic Configuration - DHCP
Optional Settings (required by some internet service providers)
Host Name:
Domain Name:
MTU:
Size: 1500
Network Setup
Router IP
IP Address: 10 . 2 . 40 . 1
Subnet Mask: 255.255.255.0
DHCP Server Settings
DHCP Server:
Enabled
Disabled
DHCP Reservation
Start IP Address: 10.2.40. 100


NetAdmin
Physical
Config
Desktop
Programming
Attributes
Web Browser
URL
http://10.3.88.25
Go
Stop
Wireless-N Broadband Router
Setup
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Basic Setup
DDNS
MAC Address Clone
Internet Setup
Internet Connection type
Automatic Configuration - DHCP
Optional Settings (required by some internet service providers)
Host Name:
Domain Name:
MTU:
Size: 1500
Network Setup
Router IP
IP Address: 10 . 3 . 40 . 1
Subnet Mask: 255.255.255.0
DHCP Server Settings
DHCP Server:
Enabled
Disabled
DHCP Reservation
Start IP Address: 10.3.40. 100
Top


B2-PC4
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.cisco.com
Pinging 209.165.202.134 with 32 bytes of data:
Reply from 209.165.202.134: bytes=32 time=64ms TTL=124
Reply from 209.165.202.134: bytes=32 time=27ms TTL=124
Reply from 209.165.202.134: bytes=32 time=21ms TTL=124
Reply from 209.165.202.134: bytes=32 time=14ms TTL=124
Ping statistics for 209.165.202.134:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 14ms, Maximum = 64ms, Average = 31ms
C:\>nslookup www.cisco.com
Server: [10.0.1.4]
Address: 10.0.1.4
Non-authoritative answer:
Name: www.cisco.com
Address: 209.165.202.134
C:\>


B3-PC4
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.cisco.com
Pinging 209.165.202.134 with 32 bytes of data:
Reply from 209.165.202.134: bytes=32 time=58ms TTL=124
Reply from 209.165.202.134: bytes=32 time=23ms TTL=124
Reply from 209.165.202.134: bytes=32 time=23ms TTL=124
Reply from 209.165.202.134: bytes=32 time=8ms TTL=124
Ping statistics for 209.165.202.134:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 8ms, Maximum = 58ms, Average = 28ms
C:\>nslookup www;cisco.com
Server: [10.0.1.4]
Address: 10.0.1.4
*** UnKnown can't find www;cisco.com: Non-existent domain.
C:\>nslookup www.cisco.com
Server: [10.0.1.4]
Address: 10.0.1.4
Non-authoritative answer:
Name: www.cisco.com
Address: 209.165.202.134
C:\>